梁思成

《圖像中國建築史》

手繪圖

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畫棟描梁 圖解營造

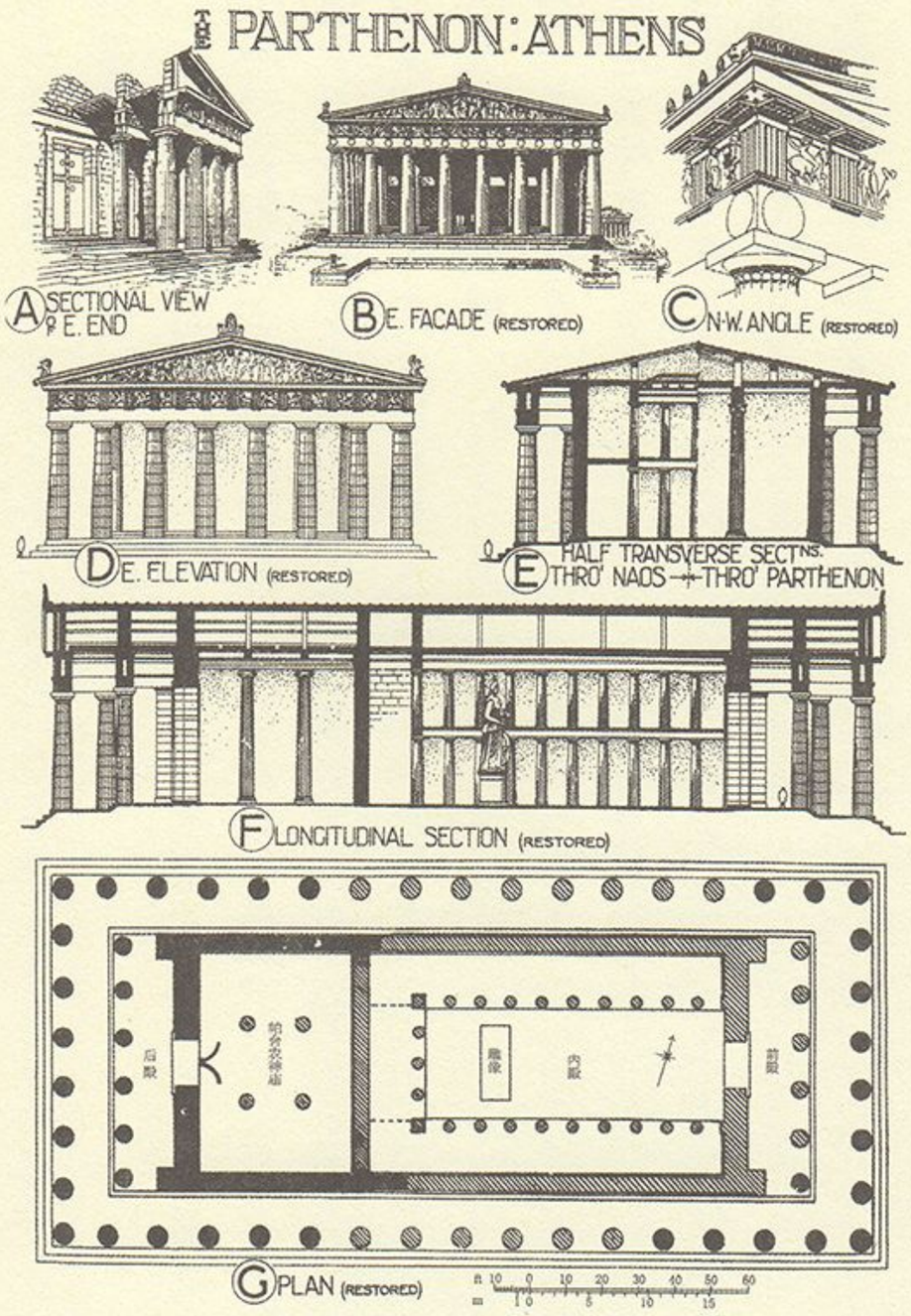
王南

本圖册向讀者呈現的數十幅手繪建築圖紙，出自梁思成先生一九四六年在四川省南溪縣李莊完成的《圖像中國建築史》（英文版，原名爲 *A Pictorial History of Chinese Architecture*）一書。這是一部向西方人介紹中國古代建築史的學術專著，雖然由于種種歷史原因，直到將近四十年後（即一九八四年）才得以在美國正式出版[1]，但依然因其杰出的學術貢獻而獲得當年「全美最優秀出版物」的榮譽。該書的最主要特色，是通過精心繪製的建築圖紙和建築實景照片，以及十分扼要的文字，給予西方讀者一個關于中國古代建築歷史發展的簡潔明晰的概括性認識，因此從某種程度上説，書中插圖的作用甚至比文字還要重要。這本圖解中國古代建築史著作的成功，很大程度上得益于其豐富而翔實的插圖，尤其是本册收録的這批手繪建築圖紙。

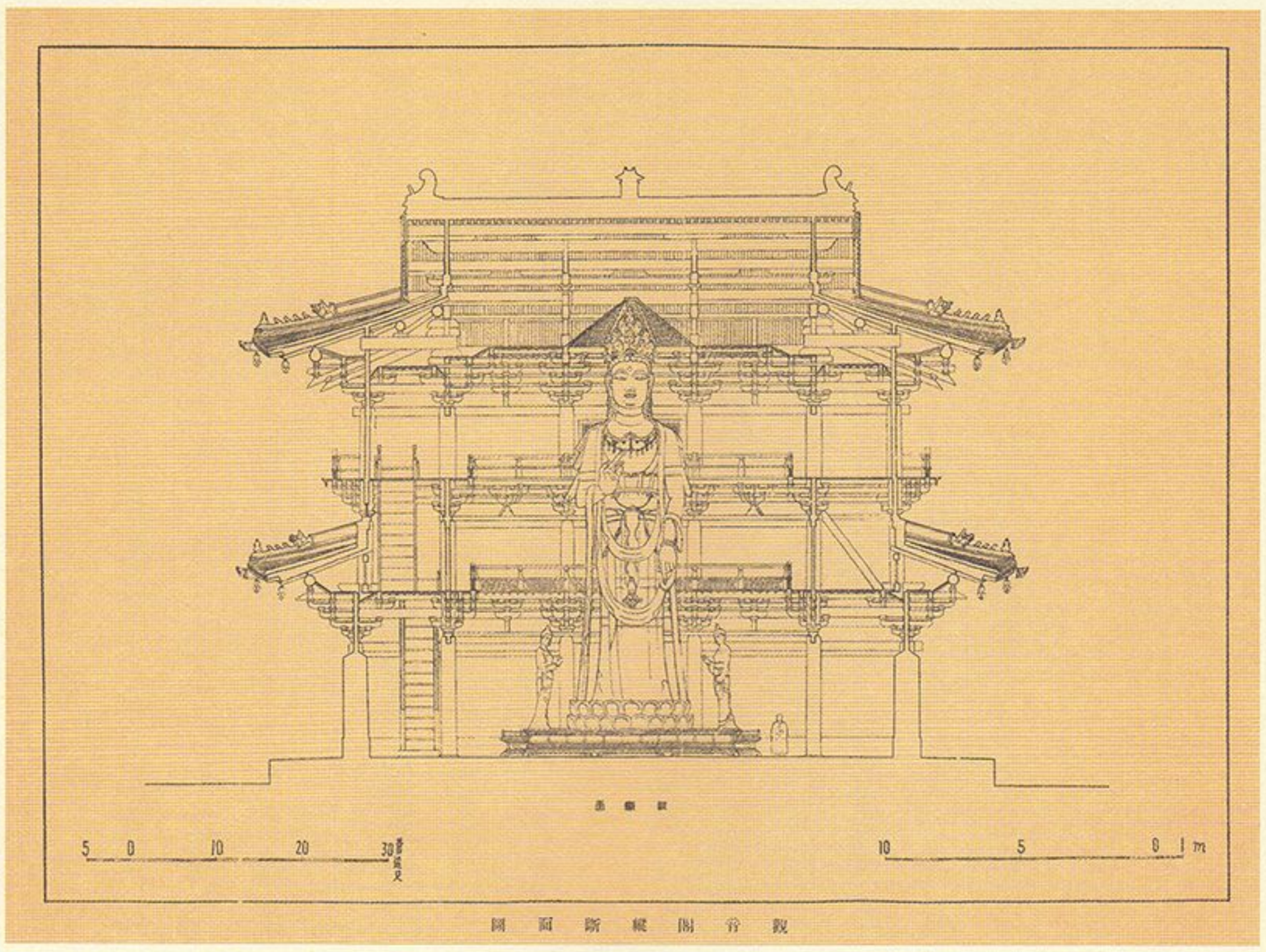
早在二十世紀二十年代赴美留學攻讀建築學專業期間，梁思成就對建築歷史學科産生了濃厚興趣。那時的他常常在賓夕法尼亞大學建築系的圖書館中飽覽西方建築史的經典著作，掩卷沉思之余，發思古之幽情。與此同時，年輕的梁思成也深爲一個文明古國竟然没有一部由中國人自己撰寫的建築史專著而感到惋惜，由此立下畢生最爲重要的一個志向：寫一部中國人自己的《中國建築史》。

從一九三一年起，梁思成加入了中國第一個專門研究中國古建築的學術機構——「中國營造學社」。在社長朱啓鈐的領導和大力支持下，梁思成與學社同仁開始對中國的田野考察，梁思成和學社同仁共計考察了全中國二百多個市、縣數以千計的古建築，并且對其中大多數建築進行了精細測繪。經過長達十多年的許多重要遺存，梁思成得以在此基礎上大致廓清了中國古代建築的發展脉絡。尤爲可貴的是，在抗日戰争時期顛沛流離、貧病交加的生活條件下，梁思成帶領學社成員在中國西南部繼續堅持着古建築考察與學術研究，并最終于一九四四年抗戰勝利前夕，在李莊這個小村里，完成了他平生第一部《中國建築史》的書稿；兩年後，又完成了英文版的《圖像中國建築史》書稿，爲的是能更方便地向西方世界介紹中國古代建築的歷史發展與輝煌成就。

《圖像中國建築史》里的手繪插圖，是專爲此書量身繪製的，全套圖紙均達到了當時的世界一流水准，即便在今天看來，依然是中國古代建築史研究插圖中的經典之作，是一座難以逾越的高峰。早在賓大求學時期，梁思成就十分推崇西方建築史的經典著作《弗萊徹建築史》一書的插圖。該書典型的插圖樣式是把一批經典建築或者單座重要建築的不同圖紙，通過精心安排的構圖，組合成一幅圖文并茂的大圖，從而增加了這部建築史



《弗萊徹建築史》中的帕提農神廟。圖片來源：《弗萊徹建築史》（原書第二十版）



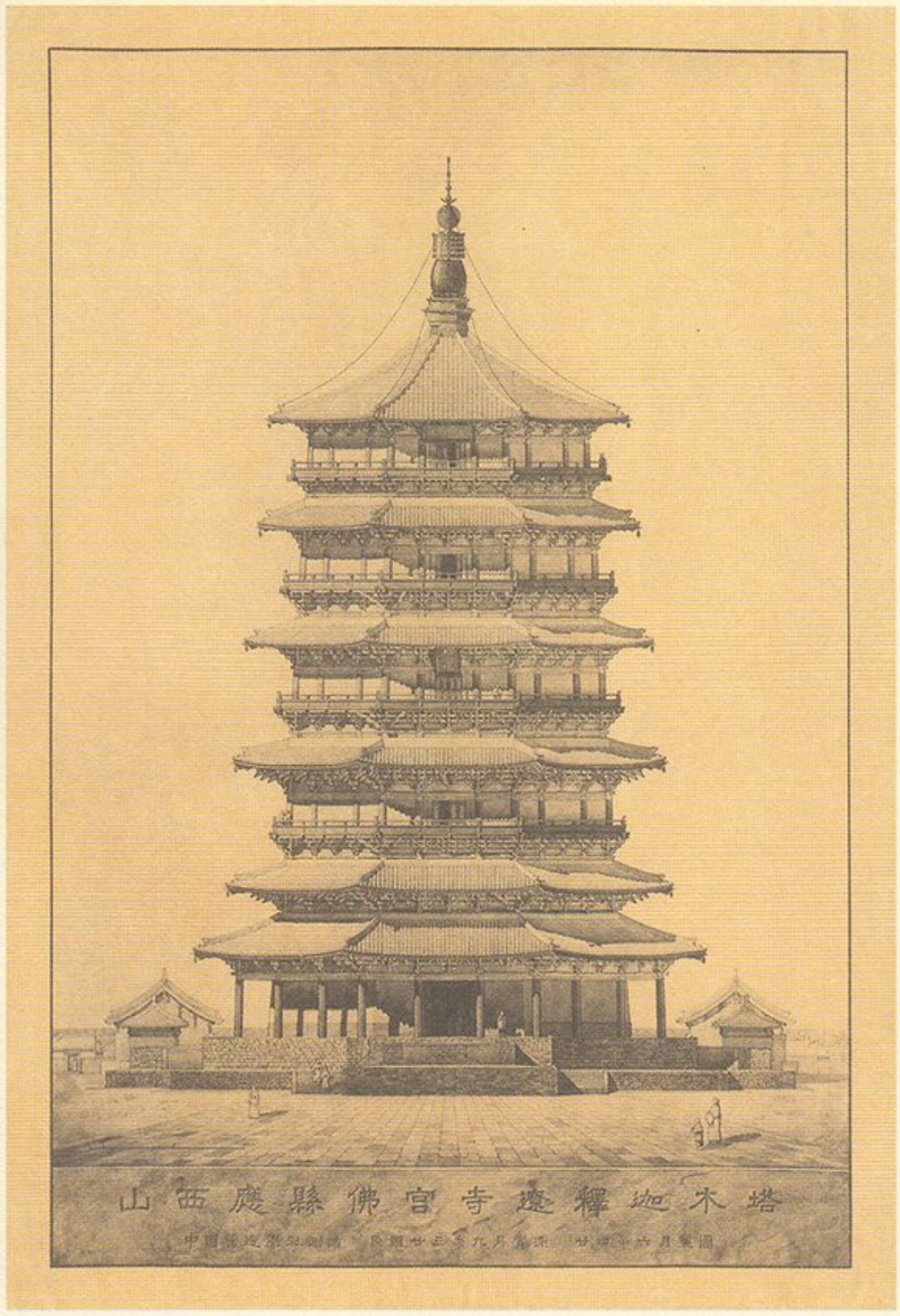
梁思成繪製的獨樂寺觀音閣剖面圖。圖片來源：《中國營造學社匯刊》第三卷第二期

專著的可讀性。梁思成對自己的專著或者論文中的建築圖紙，一直都是以達到《弗萊徹建築史》等西方經典著作的世界級水准作爲基本要求。一九三二年，梁思成完成了他的第一篇中國古建築考察報告《薊縣獨樂寺觀音閣山門考》，在這篇論文中，他已經繪製出具有世界一流水平的插圖，尤其是其中的巨幅獨樂寺觀音閣立面渲染圖和剖面圖，將觀音閣無比複雜的木構架，尤其是多達二十余種不同樣式的斗栱構造，表現得有條不紊，并且還把位于樓閣中央、高達十七米的觀音立像也畫在了建築剖面圖中，完美地呈現出這座建于公元九九八四年的遼代木結構樓閣的動人神韻。

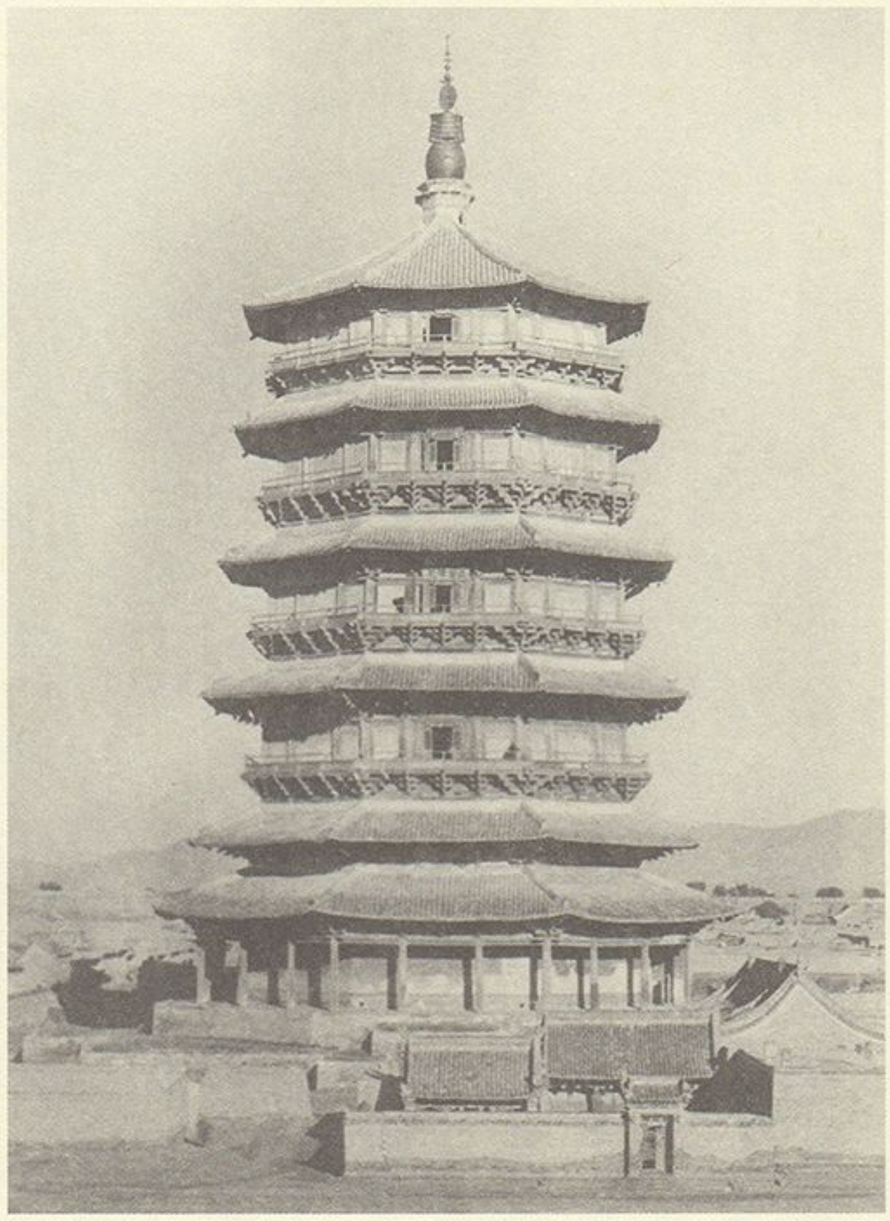
梁思成不僅對自己的繪圖嚴格要求，還特地讓專門幫他繪製插圖的助手莫宗江也去仔細研究《弗萊徹建築史》的插圖，要求自己的弟子也要達到同樣的水准。高標准的要求，加上梁思成、莫宗江二人十余年來勤勉的訓練，他們的建築繪圖技巧均達到了極高水平。他們筆下的中國古建築測繪圖，一方面秉承了西方建築學的製圖手法及其蘊含的西方古典主義美學精神[2]，一方面又創造性地融入了中國傳統工筆和白描的技巧，故而能更好地呈現中國古建築獨特的美感。這是西方和日本學者研究中國古建築時都不曾達到的藝術境界，在世界建築史經典著作的插圖風格中也可謂獨樹一幟。

中國古建築的造型極爲特殊，無論是木結構建築中由柱、梁和斗栱等構件組合而成的富于美感的木構架，還是凹曲面的大屋頂上各式各樣造型的瓦飾，以及豐富而優美的建築裝飾紋樣，甚至建築室內形態各异的雕像，所有這一切在梁、莫師徒二人的筆下，均得到生動的表現。特別是中國古建築中帶有大量複雜而微妙的曲綫，大到屋面舉折或舉架形成的坡度，或者屋角的起翹，小到柱子、月梁和斗栱等構件的「卷殺」曲綫，以及鴟吻、脊獸、瓦當、滴水等各種瓦飾的曲綫，乃至剖面圖中被剖切的木構件中呈現的木紋，所有這些曲綫造型，有時是通過精確測量或計算的一組折綫加以表現，但更多時候則要通過徒手繪製的微妙曲綫來完成——可以説本圖册中的每一幅插圖，都是尺規作圖和徒手繪畫的完美結合。這批墨綫圖中那一根根粗細得當、曲直自如的綫條下所呈現出的中國古代匠師的鬼斧神工。梁思成的藝術品來欣賞。

值得一提的是，梁、莫二人測繪生涯中完成的難度最高的一批圖紙，是山西應縣木塔的測繪圖。這座歷時千年、高約六十七米的世界現存最古老、最高大的木塔，外觀五層、内部九層，通體由數萬個木構件組成，斗栱更是多達五十六種不同樣式，每一層中央都有不同的佛教造像組合，真是中國古代匠師的空洞爬上屋面，并和莫宗江于一九三四年對其進行了全面測量，梁思成甚至冒着生命危險，從木塔頂層屋頂的抓住冰冷的鐵鏈，一路爬上金屬塔刹的頂端進行測量，没有任何保護措施。此後即着手繪製木塔的測繪圖，直到一九三六年底，調查報告的文稿和插圖才全部完成，其中木塔正立面的渲染圖和剖面圖（完成于一九三五年），可謂其繪圖生涯的巔峰之作。繪製這兩幅驚世之作時，梁思成三十四歲，莫宗江僅十九歲（相當于當今大學本科二年級學生）。可惜的是，當時這批圖紙和考察報告剛剛交付印刷廠，即爆發「盧溝橋事變」，梁思成和學社成員也開始了逃難生涯。在李莊期間，由于缺乏繪圖材料，應縣木塔測繪圖没有條件重繪，與本册收録的《圖像中國建築史》其他手繪圖的體例也有所不同。爲了彌補這一遺憾，我們特地找到了早期的應縣木塔測繪圖紙配發在這里，以饗廣大讀者。

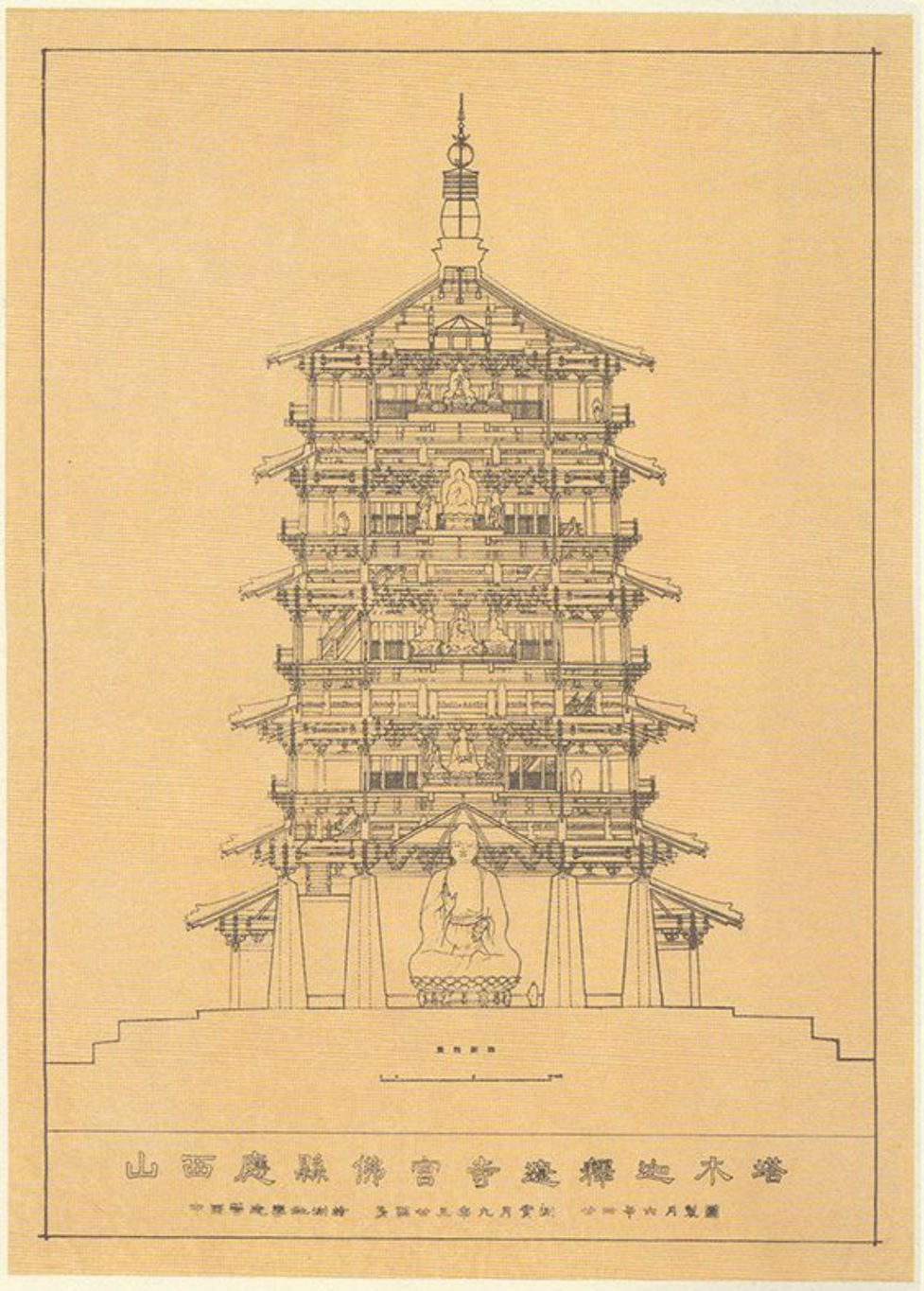


山西應縣佛宮寺釋迦木塔立面圖。



山西應縣佛宮寺釋迦木塔照片。

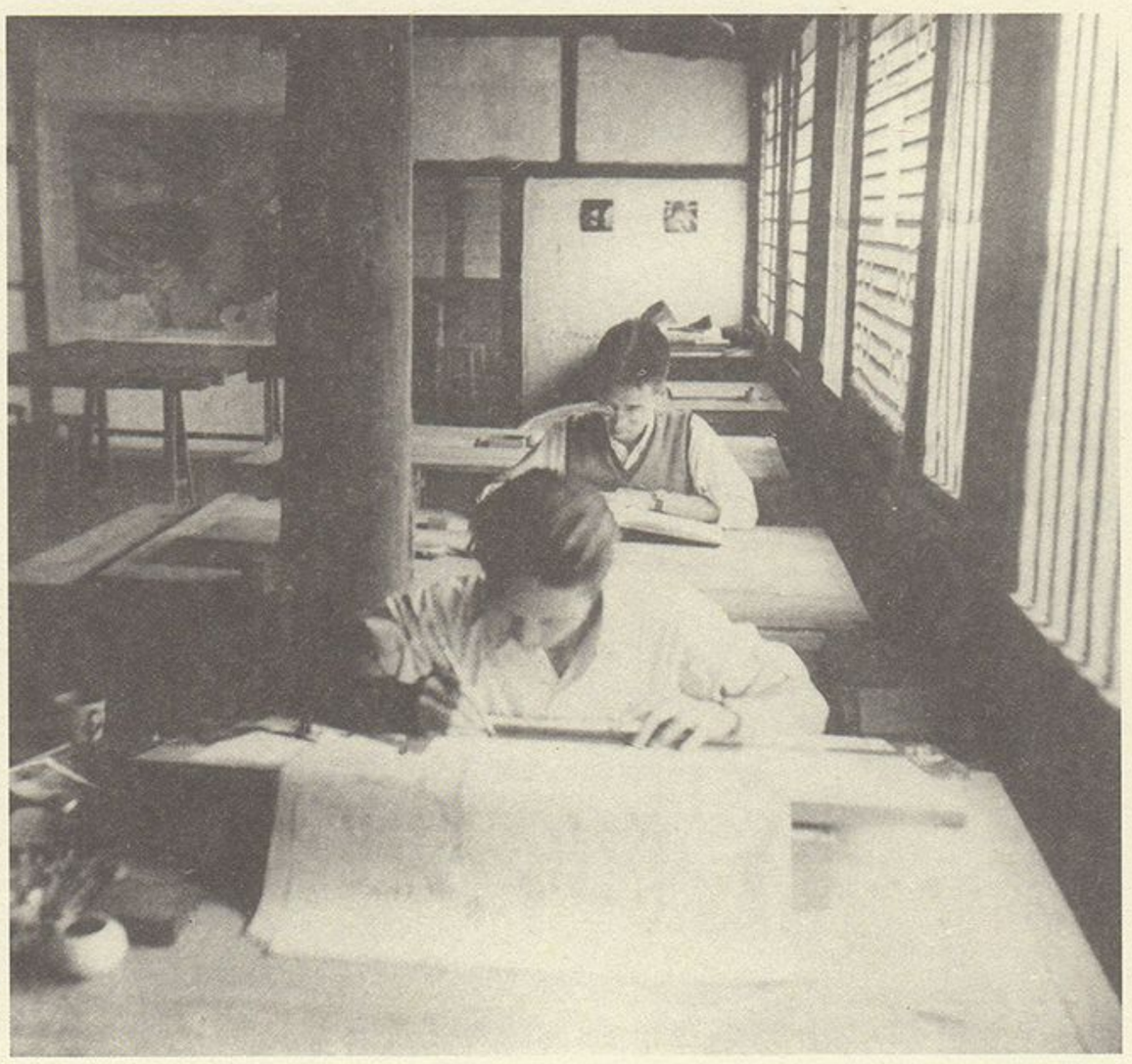
對于這批圖紙高度的藝術性，我們在贊嘆之余，還要特別説明的是，繪製這些杰作的工具，僅僅是簡陋的鴨嘴筆和黑墨水而已。現在我們購買圓規時一般還配有鴨嘴筆，位于安鉛筆芯一端的另一頭，可以將墨水滴在鴨嘴筆的「鴨嘴」之間來繪製墨綫。當年梁思成、莫宗江繪製這些墨綫圖時，就是把墨水滴在鴨嘴筆上，并通過調節筆側面的螺絲松緊來控制墨水流下的速度，從而控制墨綫的粗細，這實在是十分費事而且難以控制的工具。不僅如此，學社在李莊期間，研究經費没了着落，所有人都過着吃不飽、穿不暖的困頓生活，梁思成更是由于身體的病痛，必須穿着笨重的「鐵馬甲」支撑彎曲的腰椎來畫圖。讀者可以想象書中這些美麗的圖紙，是靠着什么樣的驚人毅力繪製出來的。今天讀建築系的學生，可以輕而易舉地買到進口的各種粗細規格的針管筆或者一次性墨水筆，再也無從體會用鴨嘴筆繪圖的艱難了。與此形成鮮明對照的是，手握高級繪圖工具的當代學子們，却再也畫不出梁、莫二人那么精彩的圖。特別是由于計算機製圖的發明，學生們連使用針管筆和繪圖板畫圖的機會都少之又少，而這本應是建築系學生區别于大學其他專業學生的重要標志之一。建築系的學生不再使用鴨嘴筆、針管筆、丁字尺、三角板、圖板和繪圖桌，不但失去了建築師用手繪圖的樂趣，同時失去的還有建築製圖中的美感體驗，以及通過心手相應來進行建築設計構思的諸般靈感和快感。



山西應縣佛宮寺釋迦木塔剖面圖。

這批圖紙除了高度的藝術性，另一個重要價值在于，它們清晰地勾勒出中國古代建築史的概要，所以即便不看任何說明文字，單是欣賞這批插圖，也能對中國古建築有個粗略的了解。尤其是當中許多專門繪製的「綜合集成式」插圖（借鑒了《弗萊徹建築史》的做法），由一批内容密切相關的小圖共同構成完整的大圖，并且在圖中加入中英文雙語解説，圖文并茂，信息量大，很像當下的「繪本」，十分適合現在這個「讀圖時代」。從這個意義上看，梁思成在二十世紀四十年代用心繪製的插圖，在不經意間契合了七十多年後的當下。

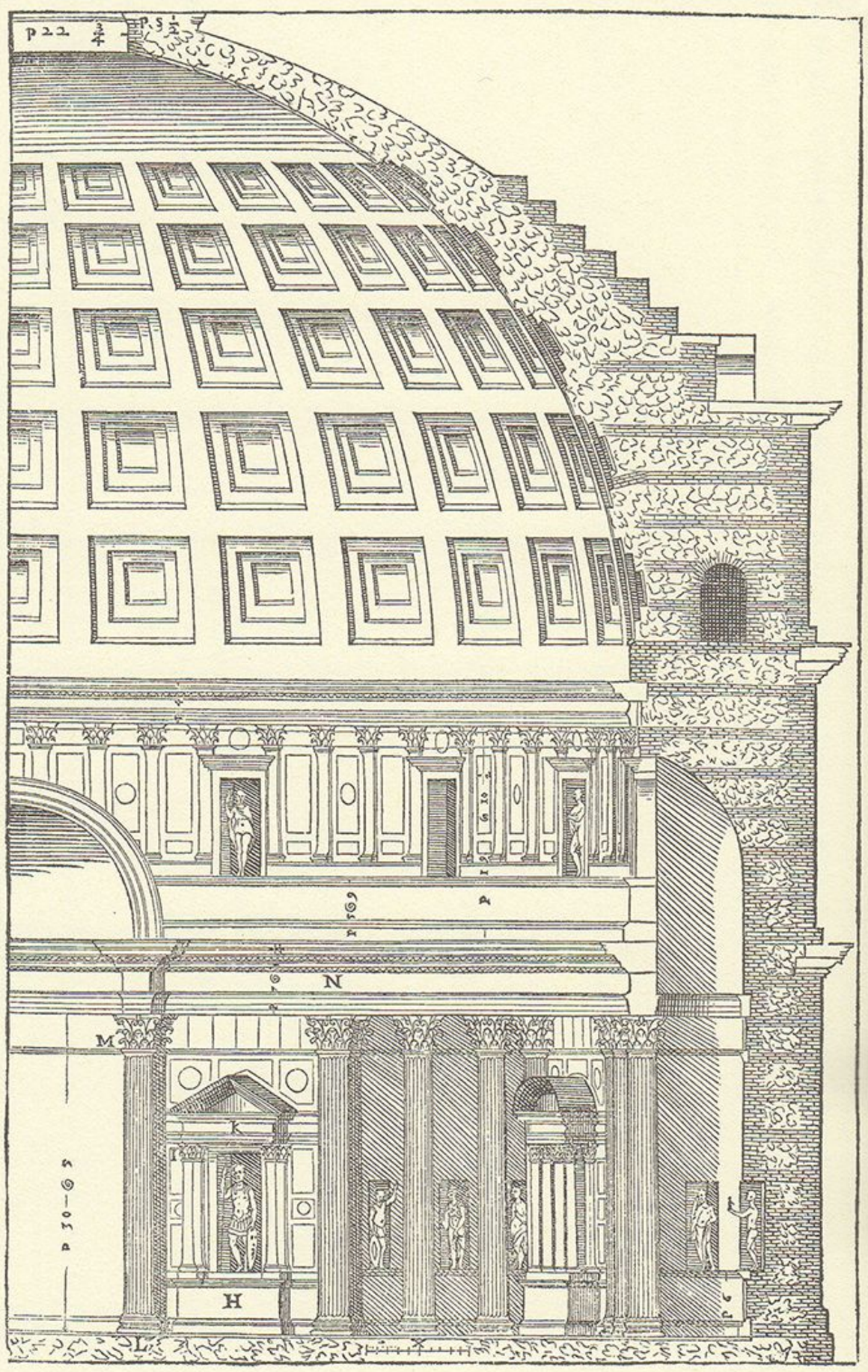
縱觀這數十幅圖紙，首先用一幅中國古建築標准正立面圖，幫助讀者直觀了解中國古代單體建築由臺基（或曰階基）、屋身（由立柱和斗栱組成）及屋頂組成的三段式經典構圖，同時對一些主要部位的名稱進行解釋。然後是一幅中國古代木結構建築基本法式（即order，多譯作「柱式」）加以類比，意在展現中國傳統木結構建築以材分（宋式）或者斗口（清式）作爲基本的模度，然後使建築各部分構件尺寸皆與基本模度成比例的「標准化」、「模數化」的營造方法，這其實是中



梁思成（後）、莫宗江在李莊營造學社繪圖的場景。圖片由清華大學建築學院中國營造學社紀念館提供

國古建築的精髓所在。圖中也以宋式爲例，解釋了木結構建築自上而下一系列主要構件的名稱。隨後的兩幅圖，分別是北宋的《營造法式》和清代的《工部工程做法》這兩部十分難得保存下來的中國古代官方建築專書中所體現的木構建築的基本法式和原則。這兩部古籍被梁思成譽爲「中國建築的兩部文法課本」，因爲在他看來，它們所體現的關于木結構建築的基本營造法則，就像是中國古建築的「語法」一樣重要，只有掌握了這些基本原則，才有可能進一步寫出關于中國建築的好「文章」。以材分或斗口作爲木結構建築基本模度的標准化、模數化的營造方法，正是以上兩部古書的核心。以上四幅圖構成一個重要的基本綱領。之後的圖紙則主要表現了中國各個重要歷史時期木結構建築的發展，其中，先秦、秦漢和魏晋南北朝時期已無任何地上木結構建築遺存，于是主要通過間接的圖像資料加以表現，包括安陽殷墟考古遺迹，出土的戰國金屬器物上的建築形象，漢代的崖墓、墓闕、石室、明器、畫像石以及畫像磚中呈現的漢代木結構建築形象，雲崗、天龍山、響堂山、敦煌等石窟中的雕刻或壁畫中展現的魏晋南北朝乃至唐代的木結構建築樣式；從唐代到清代，則皆有代表性的木結構建築實例，包括唐代的五臺山佛光寺東大殿，遼代的薊縣獨樂寺觀音閣及山門、寶坻廣濟寺三大士殿、大同善化寺（部分建築爲金代），北宋的榆次雨華宮、正定隆興寺（隆興寺始建于隋朝，時稱「龍藏寺」，後改名「龍興寺」），唐朝改名「隆興寺」）轉輪藏殿、登封少林寺初祖庵大殿，金代的五臺山佛光寺文殊殿、曲阜孔廟碑亭，元代的正定陽和樓、曲陽北岳廟大殿，明清的北京長陵祾恩殿、曲阜孔廟主要建築群、北京故宮三大殿及文淵閣等。貫穿其間的五幅大圖，即「歷代木構殿堂外觀演變圖」、「歷代殿堂平面及列柱位置比較圖」、「歷代斗栱演變圖」、「歷代要頭（梁頭）演變圖」和「歷代闌額普拍枋演變圖」，則是在大量測繪研究的基礎上總結出的關于中國古代建築發展演變歷程直觀而生動的圖解，這些綱舉目張的「演變圖」或「比較圖」，不僅可以作爲中國古建築斷代的基本依據，如其中的「歷代斗栱演變圖」，清楚揭示了由唐至清，斗栱與立柱高度的比值逐漸變小的趨勢，是中國古建築斷代最簡單易行的依據；還可以顯示出中國歷代木構建築藝術風格變遷之大略，即梁思成劃分的豪勁時期（唐、遼及北宋初）、醇和時期（北宋末至元）與羈直時期（明清）三大歷史分期，這從「歷代木構殿堂外觀演變圖」中看來尤爲直觀。梁思成通過清晰易懂的圖解分析，回應了此前一些西方學者認爲中國古代建築是「非歷史的」（《弗萊徹建築史》一書即持此種觀點）等種種偏見。對于磚石佛塔這一中國古建築的重要類型，梁思成也通過一幅「歷代佛塔類型演變圖」和一系列經典實例測繪圖對之進行了基本分類以及歷史風格的梳理。此外，還有一組爲數不多的圖紙對中國古建築的其他類型，諸如井干式民居、墓葬、磚砌無梁殿、橋梁、測影臺等進行簡要介紹，盡管不是這部圖解建築史的重點，但在某種程度上表明了梁思成和中國營造學社對中國古建築研究頗爲廣闊的視野。

最後還要强調的一點是，這些精確的測繪圖，除了可以作爲建築繪圖的杰作加以欣賞，或者當作中國古代建築史的圖解綱要之外，第三個重要的意義在于，它們其實蘊含着中國古代建築杰作背後的「匠心」和「秘訣」，比如前面提到的標准化、模數化的設計思想，還有古代能工巧匠們創造這些建築時所運用的比例關系等等。



文藝復興時期的帕拉第奧萬神廟測繪圖。圖片來源：*The Four Books on Architecture*

由于中國古代的建築專著流傳極少，而歷代匠師們依靠師徒口口相傳的秘訣，絶大多數隨着中國近現代歷史的劇變而湮没失傳。因此，如果想要探究中國古人創造的不計其數的建築藝術瑰寶背後的奧秘，就必須結合爲數不多的文獻記載，對幸存至今的珍貴建築遺迹進行精確的測繪，并通過對測繪圖紙和數據的科學分析，來破譯中國古人設計的秘密。像這樣「整理國故」的過程，西方人也曾經歷過，那就是他們所津津樂道的「文藝復興」。在西方文藝復興的歷史中，建築藝術的復興扮演着舉足輕重的角色，一批文藝復興時期的建築巨匠如布魯乃列斯基、阿爾伯蒂、布拉曼特、帕拉第奧等人，正是通過對古羅馬建築遺迹的考察、測繪和研究，重新「發現」了古羅馬偉大建築杰作（如羅馬萬神廟、大角鬥場和卡拉卡拉浴場等）所蘊藏的高度發達的建築技藝，并加以繼承和發展，才能創造出像佛羅倫薩大教堂穹頂或者梵蒂岡聖彼得大教堂這些文藝復興的建築杰作。而梁思成以及中國營造學社成員們的所作所爲，正是中國建築文藝復興的工作。如果我們希望有朝一日，中國當代城市和村鎮拔地而起的新建築不再僅僅是西方當代建築的簡單翻版或者山寨舶來品，而是像這本圖册中曾經

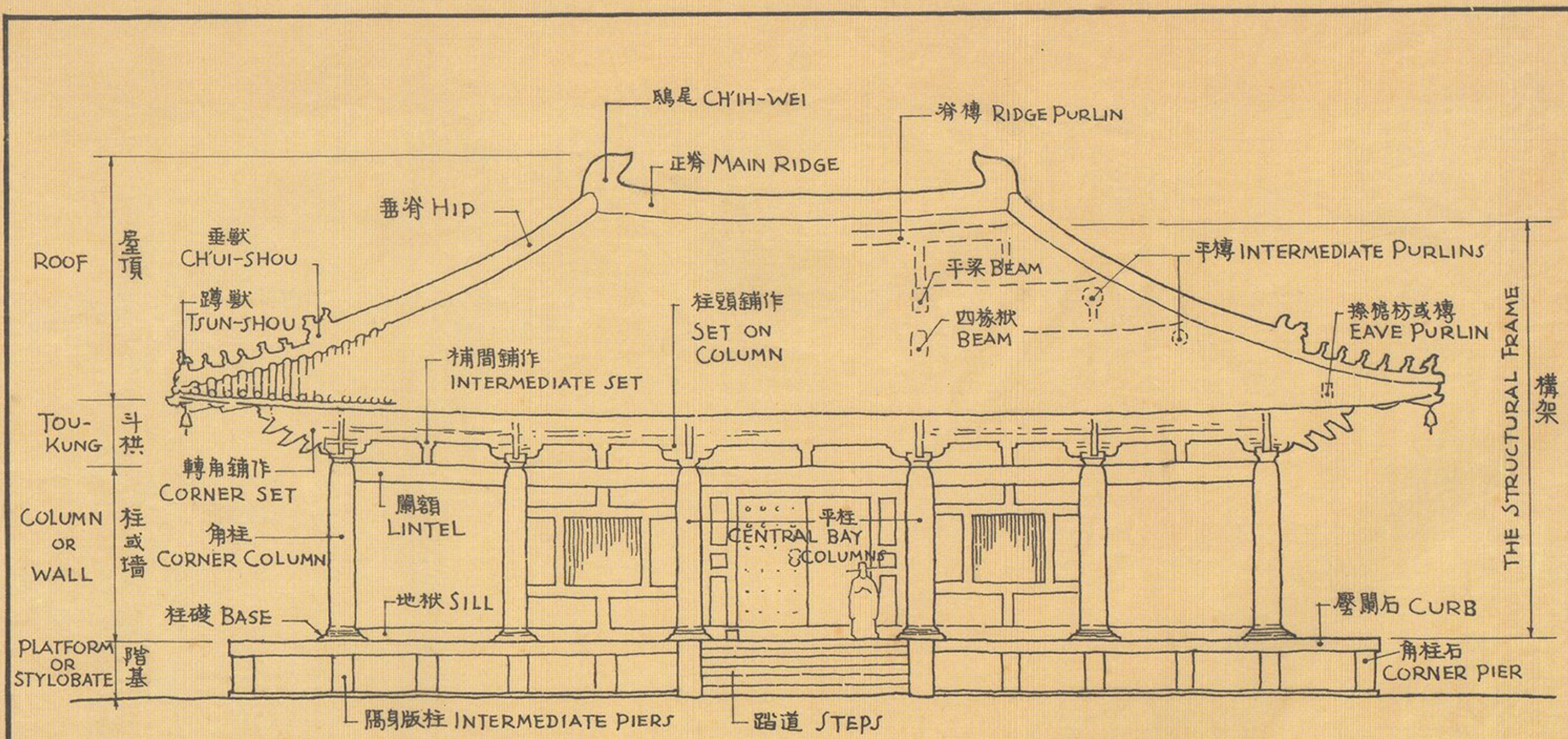
在歷史上輝煌并且令我們的民族感到驕傲的偉大建築那樣的話，我們的建築師、建築學子們就必須將先輩們研究中國古建築的工作不間斷地繼續下去。

則是中國古人習慣把對建築物精益求精地營造稱作「雕梁畫棟」，而梁思成、莫宗江的古建築測繪圖及分析圖，既是對那些雕梁畫棟的古代建築杰作重新進行精益求精的描畫。這些經由一筆一筆「畫棟描梁」繪成的圖紙，將這批寶貴的圖紙重現在讀者面前。作爲一名中國古建築的忠實愛好者，筆者希望每一個喜愛中國古建築的人都能從中獲得美的享受；同時，作爲一名從梁思成先生一手創辦的清華大學建築系畢業的建築學子和建築系教師，筆者更由衷地希望每一個中國建築學子，都能仔細描摹一遍先輩學者們辛勤測繪的圖紙，并從中汲取實現中國建築文藝復興的養分和創造力。

① 該書由梁思成生前好友、著名漢學家費慰梅（Wilma Fairbank）與林洙女士共同編輯，并由麻省理工學院出版社（MIT Press）出版。

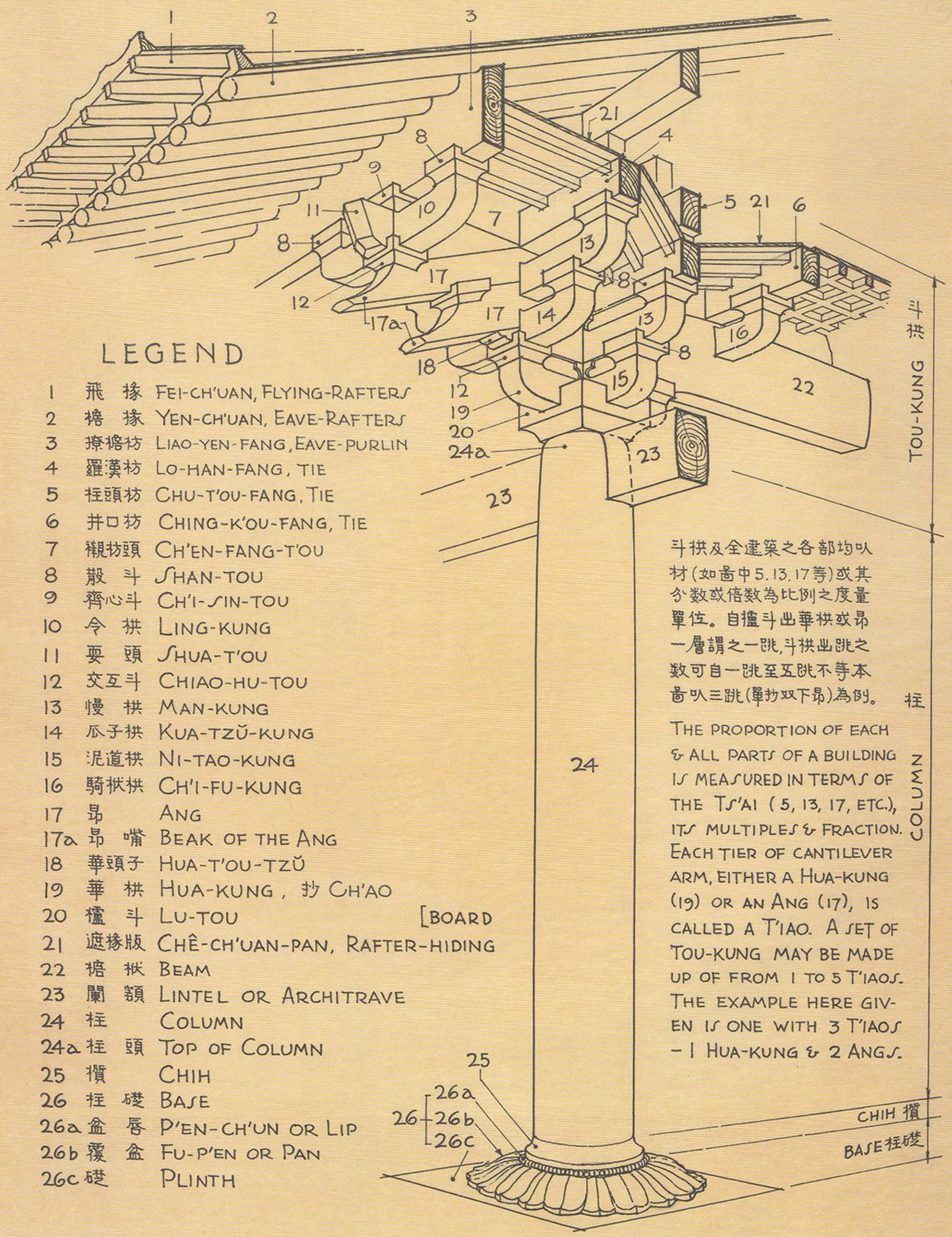
② 這主要源自巴黎美術學院的傳統，梁思成就讀的賓夕法尼亞大學建築系以繼承巴黎美術學院的傳統爲重要特色。

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NAMES OF PRINCIPAL PARTS OF A CHINESE BUILDING

中國建築主要部份名稱圖



LEGEND

1	飛椽	FEI-CH'UAN, FLYING-RAFTERS
2	檐椽	YEN-CH'UAN, EAVE-RAFTERS
3	撩檐枋	LIAO-YEN-FANG, EAVE-PURLIN
4	羅漢枋	LO-HAN-FANG, TIE
5	柱頭枋	CHU-T'OU-FANG, TIE
6	井口枋	CHING-K'OU-FANG, TIE
7	襯枋頭	CH'EN-FANG-T'OU
8	散斗	SHAN-TOU
9	齊心斗	CH'I-SIN-TOU
10	令拱	LING-KUNG
11	耍頭	SHUA-T'OU
12	交互斗	CHIAO-HU-TOU
13	慢拱	MAN-KUNG
14	瓜子拱	KUA-TZŬ-KUNG
15	泥道拱	NI-TAO-KUNG
16	騎栿拱	CH'I-FU-KUNG
17	昂	ANG
17a	昂嘴	BEAK OF THE ANG
18	華頭子	HUA-T'OU-TZŬ
19	華拱	HUA-KUNG, 抄 CH'AO
20	櫨斗	LU-TOU [BOARD
21	遮椽版	CHÊ-CH'UAN-PAN, RAFTER-HIDING
22	搪栿	BEAM
23	闌額	LINTEL OR ARCHITRAVE
24	柱	COLUMN
24a	柱頭	TOP OF COLUMN
25	攢	CHIH
26	柱礎	BASE
26a	盆唇	P'EN-CH'UN OR LIP
26b	覆盆	FU-P'EN OR PAN
26c	礎	PLINTH

斗拱及全建築之各部均以材(如畵中5, 13, 17等)或其分数或倍数為比例之度量單位。自櫨斗出華拱或昂一層謂之一跳,斗拱出跳之数可自一跳至五跳不等本畵以三跳(單抄双下昂)為例。

THE PROPORTION OF EACH & ALL PARTS OF A BUILDING IS MEASURED IN TERMS OF THE TS'AI (5, 13, 17, ETC.), ITS MULTIPLES & FRACTION. EACH TIER OF CANTILEVER ARM, EITHER A HUA-KUNG (19) OR AN ANG (17), IS CALLED A T'IAO. A SET OF TOU-KUNG MAY BE MADE UP OF FROM 1 TO 5 T'IAOS. THE EXAMPLE HERE GIVEN IS ONE WITH 3 T'IAOS - 1 HUA-KUNG & 2 ANGS.

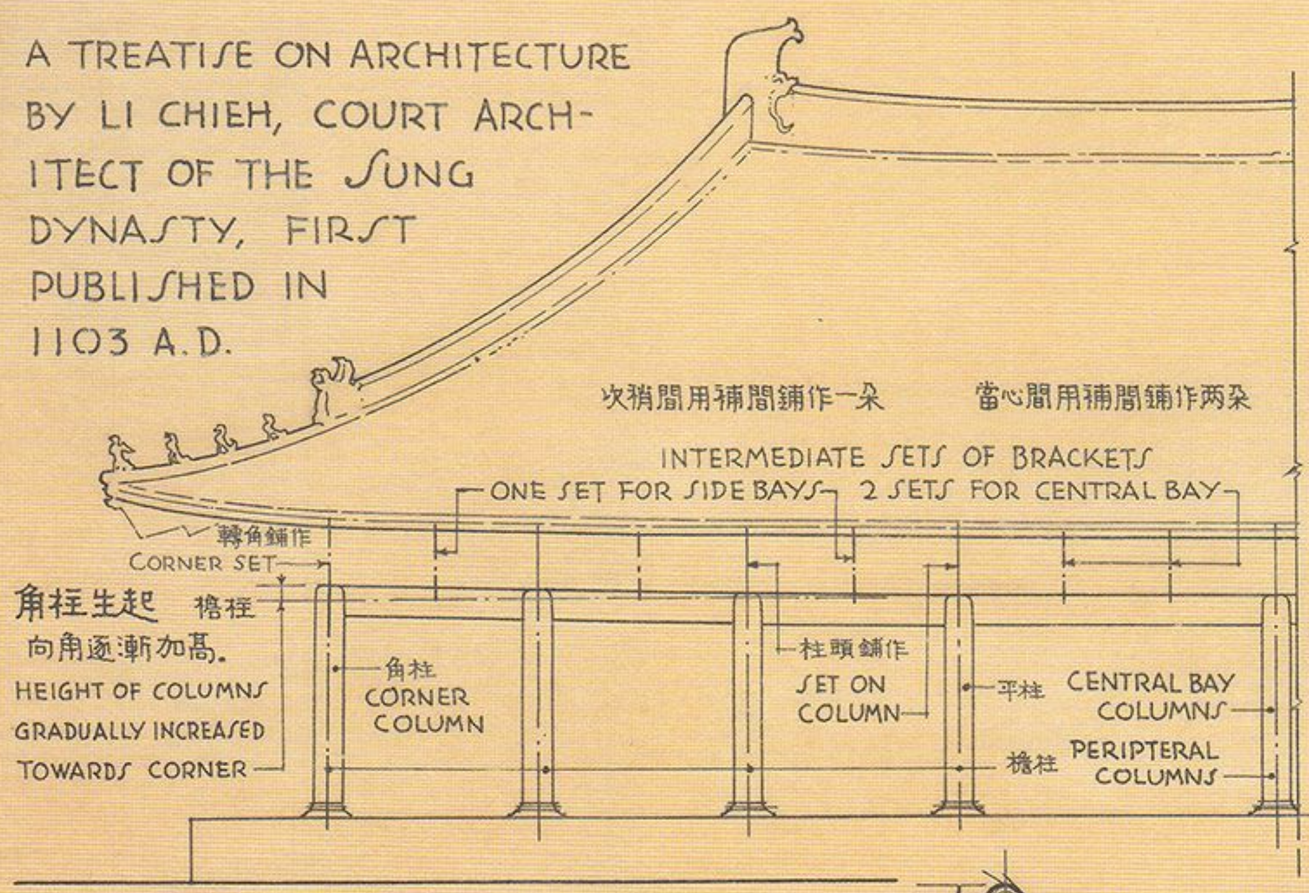
中國建築之"ORDER"·斗拱,搪柱,柱礎 THE CHINESE "ORDER"

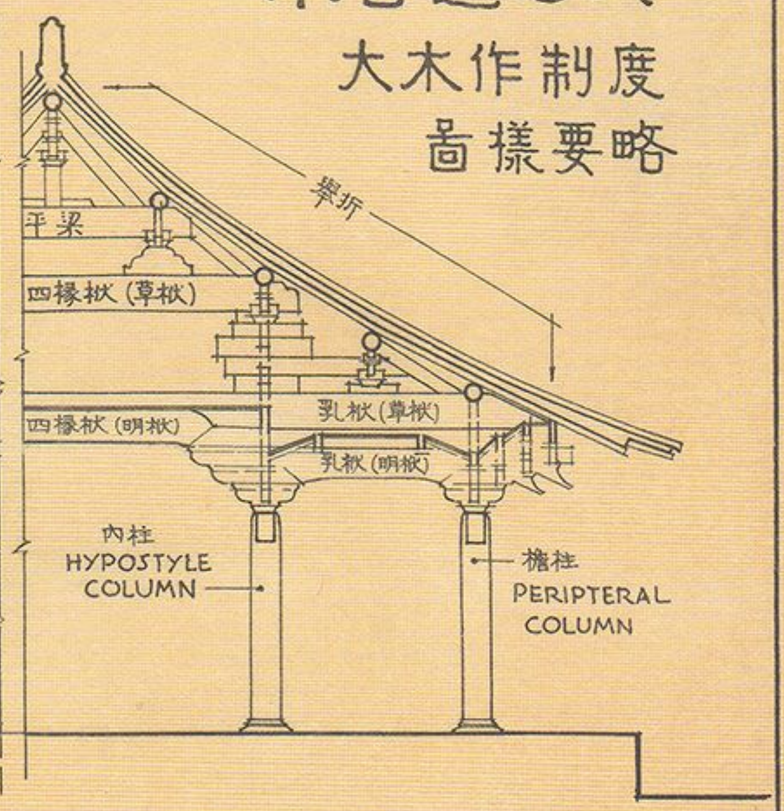
RULES FOR STRUCTURAL CARPENTRY ACCORDING TO YING-TSAO-FA-SHIH.

宋營造法式

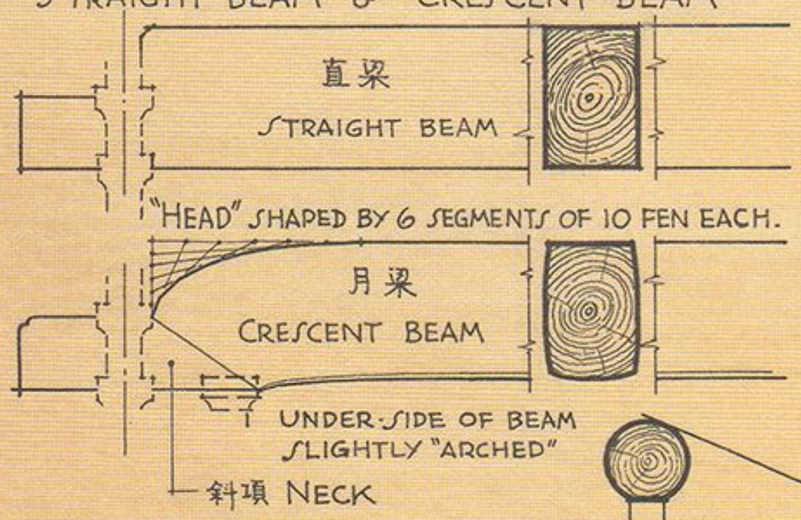
大木作制度圖樣要略

A TREATISE ON ARCHITECTURE BY LI CHIEH, COURT ARCHITECT OF THE SUNG DYNASTY, FIRST PUBLISHED IN 1103 A.D.



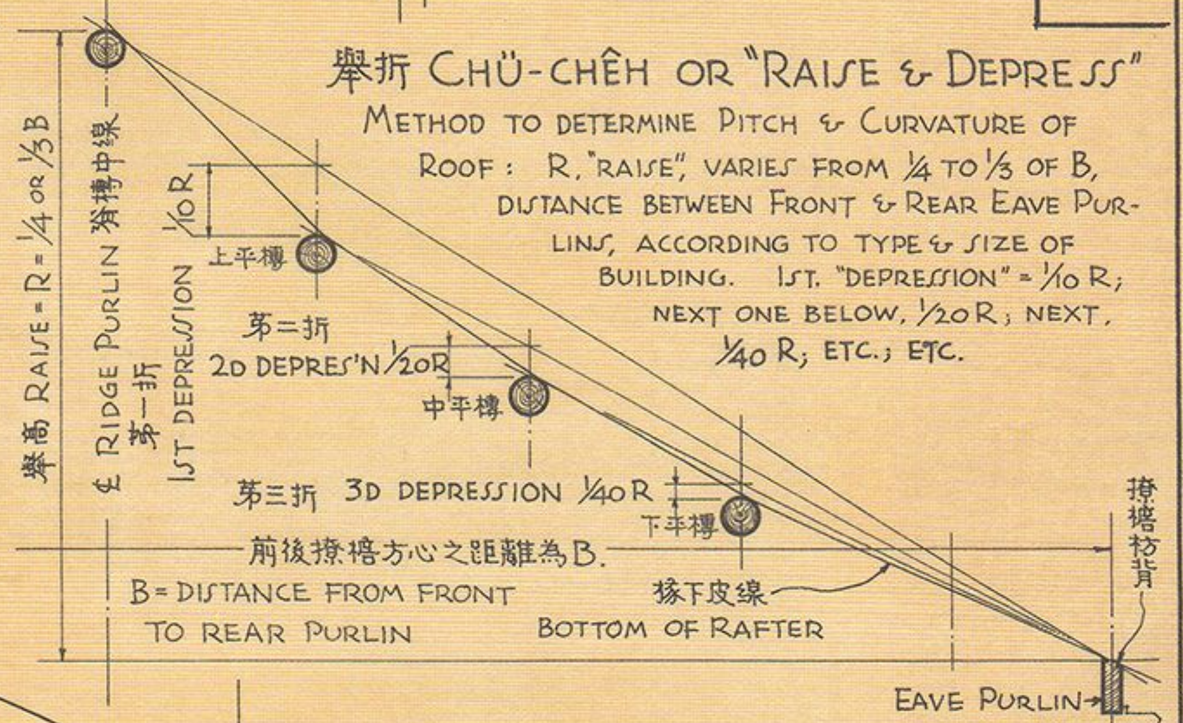


梁 BEAM 梁有直梁月梁二種。月梁梁首以六瓣卷殺,依跳数留斜項,梁底颤起。 2 TYPES OF BEAMS: STRAIGHT BEAM & "CRESCENT BEAM"



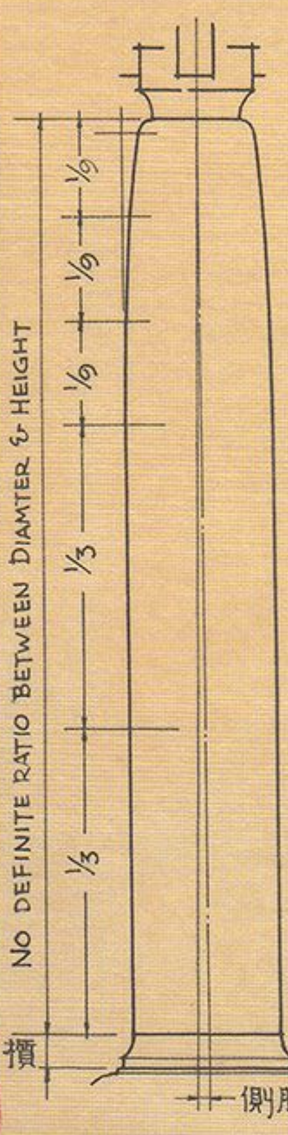
舉折 CHÜ-CHÊH OR "RAISE & DEPRESS"

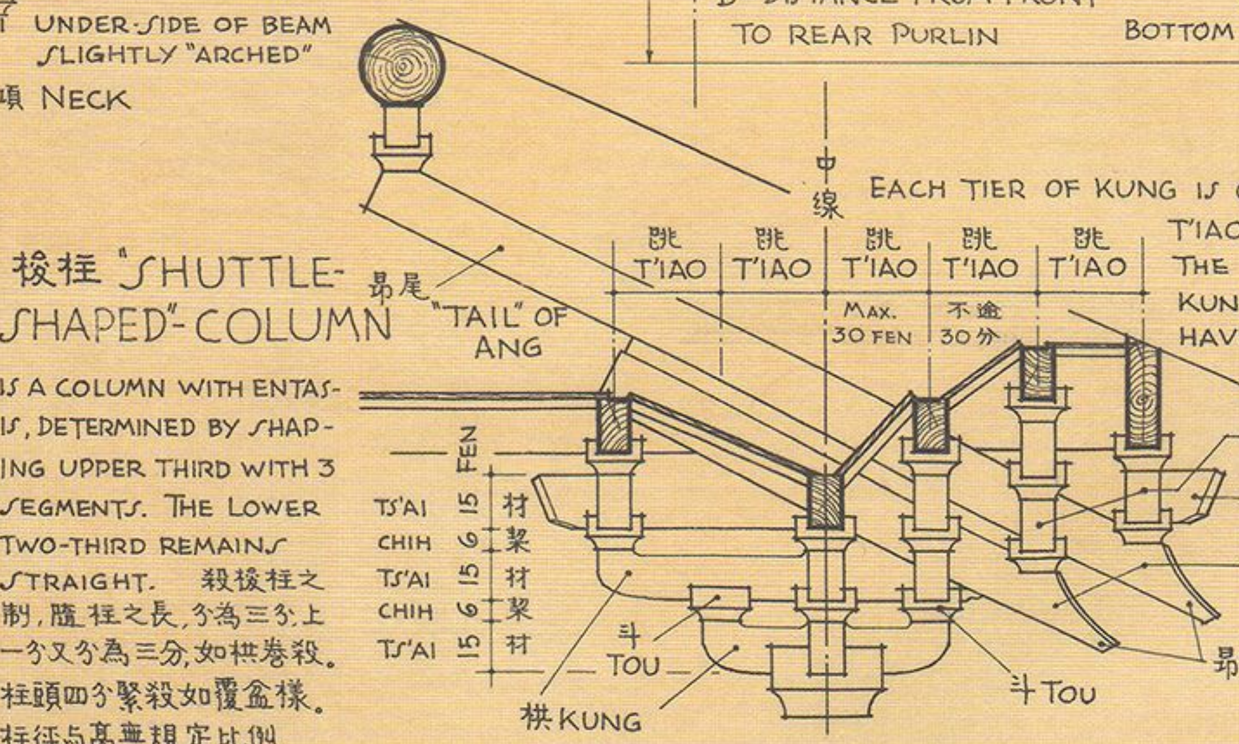
METHOD TO DETERMINE PITCH & CURVATURE OF ROOF: R, "RAISE", VARIES FROM ¼ TO ⅓ OF B, DISTANCE BETWEEN FRONT & REAR EAVE PURLINS, ACCORDING TO TYPE & SIZE OF BUILDING. 1ST. "DEPRESSION" = 1/10 R; NEXT ONE BELOW, 1/20 R; NEXT, 1/40 R; ETC.; ETC.



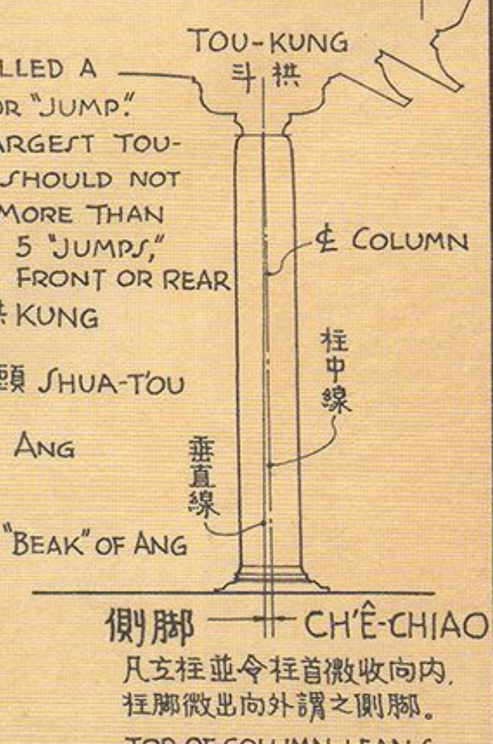
梭柱 "SHUTTLE-SHAPED"-COLUMN

IS A COLUMN WITH ENTASIS, DETERMINED BY SHAPING UPPER THIRD WITH 3 SEGMENTS. THE LOWER TWO-THIRD REMAINS STRAIGHT. 殺梭柱之制,隨柱之長,分為三分,上一分又分為三分,如栱卷殺。柱頭四分緊殺如覆盆樣。柱徑与高無規定比例。





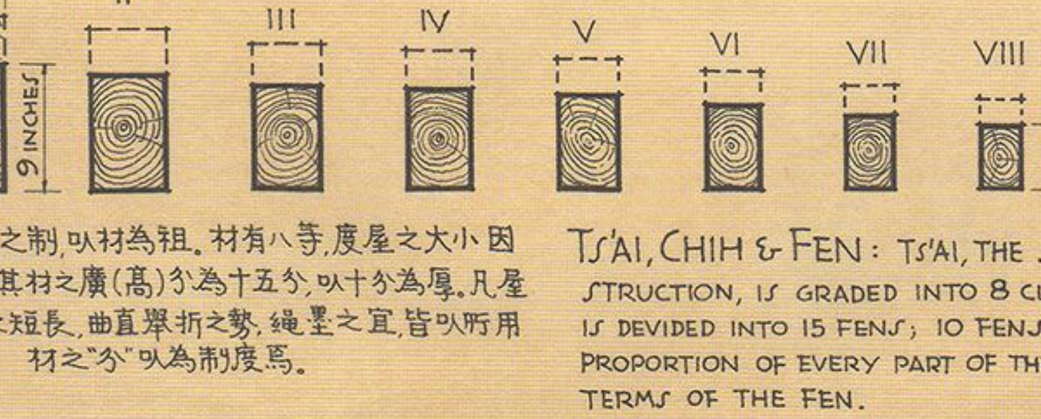
斗栱 TOU-KUNG OR BRACKET SET



側脚 CH'Ê-CHIAO

凡立柱並令柱首微收向內,柱脚微出向外謂之側脚。

TOP OF COLUMN LEANS INWARD 1% OF HEIGHT.



材栔分 造屋之制,以材為祖。材有八等,度屋之大小因而用之。各以其材之廣(高)分為十五分,以十分為厚。凡屋宇之高深,名物之短長,曲直舉折之勢,繩墨之宜,皆以所用材之"分"以為制度焉。

TS'AI, CHIH & FEN: TS'AI, THE STANDARD TIMBER FOR ALL CONSTRUCTION, IS GRADED INTO 8 CLASSES. THE DEPTH OF EACH TS'AI IS DEVIDED INTO 15 FENS; 10 FENS GIVES THICKNESS OF TS'AI. THE PROPORTION OF EVERY PART OF THE BUILDING IS THUS MEASURED IN TERMS OF THE FEN.

RULES FOR STRUCTURAL CARPENTRY ACCORDING TO KUNG-CH'ENG-TSO-FA

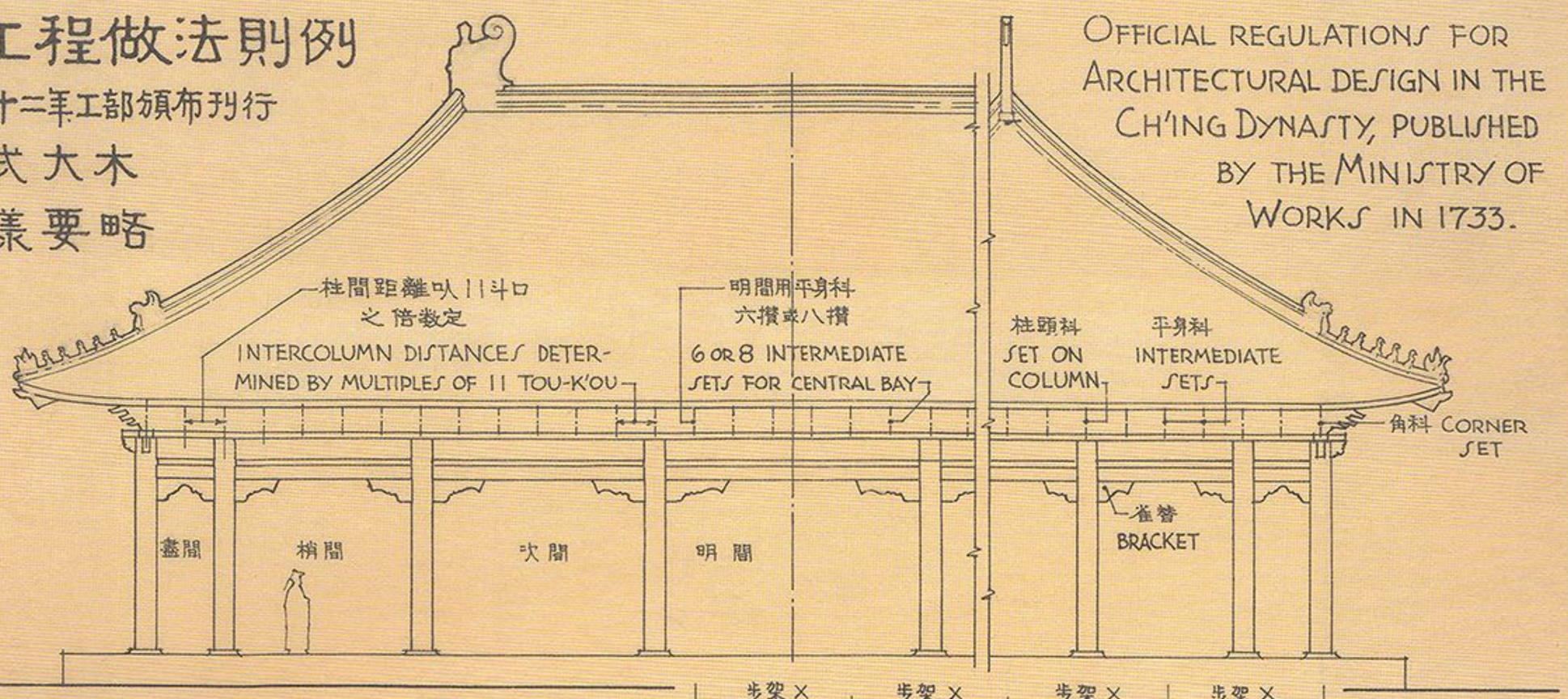
清工程做法則例

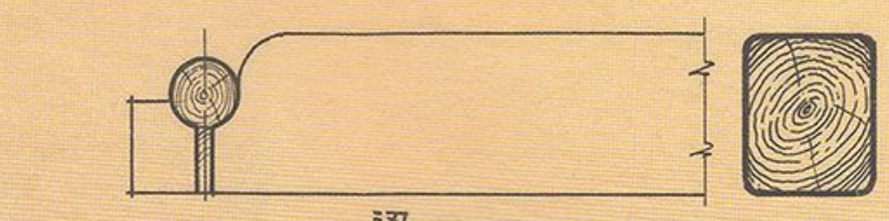
雍正十二年工部頒布刊行

大木大式

畫樣要略

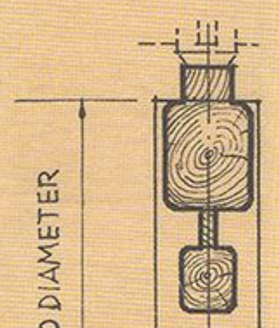
OFFICIAL REGULATIONS FOR ARCHITECTURAL DESIGN IN THE CH'ING DYNASTY, PUBLISHED BY THE MINISTRY OF WORKS IN 1733.



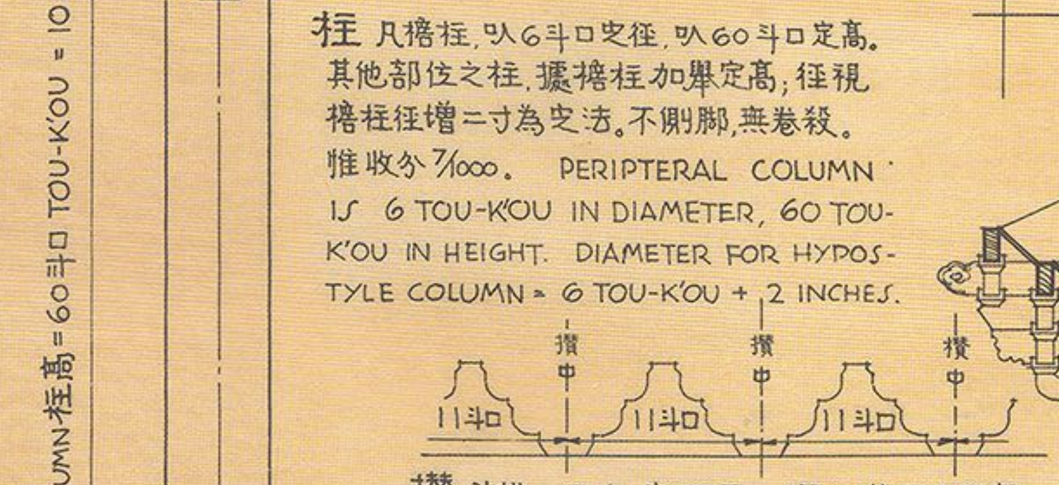


梁 按柱徑加二寸定梁厚,以厚之五分之六定高。斷面高与厚成6:5或5:4之比。

WIDTH OF BEAM = DIAMETER OF COLUMN + 2 INCHES; DEPTH = 6/5 WIDTH. THUS RATIO BETWEEN DEPTH & WIDTH OF BEAM IS AROUND 6:5 OR 5:4.

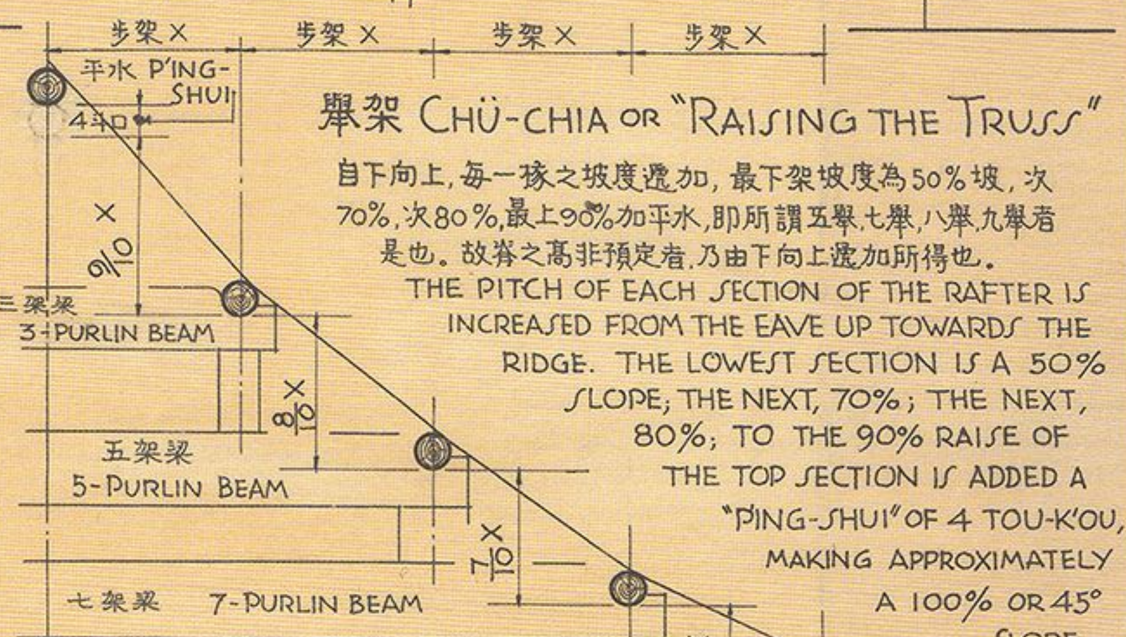


柱 凡搶柱,以6斗口定徑,以60斗口定高。其他部位之柱,據搶柱加舉定高;徑視搶柱徑增二寸為定法。不侧脚,無卷殺。惟收分7/1000。 PERIPTERAL COLUMN IS 6 TOU-K'OU IN DIAMETER, 60 TOU-K'OU IN HEIGHT. DIAMETER FOR HYPOSTYLE COLUMN = 6 TOU-K'OU + 2 INCHES.



攢 斗拱一組也,宋稱朵。攢与攢間之距離定為十一斗口,開間面濶以攢数定之。

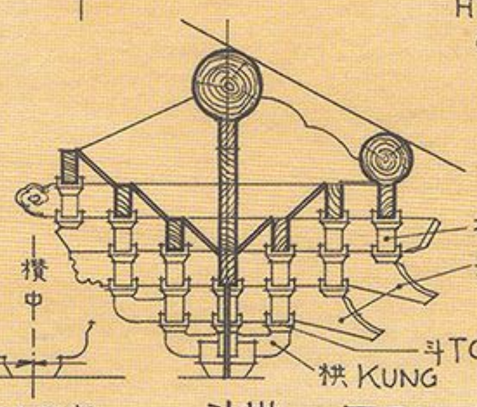
A SET OF TOU-KUNG IS CALLED A TSAN. SETS ARE SPACED AT INTERVALS OF 11 TOU-K'OU, MULTIPLES OF WHICH GIVES WIDTHS OF BAYS.



舉架 CHÜ-CHIA OR "RAISING THE TRUSS"

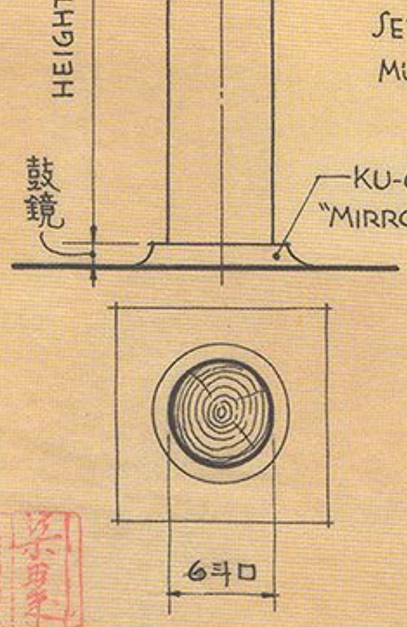
自下向上,每一椽之坡度遞加,最下架坡度為50%坡,次70%,次80%,最上90%加平水,即所謂五舉,七舉,八舉,九舉者是也。故脊之高非預定者,乃由下向上遞加所得也。

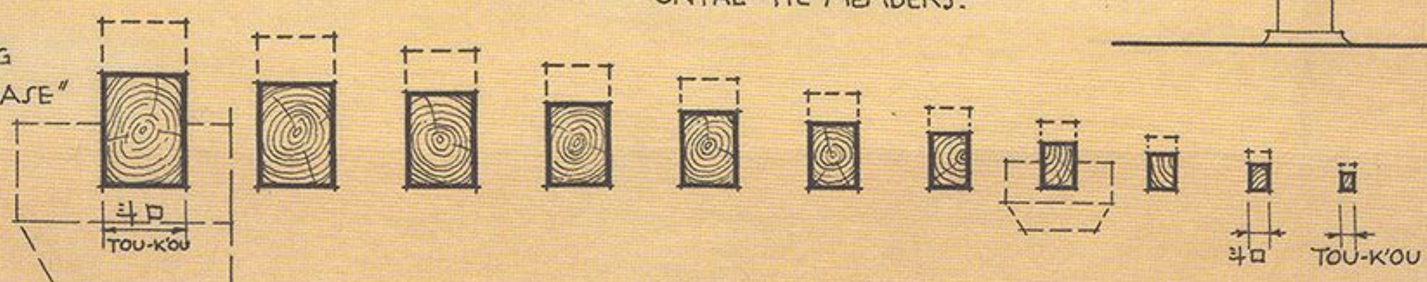
THE PITCH OF EACH SECTION OF THE RAFTER IS INCREASED FROM THE EAVE UP TOWARDS THE RIDGE. THE LOWEST SECTION IS A 50% SLOPE; THE NEXT, 70%; THE NEXT, 80%; TO THE 90% RAISE OF THE TOP SECTION IS ADDED A "P'ING-SHUI" OF 4 TOU-K'OU, MAKING APPROXIMATELY A 100% OR 45° SLOPE.



斗拱 TOU-KUNG

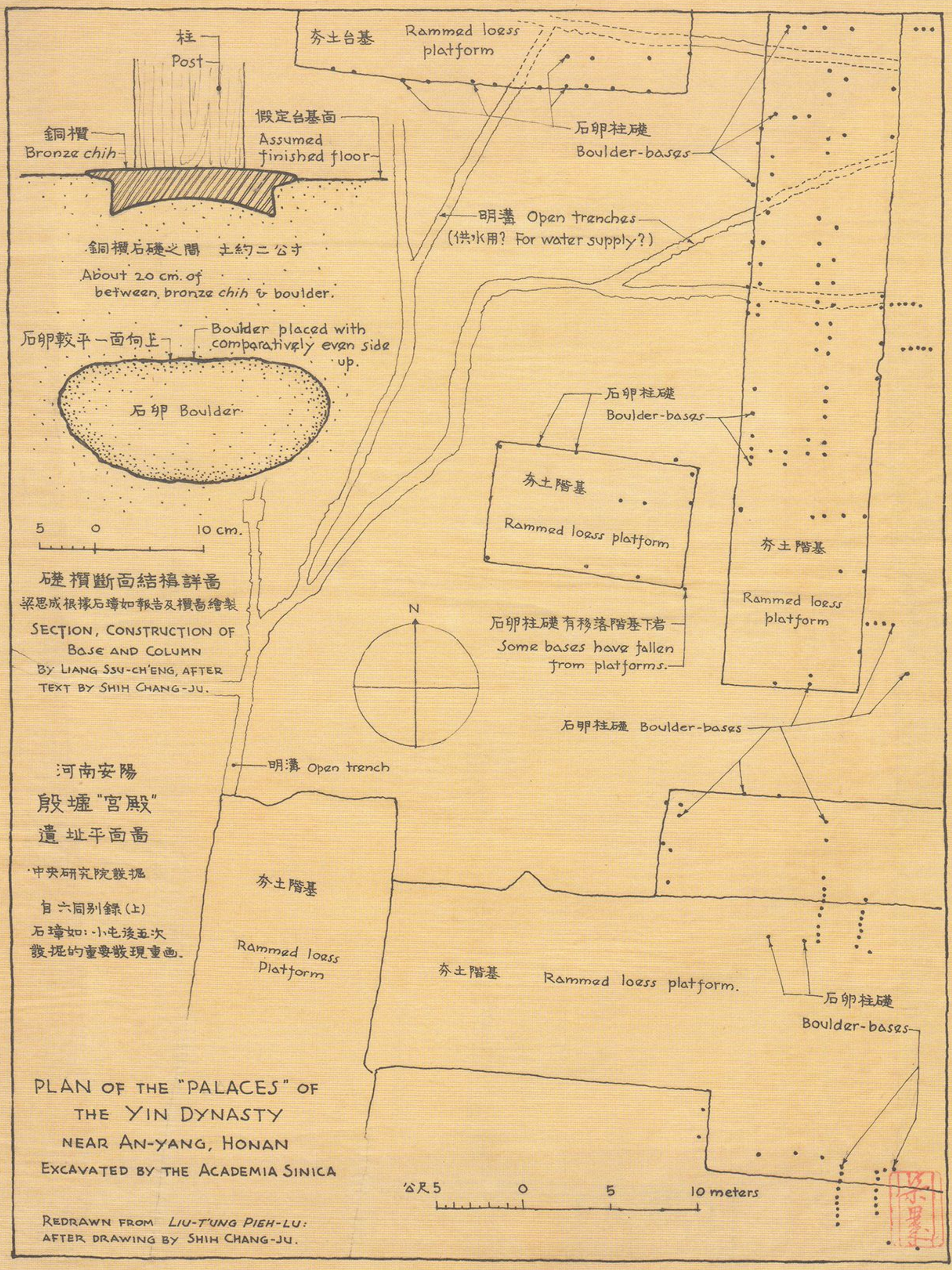
在比例上小於宋式甚多。用材以足材為主,各層枋間均不用斗。PROPORTIONALLY MUCH SMALLER THAN SUNG TOU-KUNG. TOU NO LONGER USED BETWEEN HORIZONTAL TIE MEMBERS.





斗口 TOU-K'OU 清式祘材厚曰斗口,即宋之十分也。斗口自一寸至六寸,共十一等,但実物所見,最大者僅至四寸。用材均高二斗口,單材僅用於跳頭横拱,高為1.4斗口。THE WIDTH OF A TS'AI IS KNOWN AS A TOU-K'OU, RANGING FROM 1 TO 6 INCHES; DEPTH OF TS'AI = 2 TOU-K'OU. TAN-TS'AI, OR A LIGHT TS'AI = 2 × 1.4 TOU-K'OU, USED ONLY FOR KUNGS EMPLOYED ON T'IAOS.

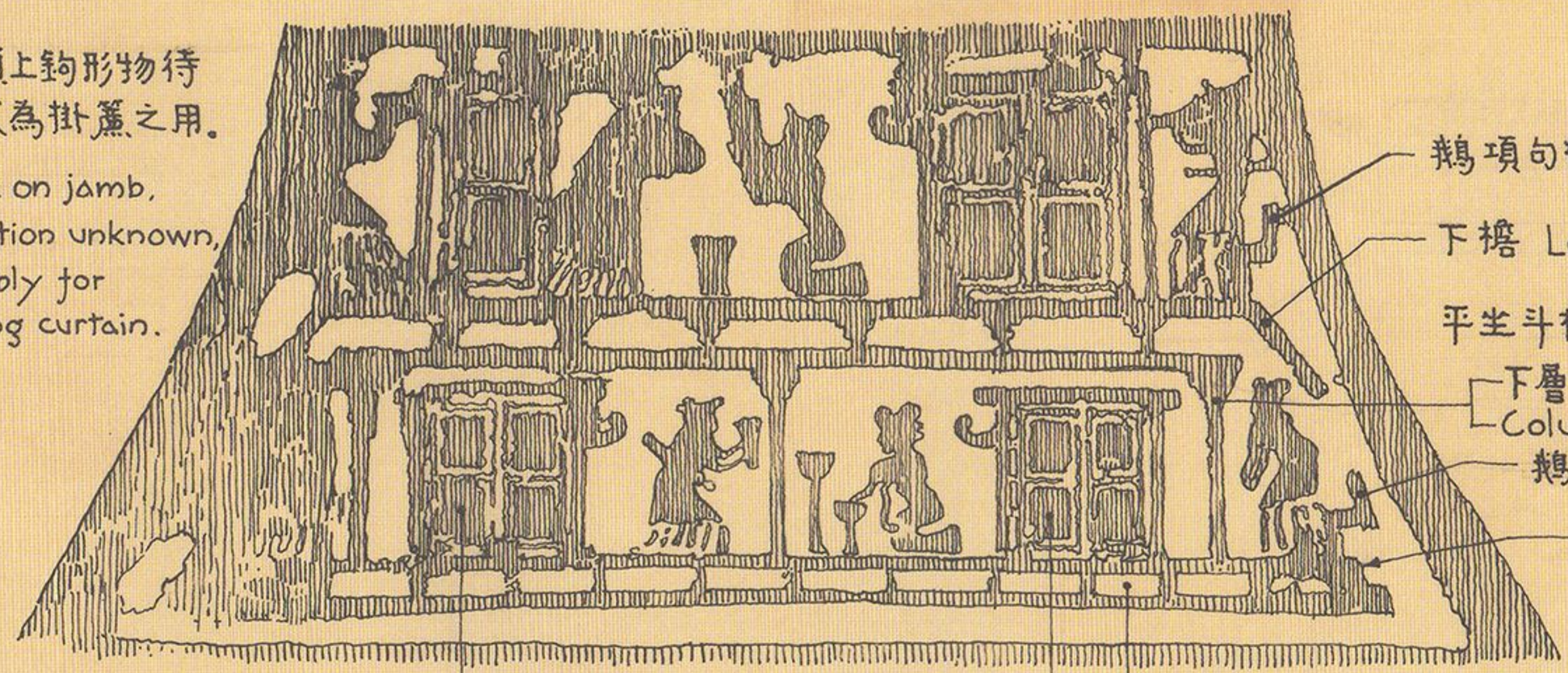
20020101-1846



1845

採桑獵紡拓本宮室畫 戰國時代

RUBBING FROM VASE PERIOD OF WARRING KINGDOMS 468-221 B.C.



門類上鉤形物待攷,或為掛簾之用。

Hook on jamb, function unknown, possibly for lifting curtain.

鵝項勾欄 Seat-railing

下檐 Lower eave

平坐斗栱 Balcony tou-kung

下層柱及斗栱 Column & tou-kung

鵝項勾欄 Seat-railing

踏步 Steps

雙扇版門,並見頰及額 Paneled door, with jamb & head

橫柱台基 Platform with (or on) struts

故宮博物院藏

COLLECTION, PALACE MUSEUM.

5

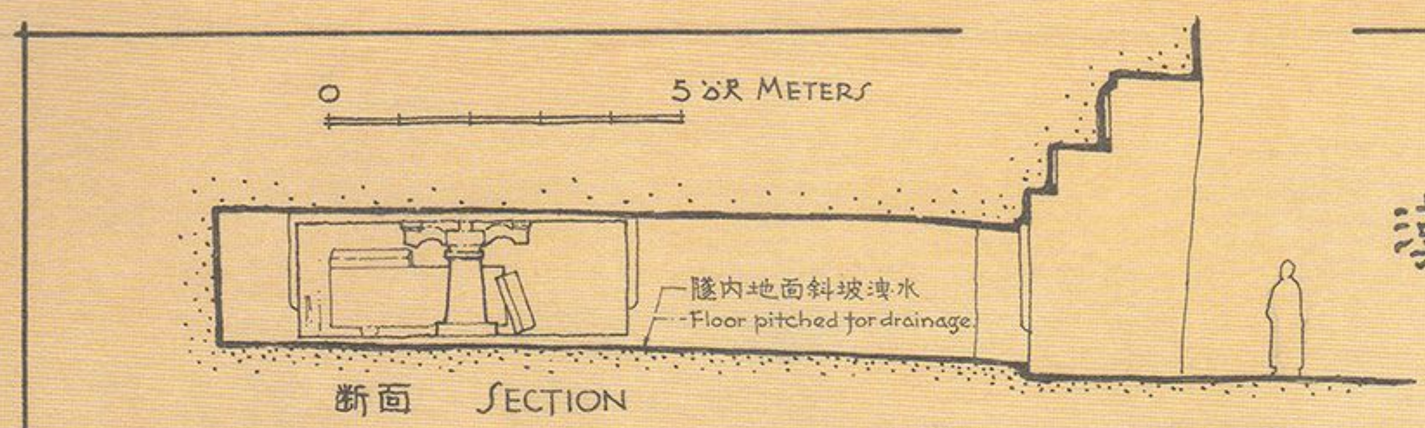
20020101-1848

四川彭山縣江口鎮附近 漢崖墓建築及雕飾

選自國立中央博物院
陳明達"彭山崖墓報告"
未刊稿

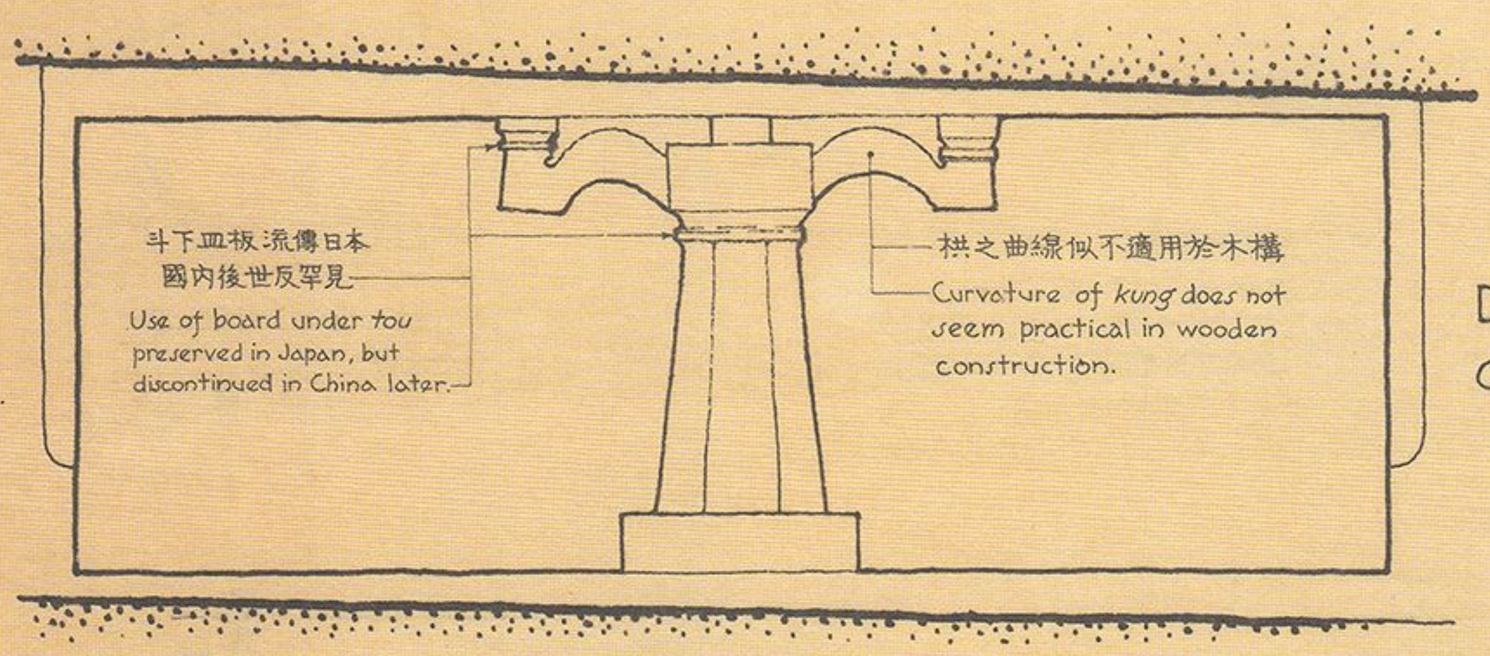
ROCK-CUT TOMBS NEAR CHIANG-K'OU, P'ENG-SHAN HSIEN, SZE-CH'UAN.

FROM CH'EN, M.-T.
"REPORT ON THE ROCK-CUT TOMBS OF P'ENG-SHAN."
NATIONAL CENTRAL MUSEUM
(UNPUBLISHED)



王家坨"第460號"崖墓

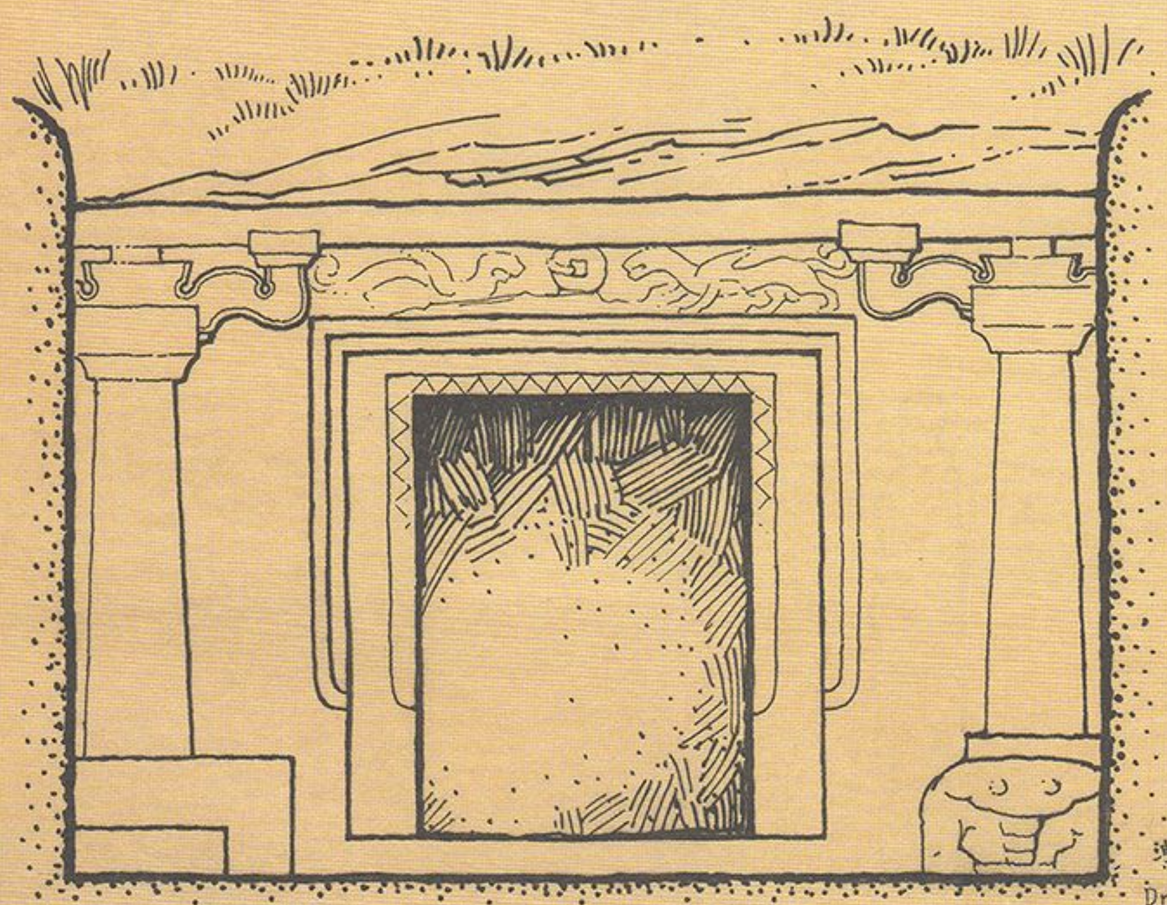
TOMB "NO. 460" WANG-CHIA-T'O



"第460號"墓室及斗栱詳圖

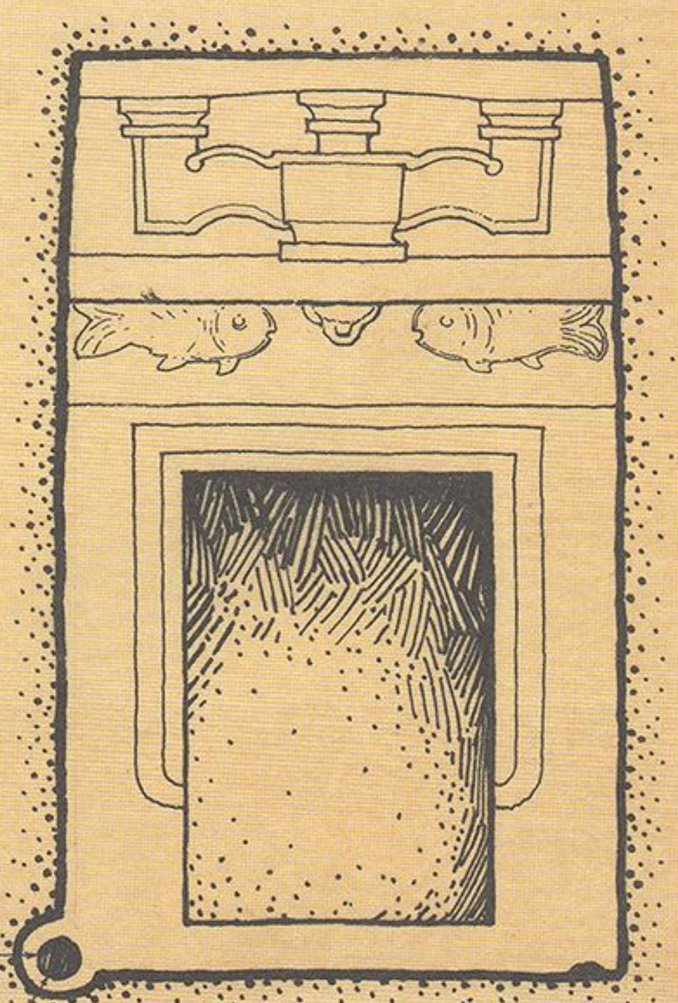
DETAIL OF FUNERARY-CHAMBER & "ORDER" TOMB "No. 460"

0 1 FOOT M.



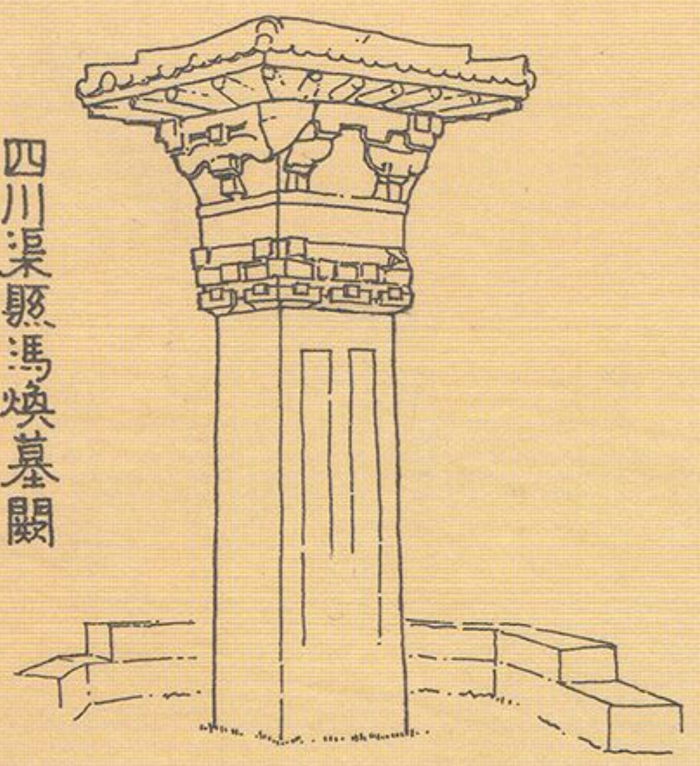
江口鎮"第355號"墓門

TOMB "No. 355", CHIANG-K'OU, DETAIL OF ENTRANCE



豆芽溝"第176號"墓門

ENTRANCE, TOMB "No. 176, TOU-YA-KOU.



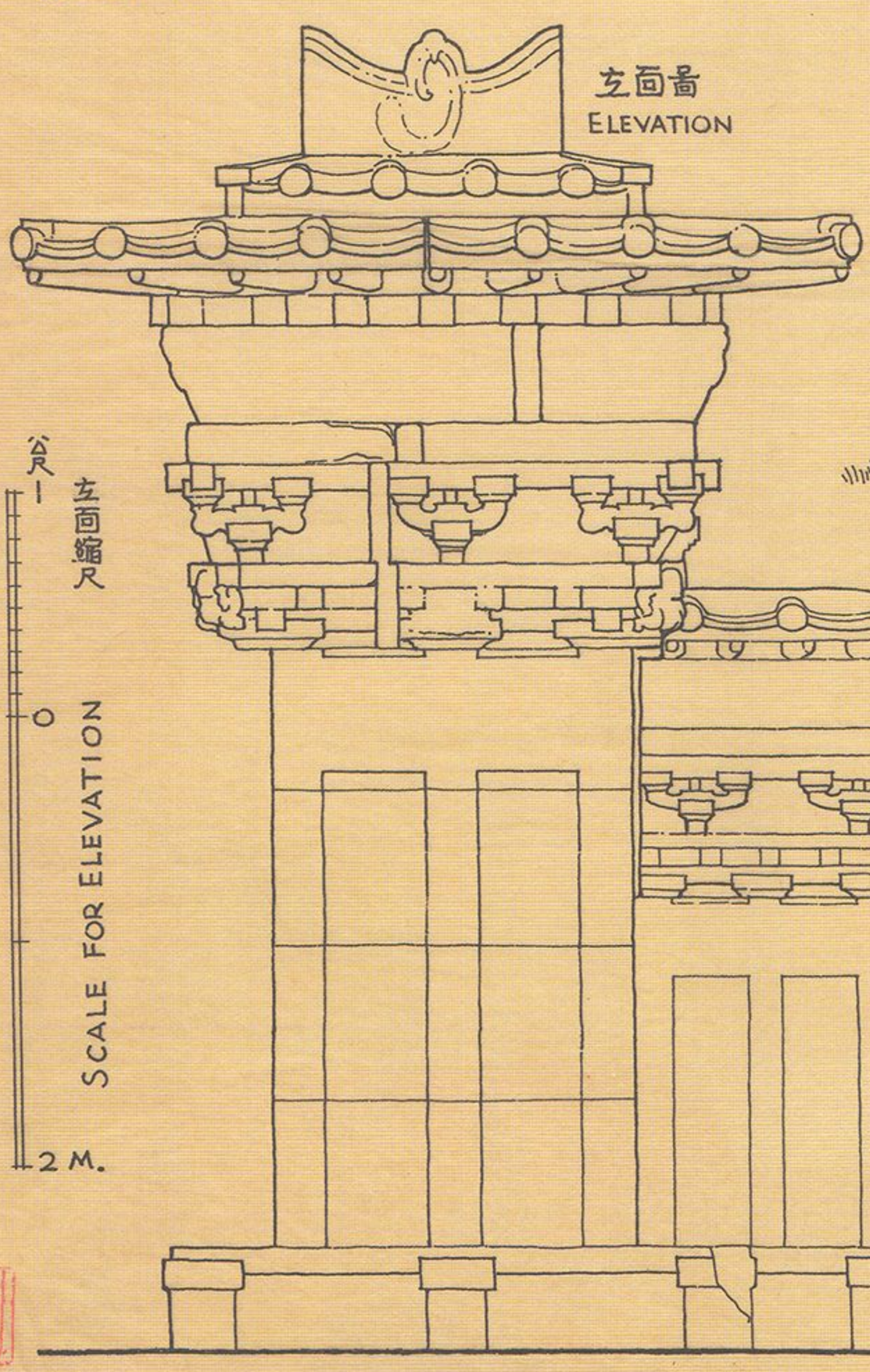
四川渠縣馮煥墓闕

CH'ÜEH AT THE TOMB OF
FENG HUAN, CH'Ü HSIEN
SZE CH'UAN



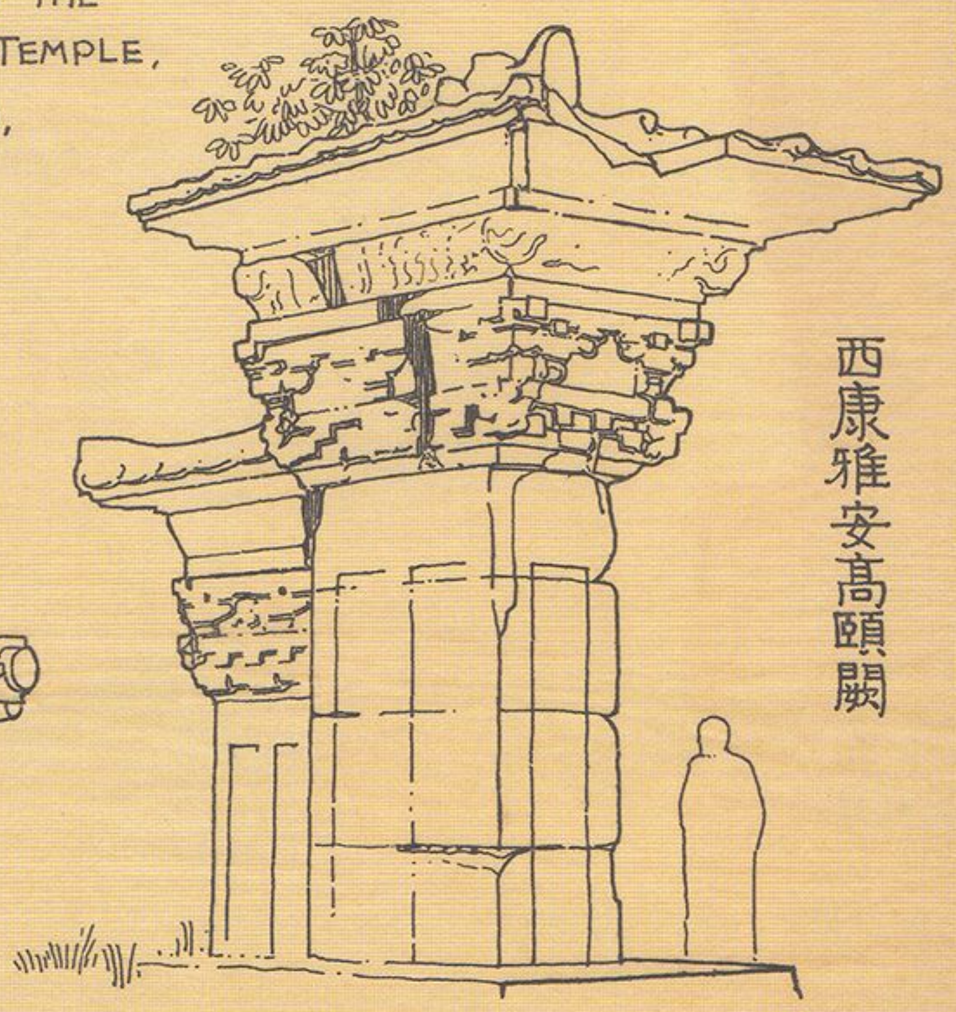
河南嵩山少室石闕

CH'ÜEH AT THE
SHAO-SHIH TEMPLE,
SUNG-SHAN,
HONAN.



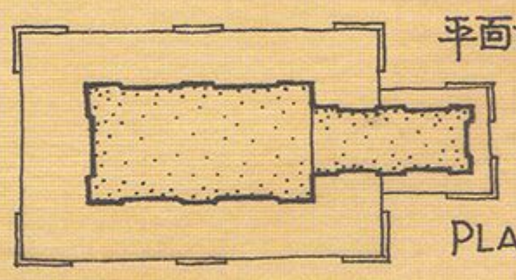
立面圖
ELEVATION

立面縮尺 1公尺 0 2 M.
SCALE FOR ELEVATION



西康雅安高頤闕

CH'ÜEH AT THE TOMB OF
KAO YI, YA-AN, SI-KANG



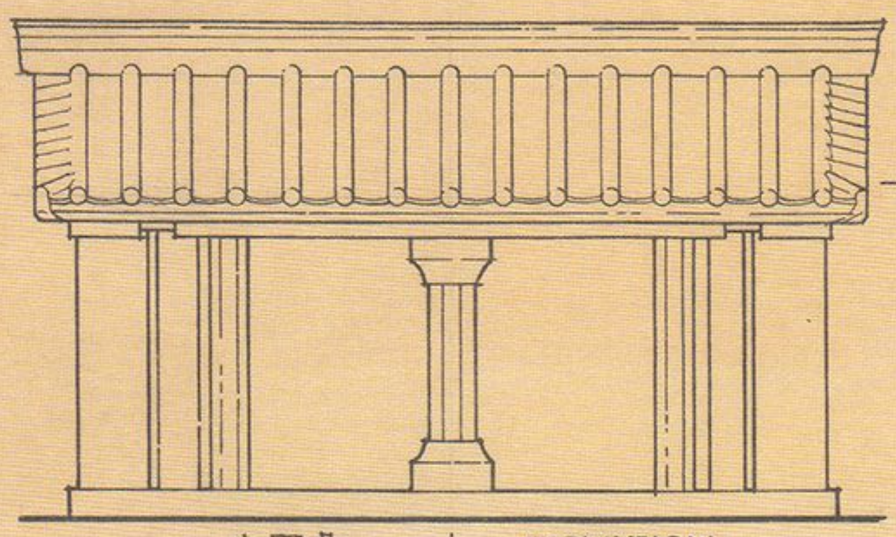
漢石闕數種

CH'ÜEH - MONUMENTAL
PIERS IN FRONT OF TEMPLES &
TOMBS OF THE HAN-DYNASTY
205 B.C. - 220 A.D.

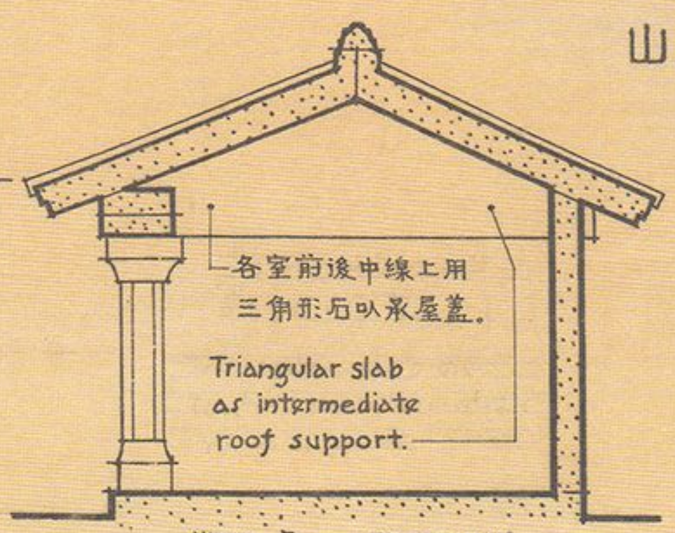
1850

山東肥城縣孝里鎮 郭巨祠石室

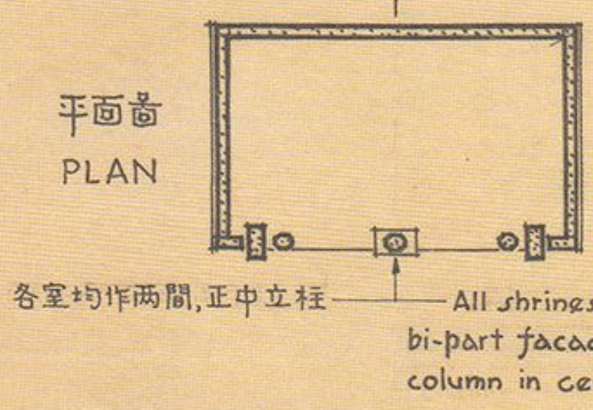
繪測 劉敦楨



立面圖 ELEVATION



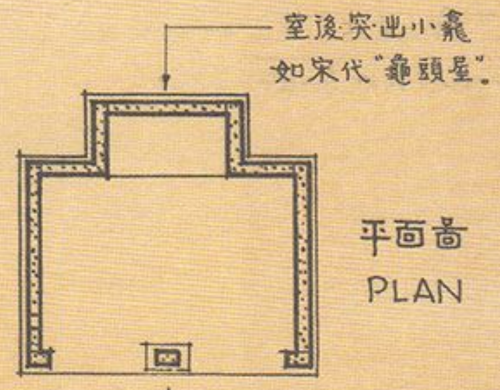
斷面圖 SECTION



平面圖 PLAN

SHRINE OF "KUO CHÜ"

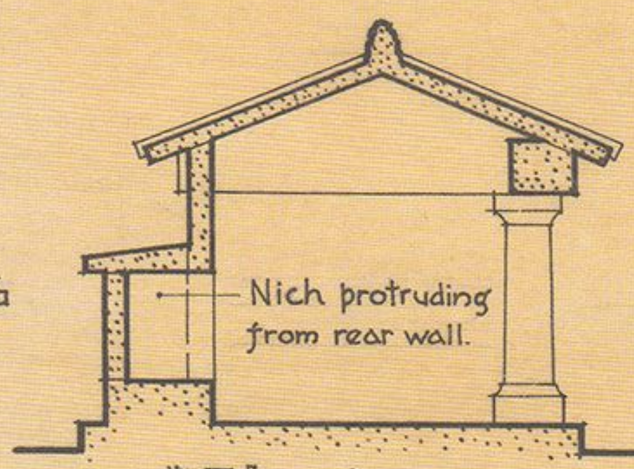
HSIAO-LI (HSIAO-T'ANG SHAN)
FEI-CH'ENG, SHANTUNG
MEASURED BY LIU TUN-TSENG



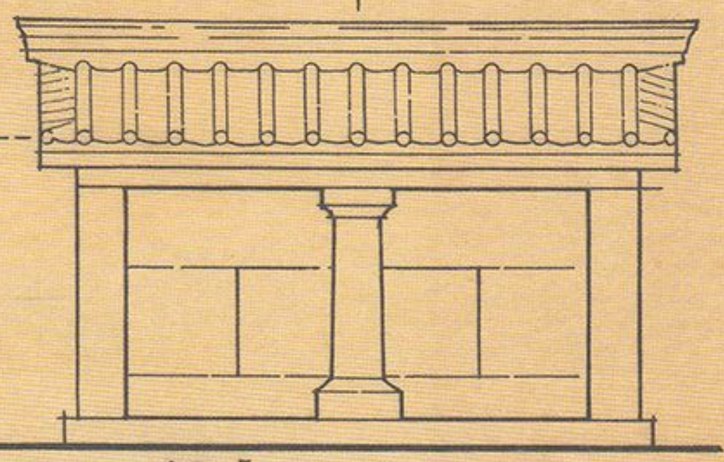
平面圖 PLAN

山東嘉祥縣 武梁祠左石室

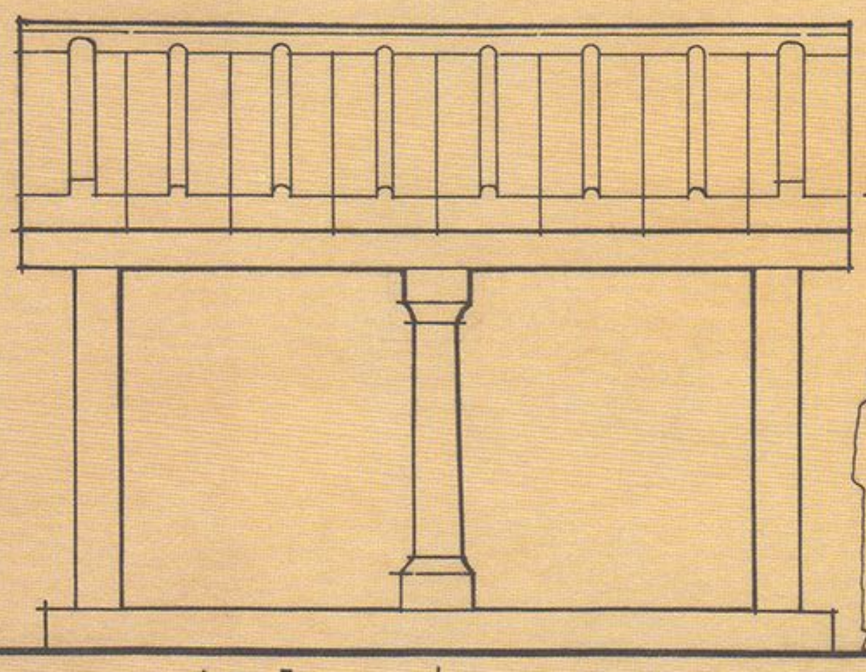
LEFT SHRINE
WU LIANG TZ'Ŭ
CHIA-HSIANG, SHAN-TUNG
CIRCA 147 A.D.
RECONSTRUCTED BY
WILMA FAIRBANK 復原圖



斷面圖 SECTION



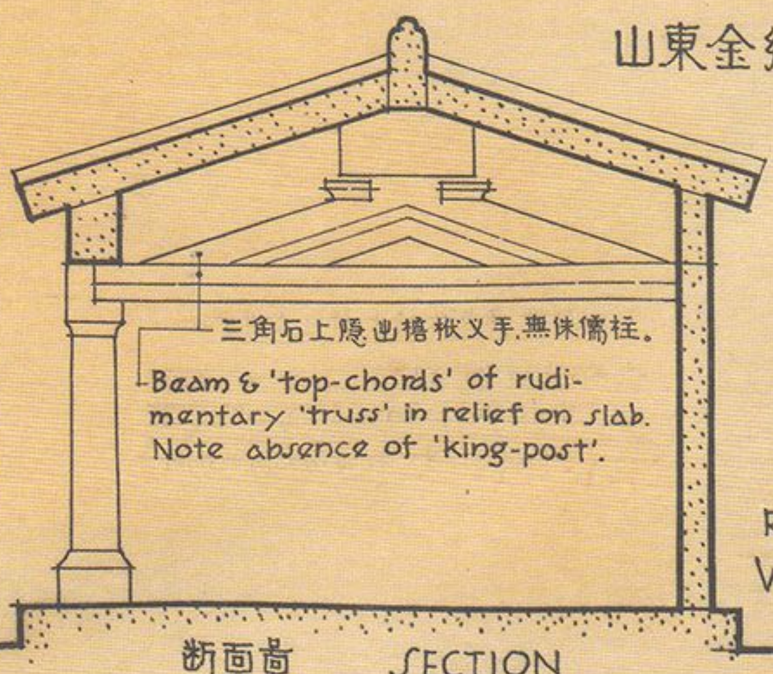
立面圖 ELEVATION



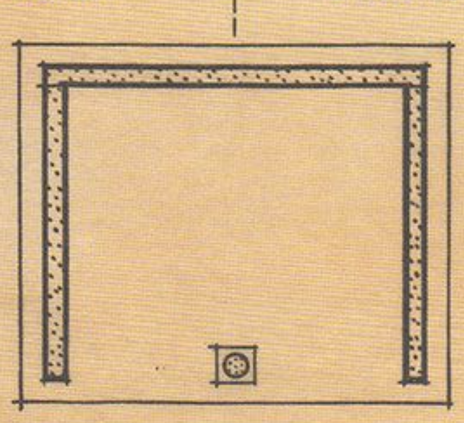
立面圖 ELEVATION

山東金鄉縣朱鮪墓 石室

SHRINE
CHU WEI'S
TOMB
CHIN-HSIANG
SHAN-TUNG.
CIRCA 50 A.D.
RECONSTRUCTED BY
WILMA FAIRBANK.
復原圖



斷面圖 SECTION



平面圖 PLAN

漢墓石室 STONE TOMB SHRINES OF THE HAN DYNASTY

公分 100 0 500 CM.
平面縮尺 SCALE FOR PLAN

公分 100 0 300 CM.
立面,斷面縮尺 SCALE FOR ELEVATION & SECTION

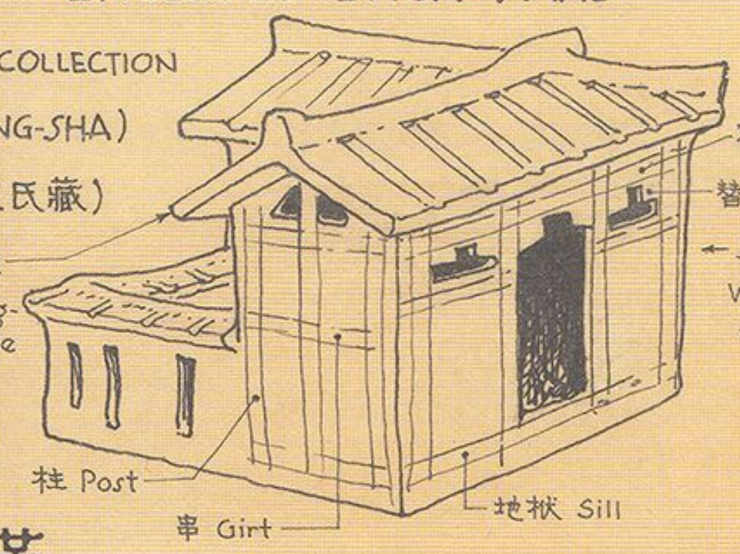
9

漢明器建築物數種

住宅 RESIDENCE WITH ENCLOSED BACK-YARD

(TSO'S COLLECTION CH'ANG-SHA)

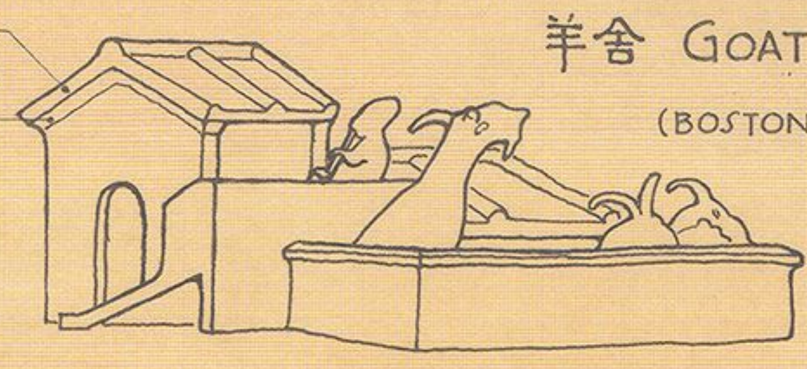
(長沙左氏藏)



硬山頂
Flushed gable

羊舍 GOAT HOUSE

(BOSTON MUSEUM OF FINE ARTS)

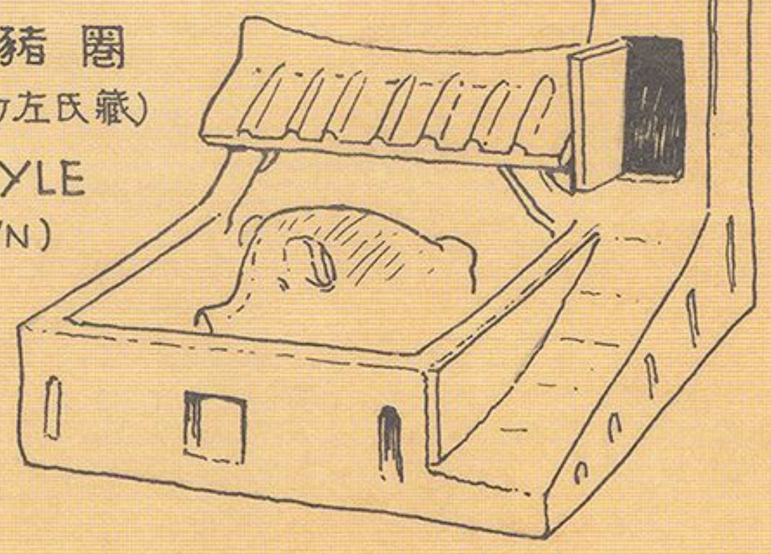


豬圈

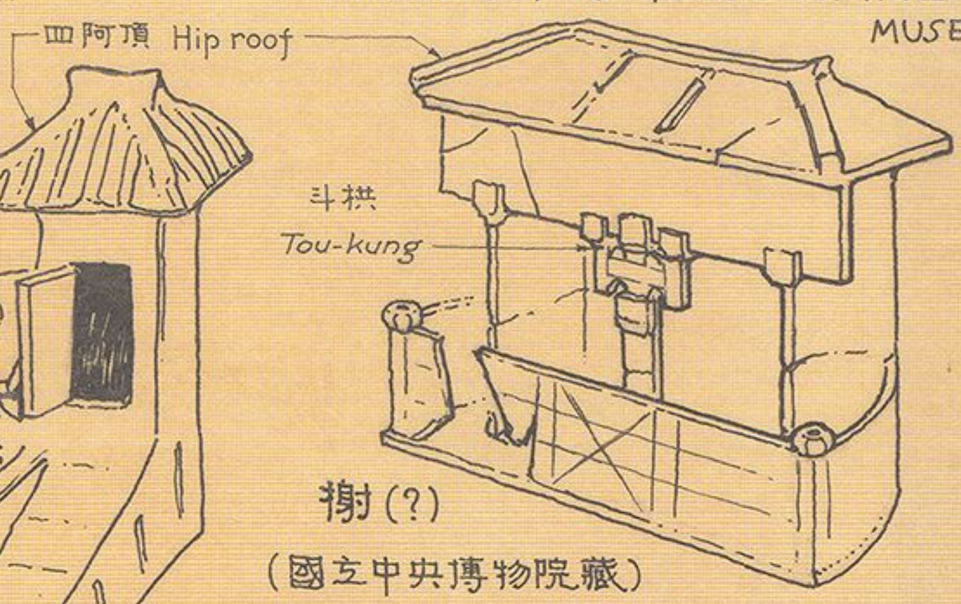
(長沙左氏藏)

PIG STYLE

(TSO'S COL'N)

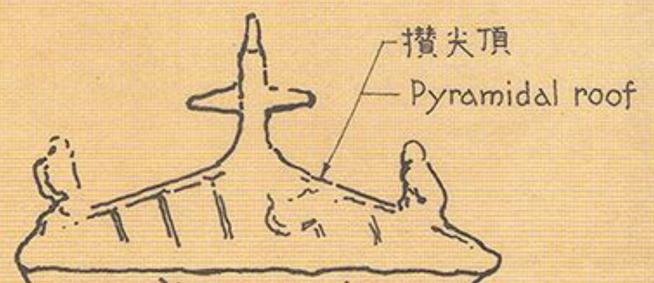


PAVILION (NATIONAL CENTRAL MUSEUM)



榭(?)

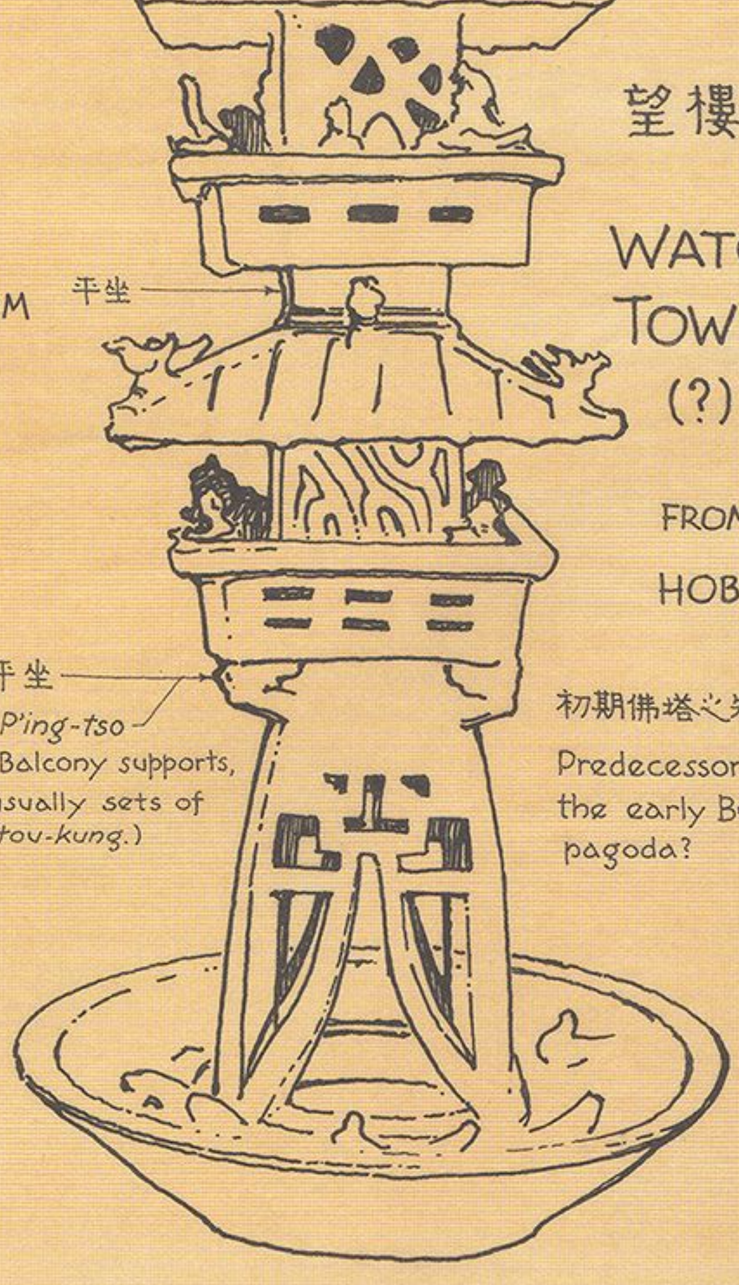
(國立中央傳物院藏)



望樓(?)

WATCH TOWER (?)

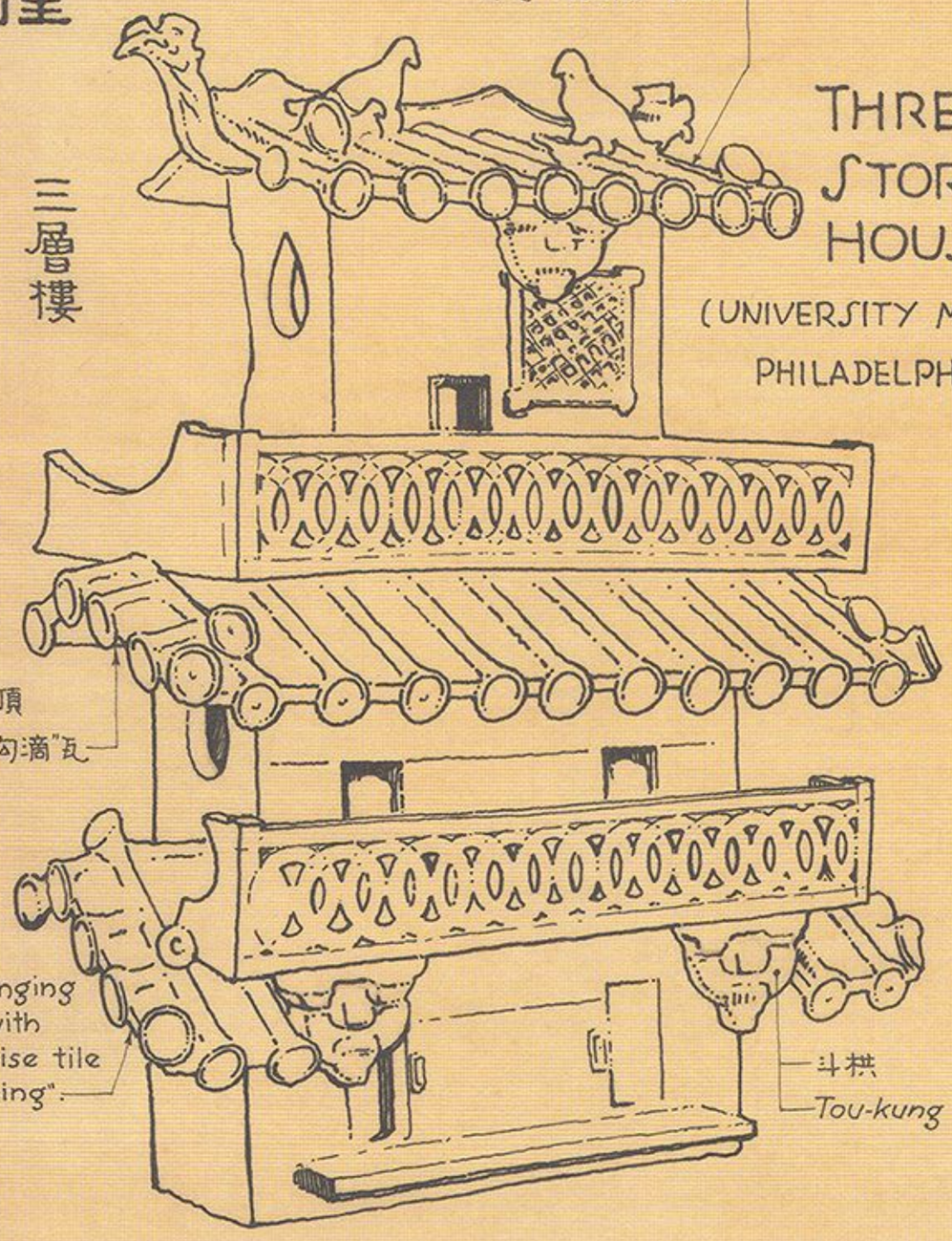
FROM HOBSON



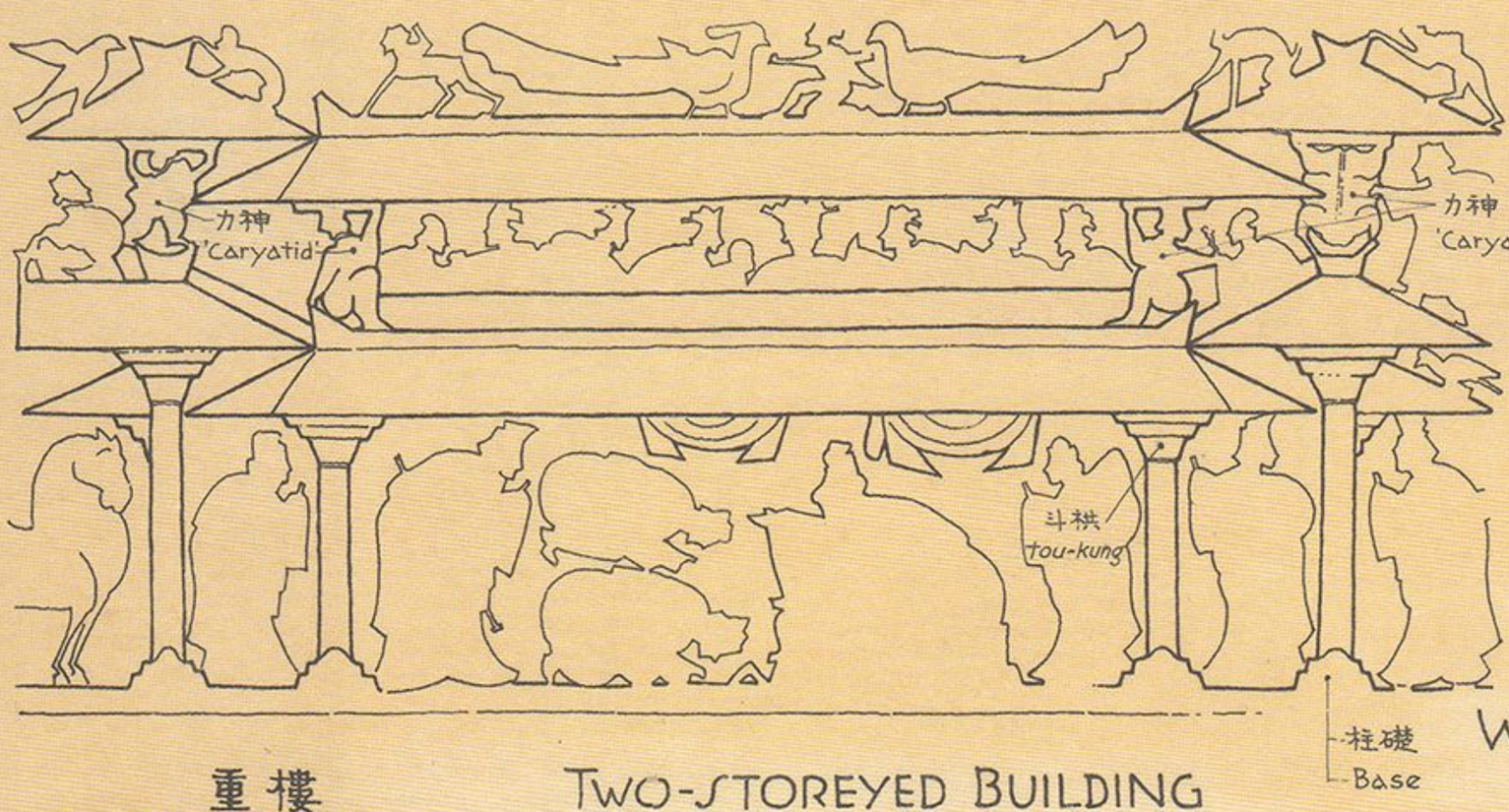
三層樓

THREE STOREY HOUSE

(UNIVERSITY MUSEUM PHILADELPHIA)



CLAY FUNEREAL HOUSE MODELS, HAN DYNASTY



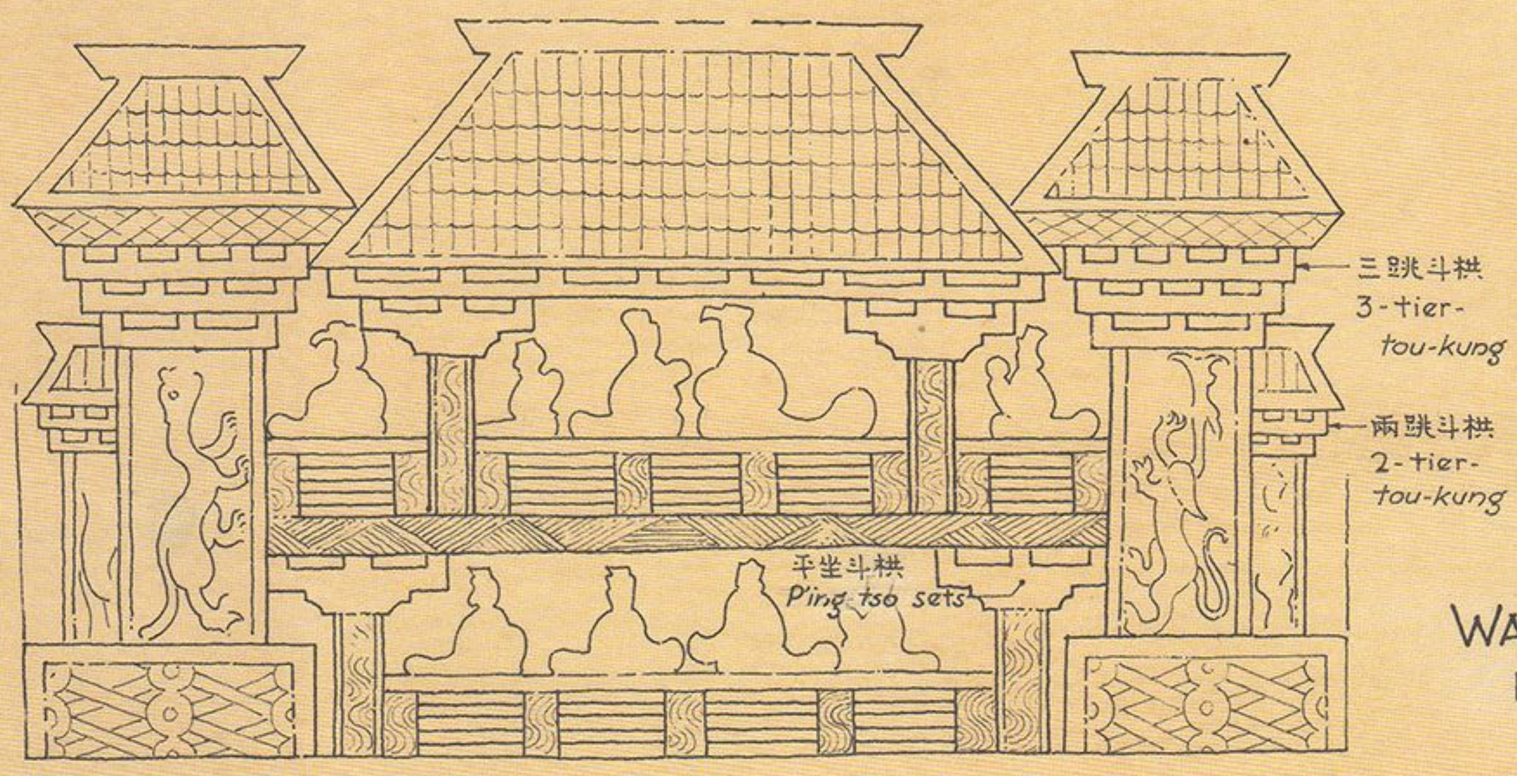
重樓
武氏祠画像石

TWO-STOREYED BUILDING
FROM THE WU FAMILY SHRINES



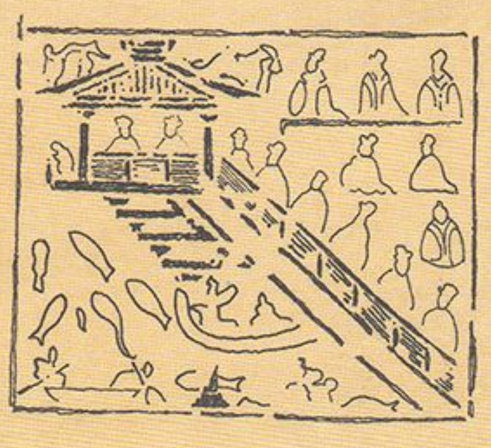
臨水亭榭（其一）
两城山画象石

WATER-FRONT PAVILION
FROM LIANG-CH'ENG SHAN



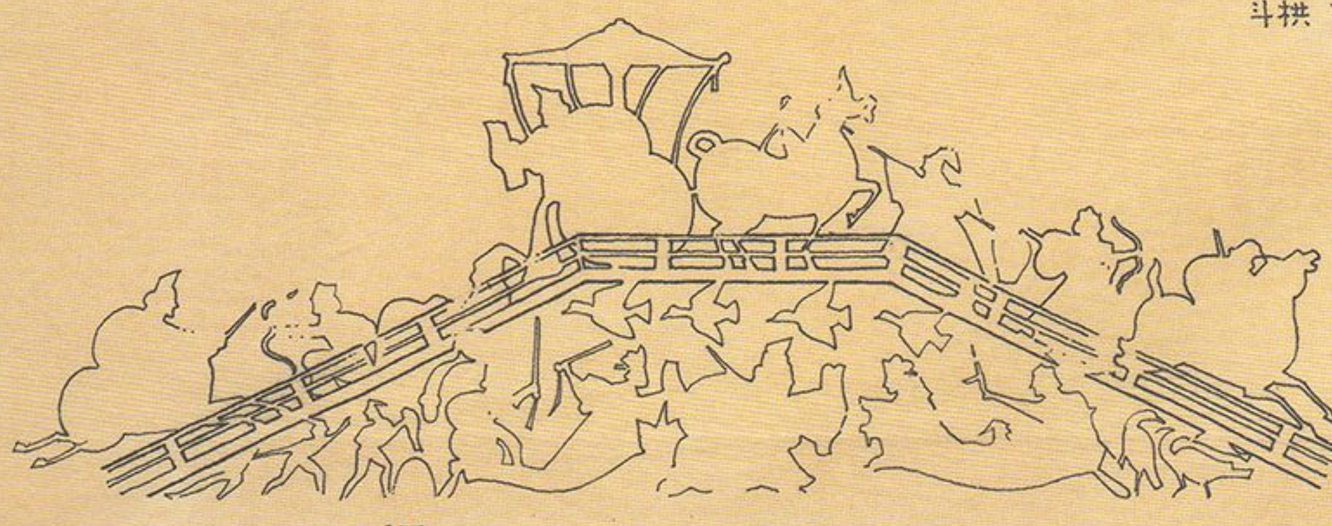
重樓並雙闕
紐約博物館藏石

TWO-STOREYED BUILDING WITH CH'ÜEH
(METROPOLITON MUSEUM, NEW YORK.)



臨水亭榭（其二）

WATER-FRONT PAVILION
FROM LIANG-CH'ENG SHAN



橋
武氏祠画象石

BRIDGE
FROM THE WU FAMILY SHRINES



城門 盩厔(函)谷關東門
CITY-GATE
EAST GATE OF HAN-KU KUAN
(BOSTON MUSEUM OF FINE ARTS)

漢画象石中建築數種 ARCHITECTURE FOUND IN ENGRAVED STONES (OR RELIEFS) OF THE HAN DYNASTY 205 B.C.-220 A.D.

歷代木構殿堂外觀演變圖

EVOLUTION OF THE GENERAL APPEARANCE OF TIMBER-FRAMED HALLS

公尺 10 0 50 meters

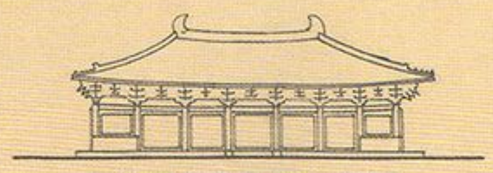
PERIOD OF VIGOUR 豪勁時期 約 APPROX. 600?-1050

PERIOD OF ELEGANCE 醇和時期 約 APPROX. 1000-1400

PERIOD OF RIGIDITY 羈直時期 約 APPROX. 1400-1900

吳殿(四阿殿·廡殿)
HIP-ROOFED

唐
T'ANG

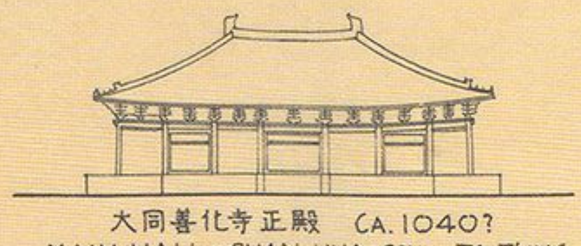


五台佛光寺正殿 857
MAIN HALL · FO-KUANG SSU · WU-T'AI

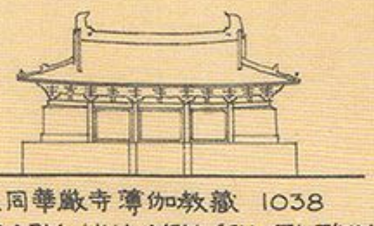
曹殿(九脊殿·歇山殿)
GABLE AND HIP-ROOFED

前向(山歇)殿厦
GABLE FACING FRONT

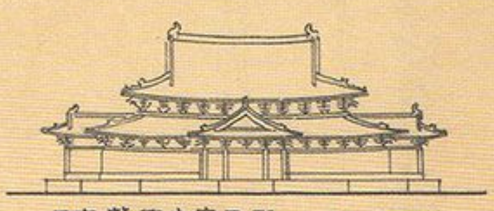
遼及宋初
LIAO & EARLY SUNG



大同善化寺正殿 CA. 1040?
MAIN HALL · SHAN-HUA SSU · TA-T'UNG



大同華嚴寺薄伽教藏 1038
LIBRARY · HUA-YEN SSU · TA-T'UNG



正定龍興寺摩尼殿 CA. 970?
MO-NI TIEN · LUNG-HSING SSU · CHENG-TING

北宋末
LATE NORTH-SUNG

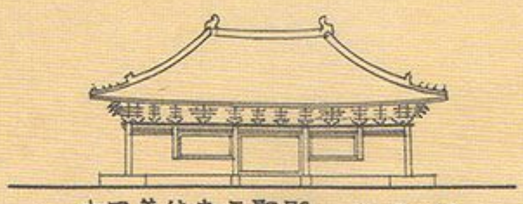


營造法式 1103
ACCORDING TO YING-TSAO FA-SHIH



✲ 嵩山少林寺初祖庵 1125
CH'U-TSU AN · SHAO-LIN SSU · SUNG MT.

金
CHIN

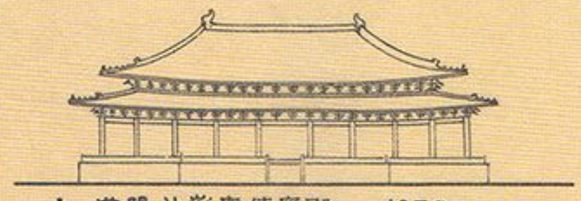


大同善化寺三聖殿 1118-43
FRONT HALL · SHAN-HUA SSU · TA-T'UNG

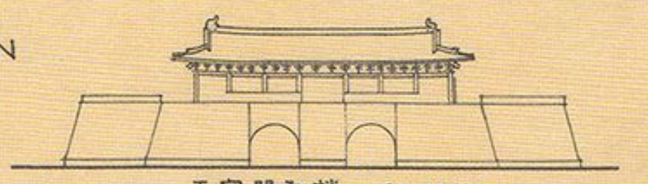


▲ 應縣淨土寺正殿 1124
MAIN HALL · CHING-T'U SSU · YING HSIEN

元
YUAN



✲ 曲陽北嶽廟德寧殿 1270
MAIN HALL · PEI-YUEH MIAO · CH'Ü-YANG

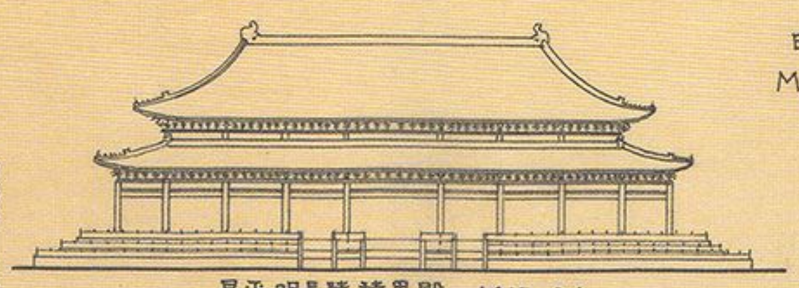


正定陽和樓 CA. 1260
YANG-HO LOU · CHENG-TING

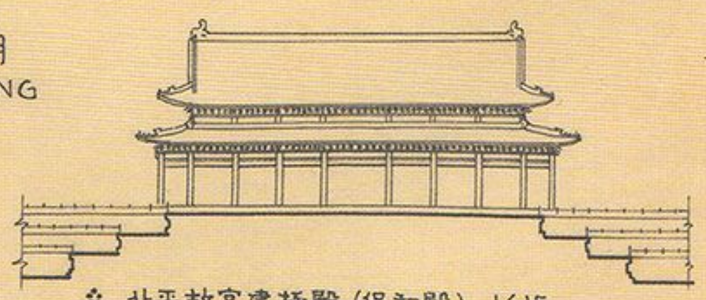


✲ 文水聖母廟 1283
SHENG-MU MIAO · WEN-SHUI

明
MING

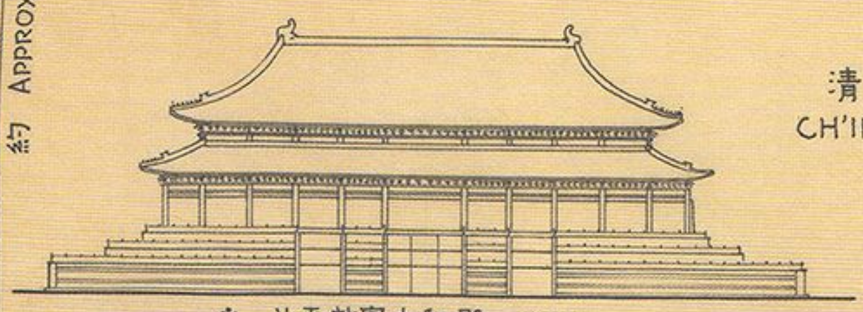


昌平明長陵稜恩殿 1415-26
SACRIFICIAL HALL · TOMB OF EMP'R YUNG-LO · CH'ANG-P'ING

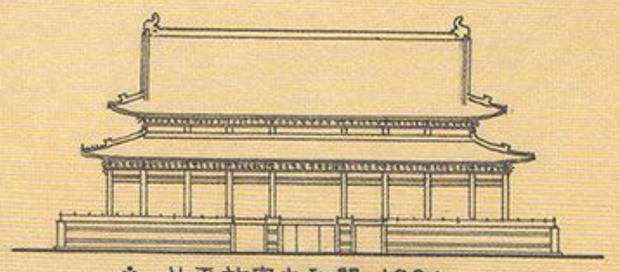


✲ 北平故宮皇極殿(保和殿) 1615
REAR AUDIENCE HALL · IMPERIAL PALACES · PEIPING

清
CH'ING



✲ 北平故宮太和殿 1697
MAIN AUDIENCE HALL · IMPERIAL PALACES · PEIPING



✲ 北平故宮太和門 1894
T'AI-HO MEN · IMPERIAL PALACES · PEIPING

✲立面畫根據実測平面及照片繪成.
Elevation drawn from measured plan and photograph.

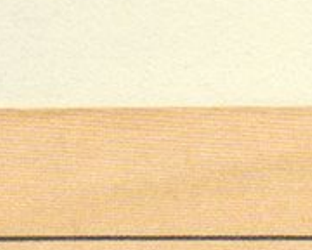
▲未經実測,立面畫完全自照片摹畫.
Not measured, elevation drawn solely from photograph.

無標誌各畫均經全部測繪.
All other elevations drawn from complete measurements.

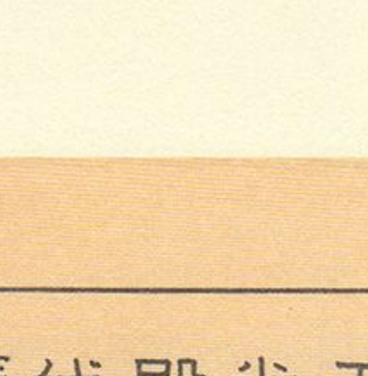
梁學

歷代殿堂平面及列柱位置比較圖

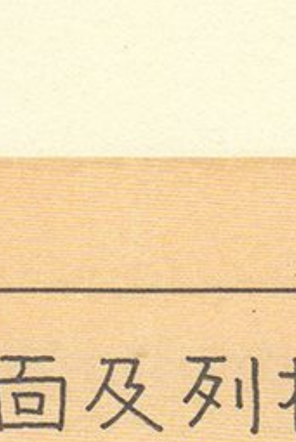
COMPARISON OF PLAN SHAPES AND COLUMNIATION OF TIMBER-FRAMED HALLS



五台佛光寺正殿
MAIN HALL · FO-KUANG SSU
WU-T'AI · 857



正定龍興寺摩尼殿
MO-NI TIEN · LUNG-HSING SSU
CHENG-TING · CA. 970



濟源
濟瀆廟淵德殿及寢殿
MAIN HALL & REAR HALL
TSI-TU MIAO · TSI-YUAN
973 (?)



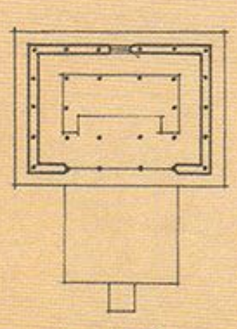
正定龍興寺
轉輪藏殿
LIBRARY
LUNG-HSING SSU
CHENG-TING
CA. 1040 ?



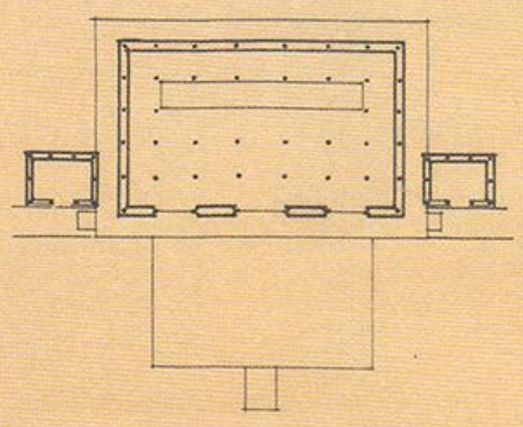
五台佛光寺文殊殿
WEN-SHU TIEN
FO-KUANG SSU
WU-T'AI · CA. 1050 ?



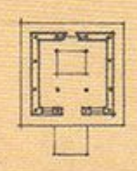
寶坻廣濟寺
三大士殿
MAIN HALL
KUANG-TSI SSU · PAO-TI
1025



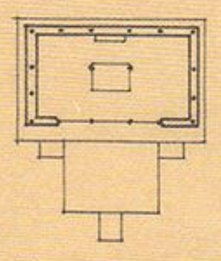
大同華嚴寺
薄伽教藏
LIBRARY · HUA-YEN SSU
TA-T'UNG · 1038



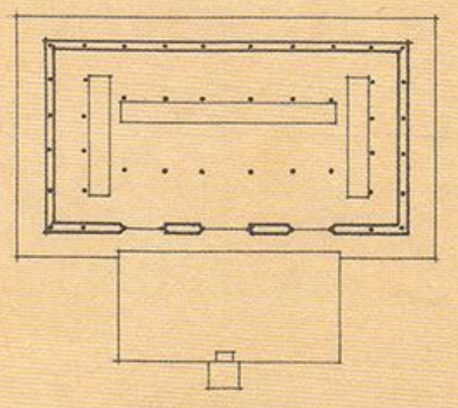
大同善化寺正殿及朵殿
MAIN HALL & 'EAR HALLS'
SHAN-HUA SSU · TA-T'UNG · CA. 1050?



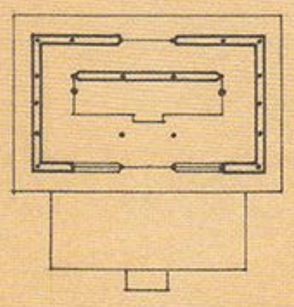
嵩山少林寺
初祖庵
CH'U-TSU AN
SHAO-LIN SSU
SUNG MT.
1125



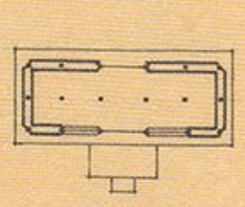
濟源奉仙觀大殿
MAIN HALL
FENG-SIEN KUAN
TSI-YUAN
CA. 1130 (?)



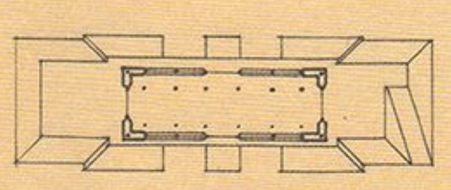
大同華嚴寺大殿
MAIN HALL · HUA-YEN SSU
TA-T'UNG · CA. 1130 (?)



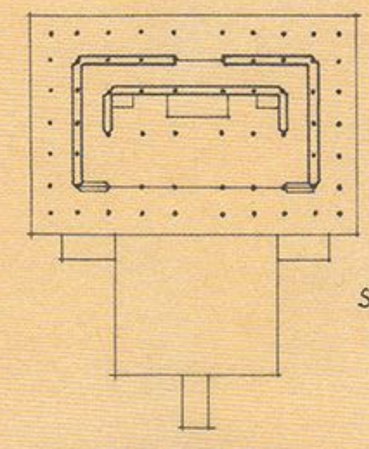
大同善化寺三聖殿
FRONT HALL · SHAN-HUA SSU
TA-T'UNG · 1118-43



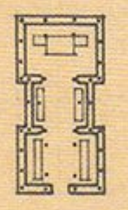
大同善化寺山門
MAIN GATE
SHAN-HUA SSU
TA-T'UNG
1118-43



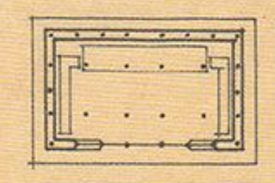
正定陽和樓
YANG-HO LOU · CHENG-TING
CA. 1260



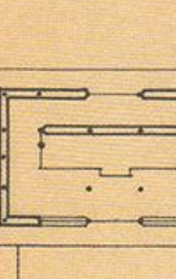
曲陽北嶽廟德寧殿
MAIN HALL · PEI-YUEH MIAO
CH'Ü-YANG · 1270



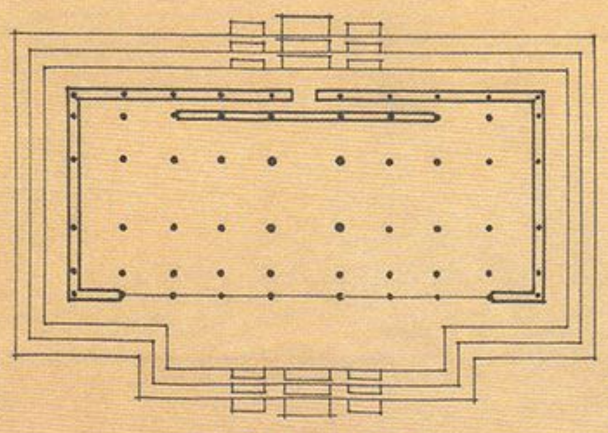
安平聖姑廟
SHENG-KU MIAO
AN-P'ING
1306



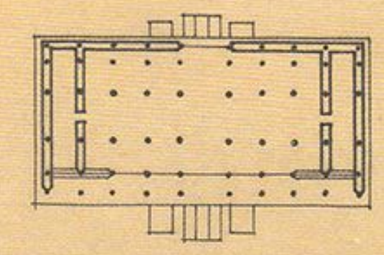
趙城廣勝下寺大殿
MAIN HALL
LOWER TEMPLE
KUANG-SHENG SSU
CHAO-CH'ENG
1319



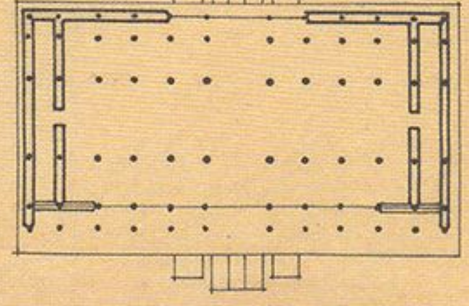
趙城廣勝寺
明應王殿
HALL OF
DRAGON KING
KUANG-SHENG SSU
CHAO-CH'ENG
1320



昌平明長陵棱恩殿
SACRIFICIAL HALL · TOMB OF EMP'R YUNG-LO
CH'ANG-P'ING · 1415-26

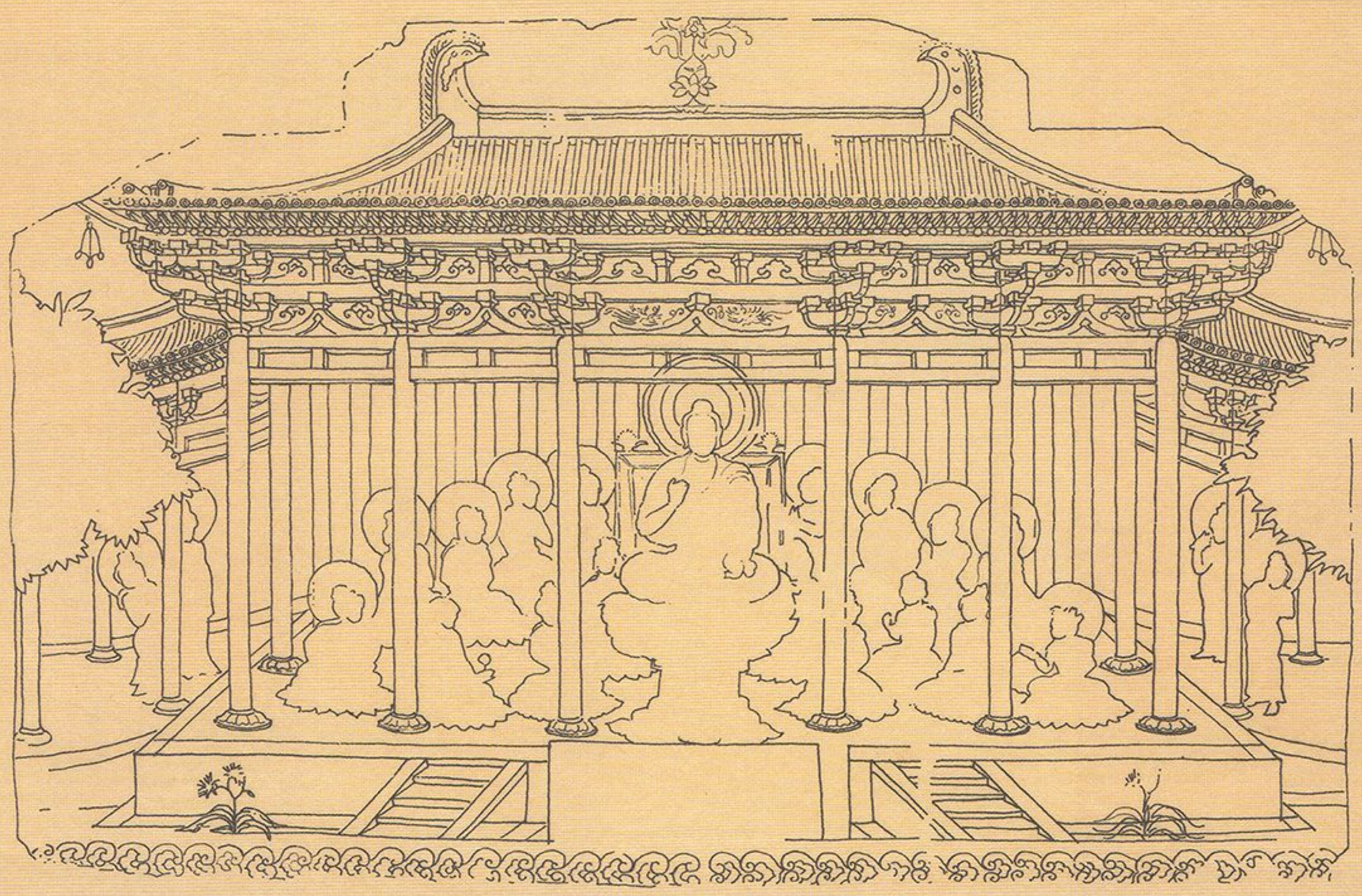


北平故宮建極殿(保和殿)
CHIEN-CHI TIEN (PAO-HO TIEN)
IMPERIAL PALACES
PEIPING · 1615



北平故宮太和殿
T'AI-HO TIEN · IMPERIAL PALACES
(PRINCIPAL HALL OF AUDIENCE)
PEIPING · 1697

公尺 10 0 20 40 60 80 meters



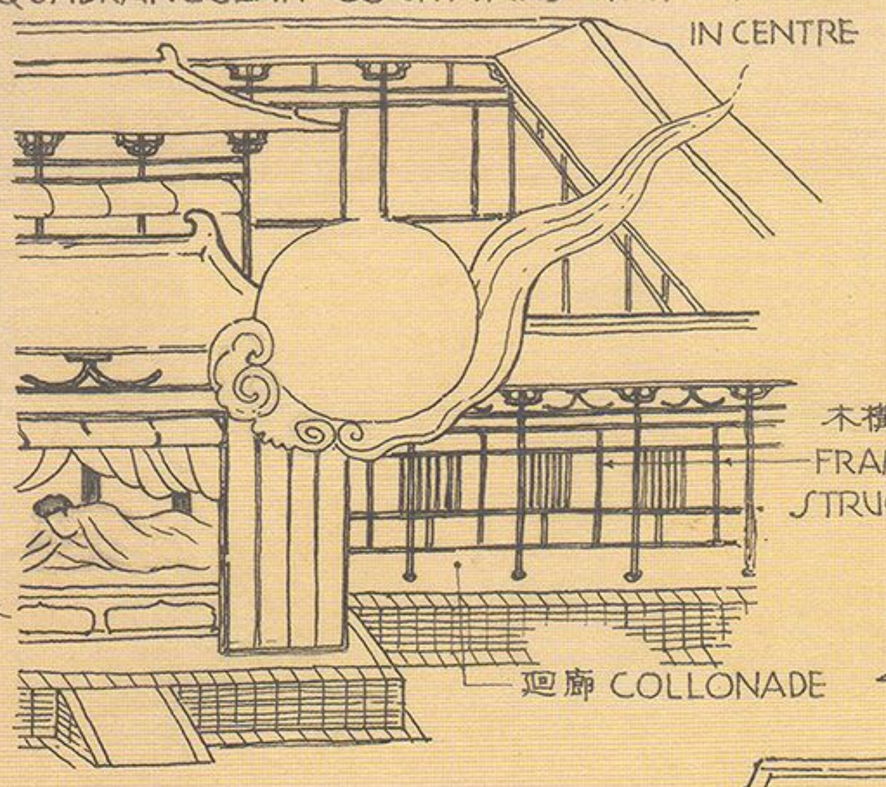
A TEMPLE HALL OF THE T'ANG DYNASTY

AFTER A RUBBING OF THE ENGRAVING ON THE TYMPANIUM OVER THE WEST GATEWAY OF TA-YEN T'A, TZ'U-EN SSŬ, SI-AN, SHENSI

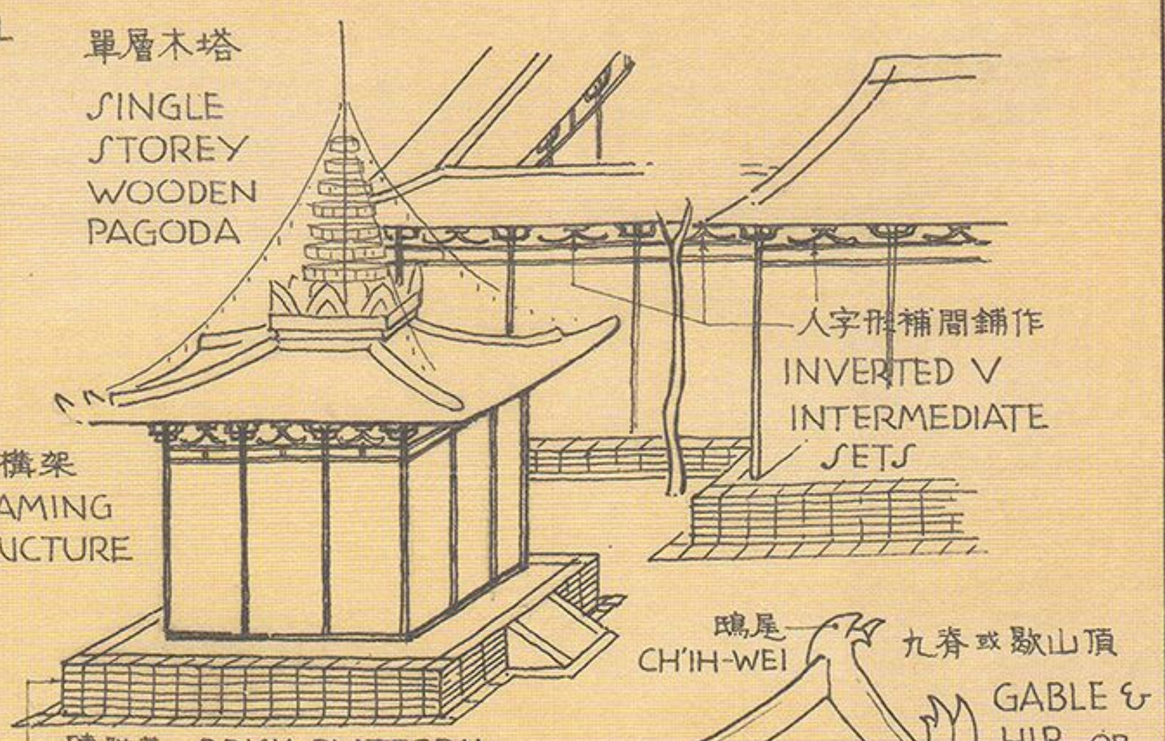
唐代佛殿圖　摹自陝西長安大雁塔西門門楣石畫像

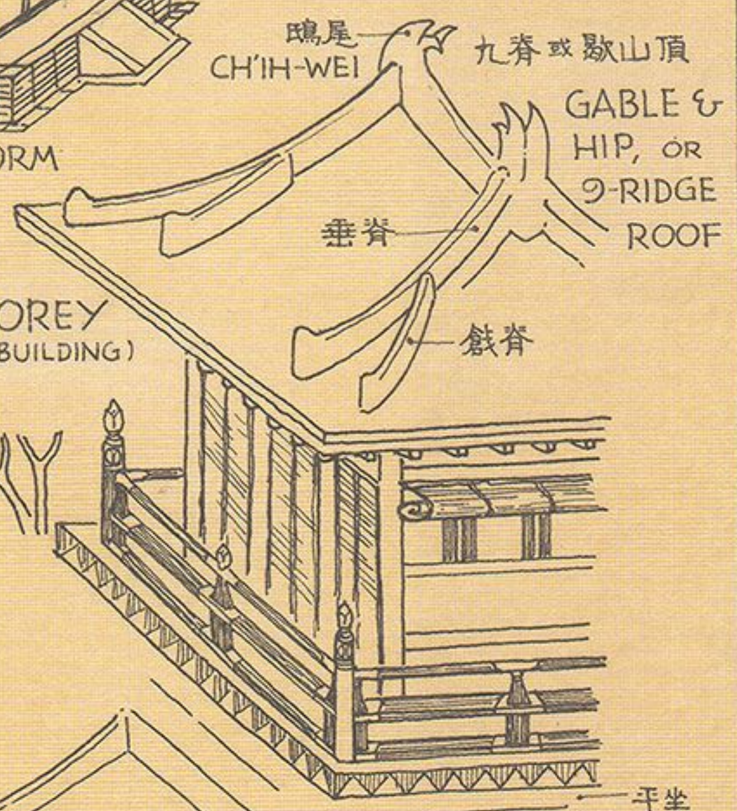
QUADRANGULAR COURTYARD WITH MAIN-HALL IN CENTRE

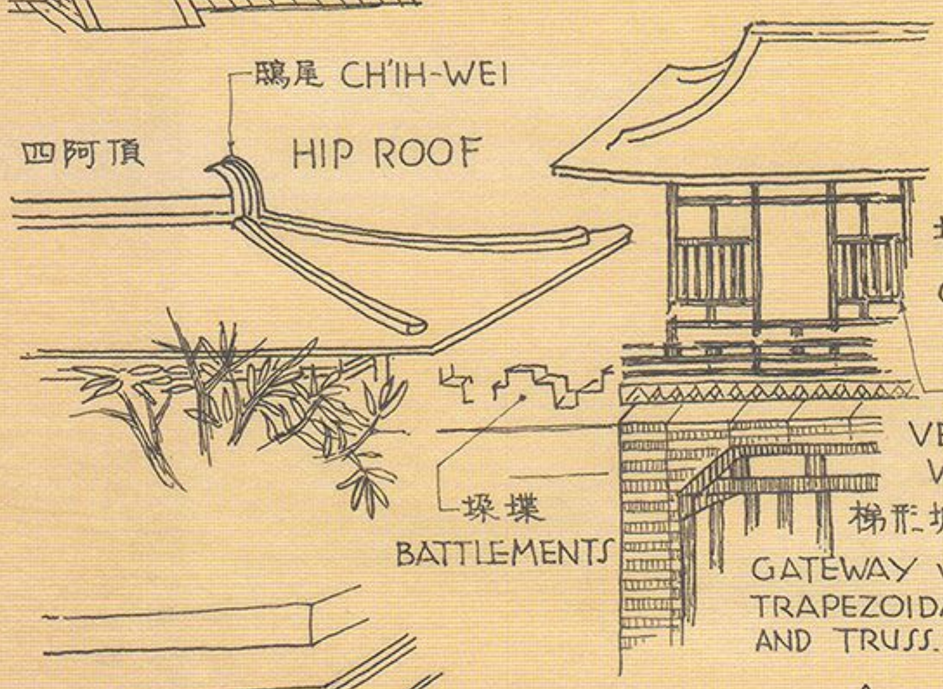
迴廊圍繞大殿在中

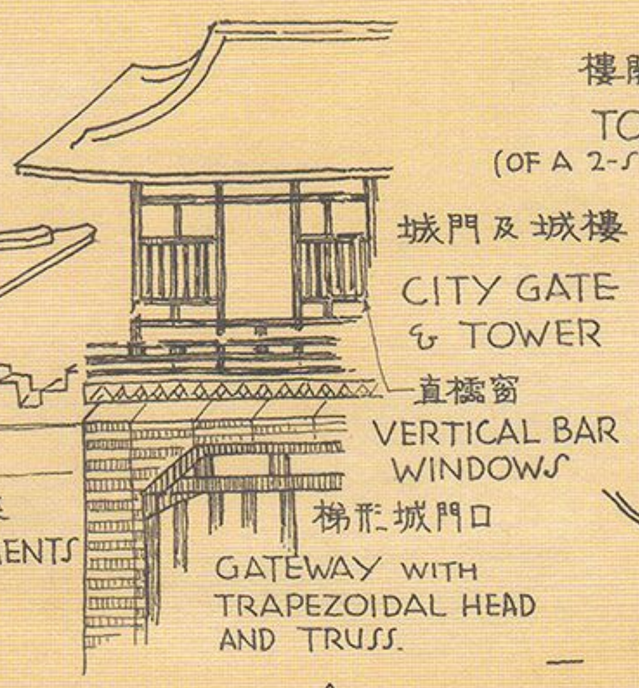


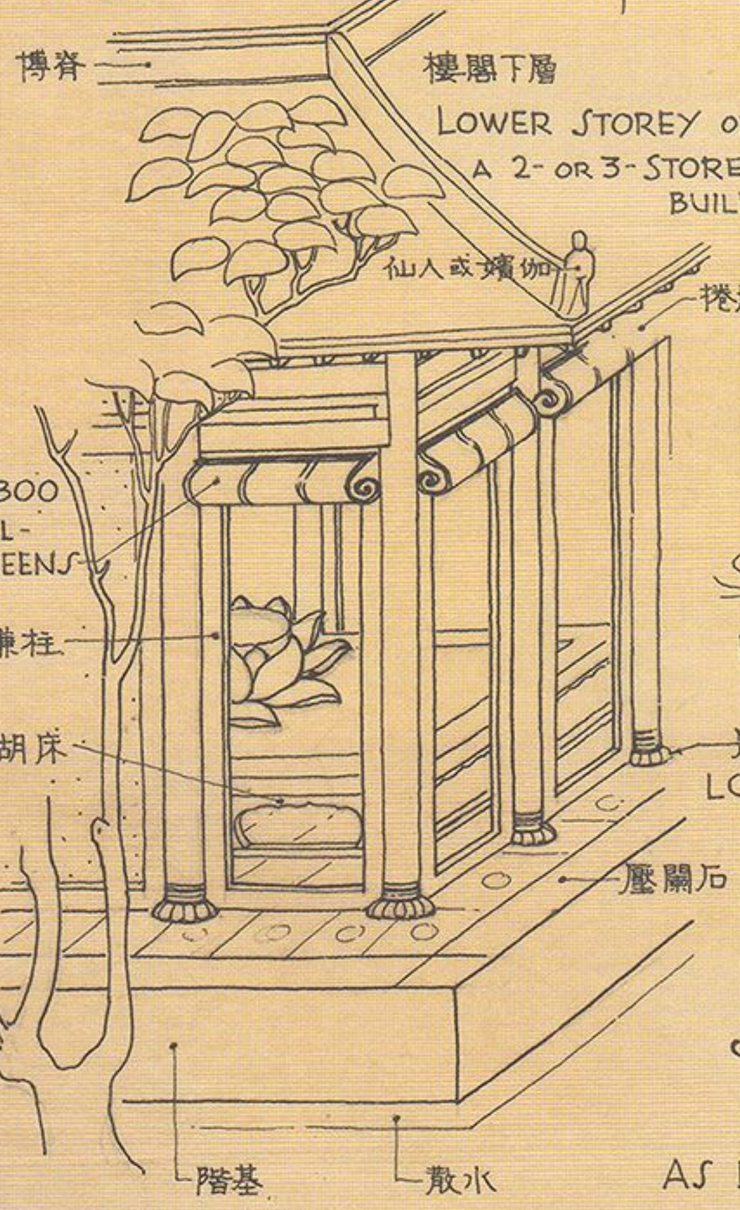
單層木塔
SINGLE STOREY WOODEN PAGODA



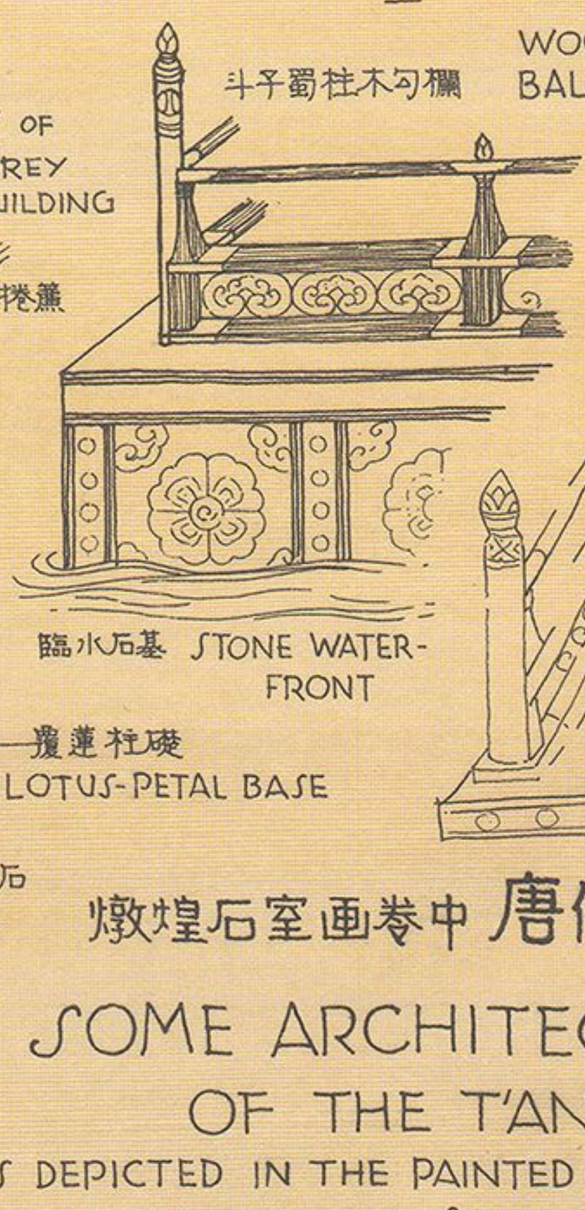








PLATFORM OR PODIUM.



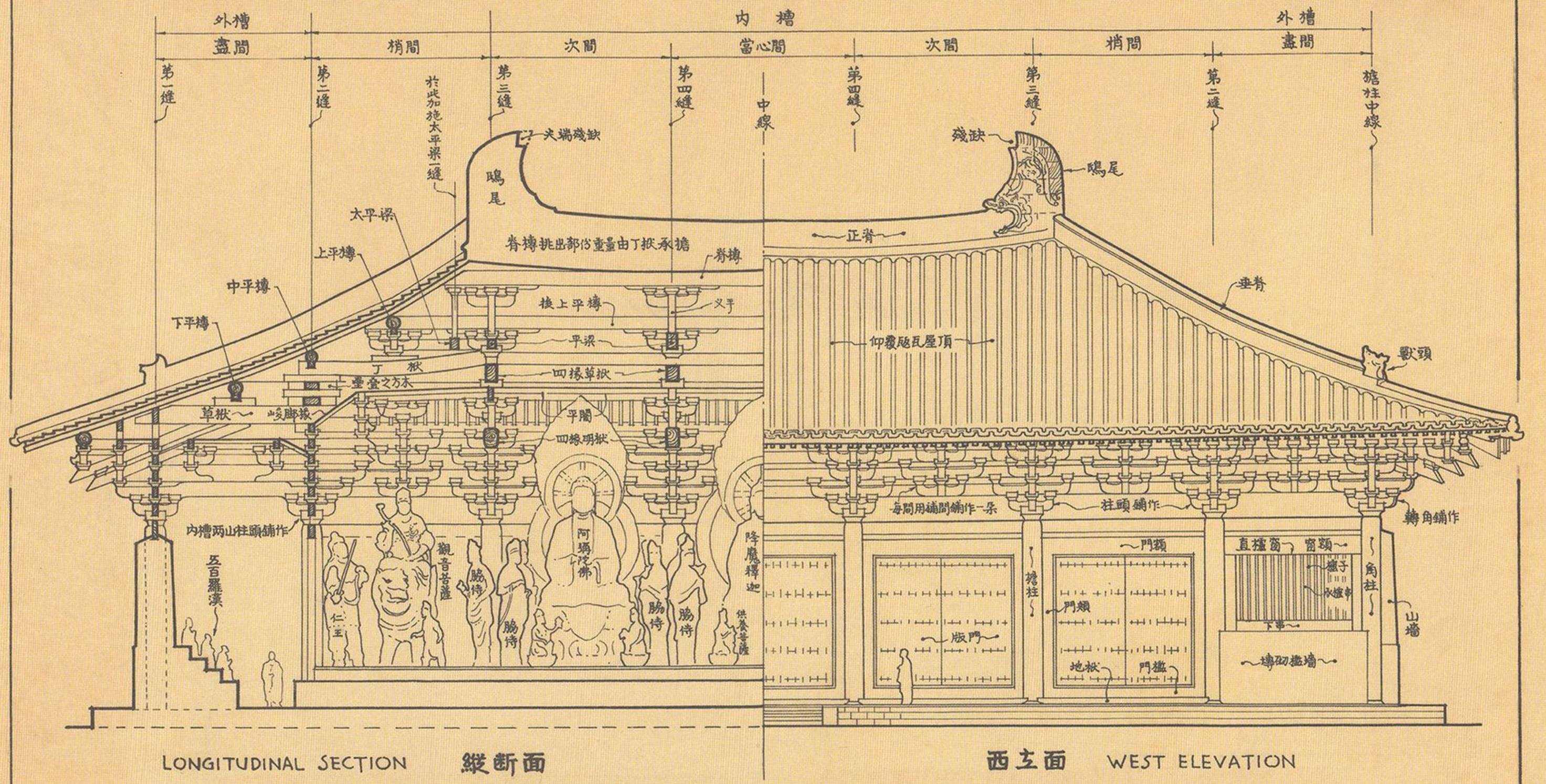


唐代建築部分詳圖 中卷画室石皇灰燉

SOME ARCHITECTURAL DETAILS OF THE T'ANG DYNASTY

AS DEPICTED IN THE PAINTED SCROLLS FROM TUN-HUANG

REDRAWN FROM STEIN: THOUSAND BUDDHAS.



LONGITUDINAL SECTION 縱斷面

1 0 5公尺

西立面 WEST ELEVATION

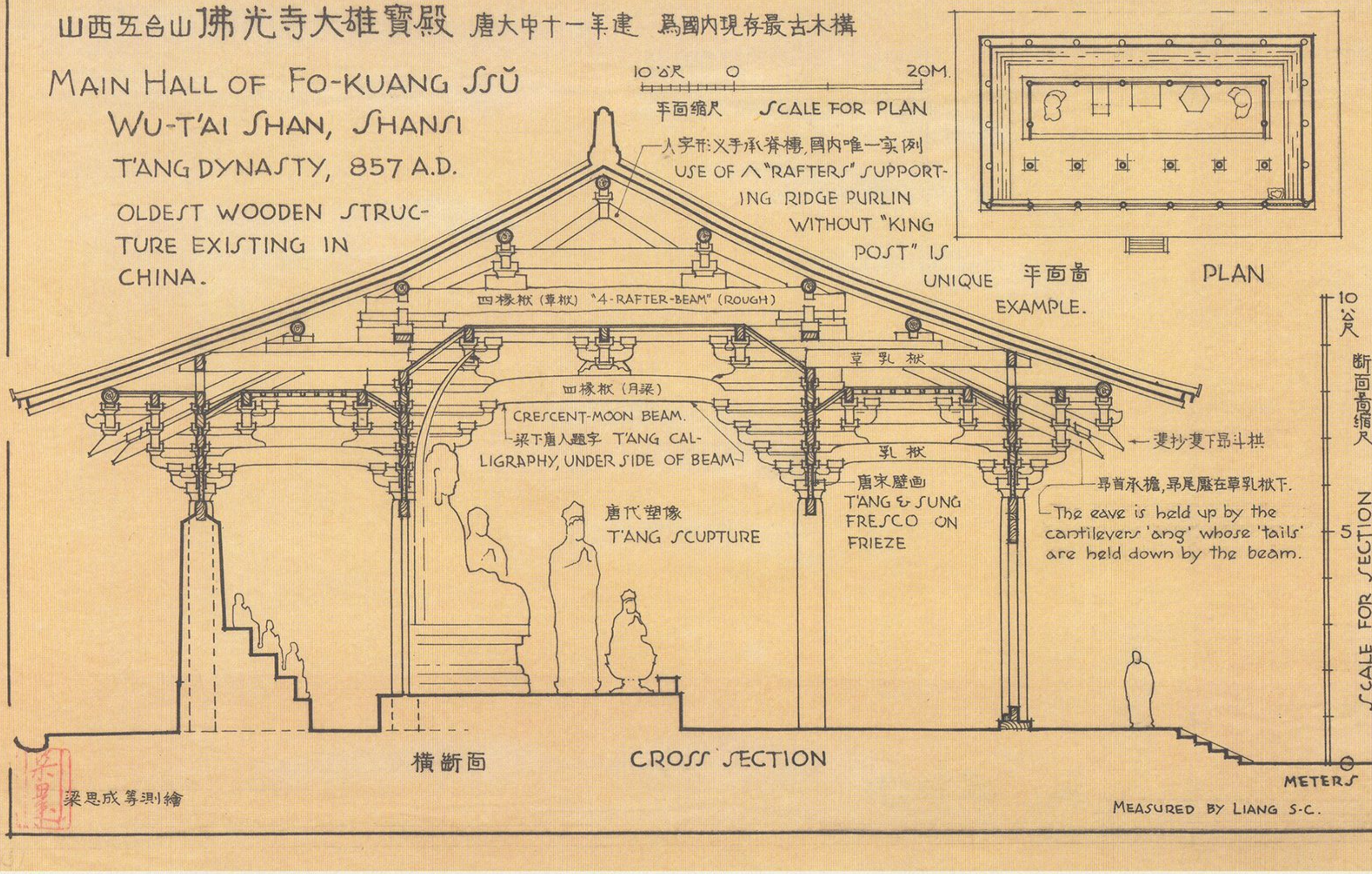
山西五台山 佛光寺大殿 唐大中十一年建 857 A.D.

MAIN HALL OF FO-KUANG SSU · WU-T'AI SHAN · SHANSI

山西五台山佛光寺大雄寶殿 唐大中十一年建 爲國內現存最古木構

MAIN HALL OF FO-KUANG SSŬ WU-T'AI SHAN, SHANSI T'ANG DYNASTY, 857 A.D.

OLDEST WOODEN STRUCTURE EXISTING IN CHINA.



河北 薊縣

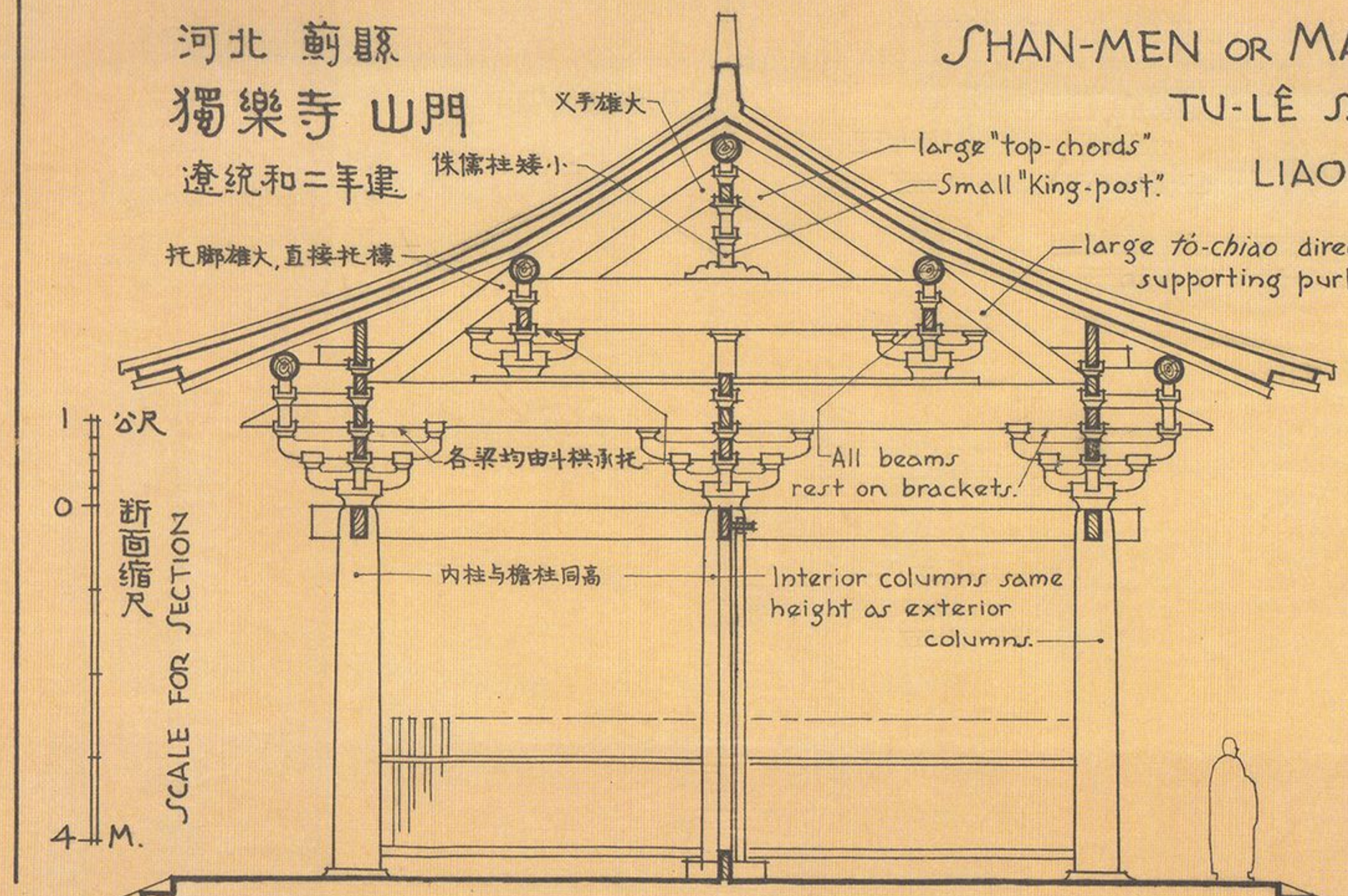
獨樂寺山門

遼統和二年建

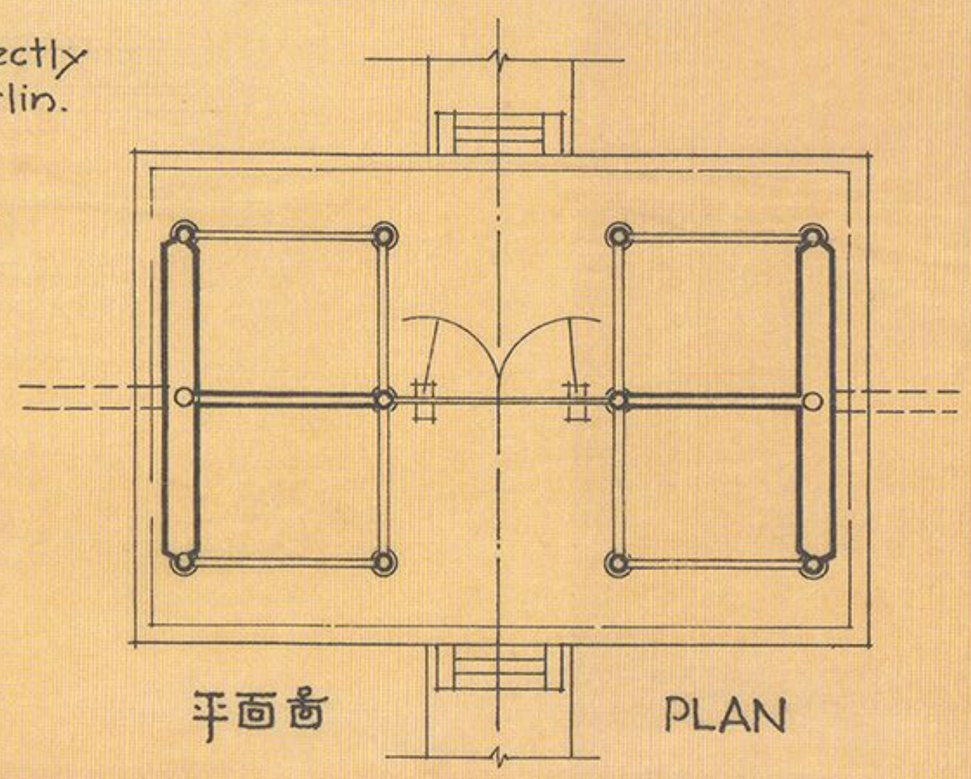
SHAN-MEN OR MAIN ENTRANCE GATEWAY

TU-LÊ SSU, CHI HSIEN, HOPEI.

LIAO DYNASTY, 984 A.D.



斷面圖 CROSS SECTION



平面圖 PLAN

公尺 5 0 10M

平面縮尺 SCALE FOR PLAN

SAN-TA-SHIH TIEN

OR THE HALL OF THE THREE BODDHISATVAS

KUANG-CHI SSU

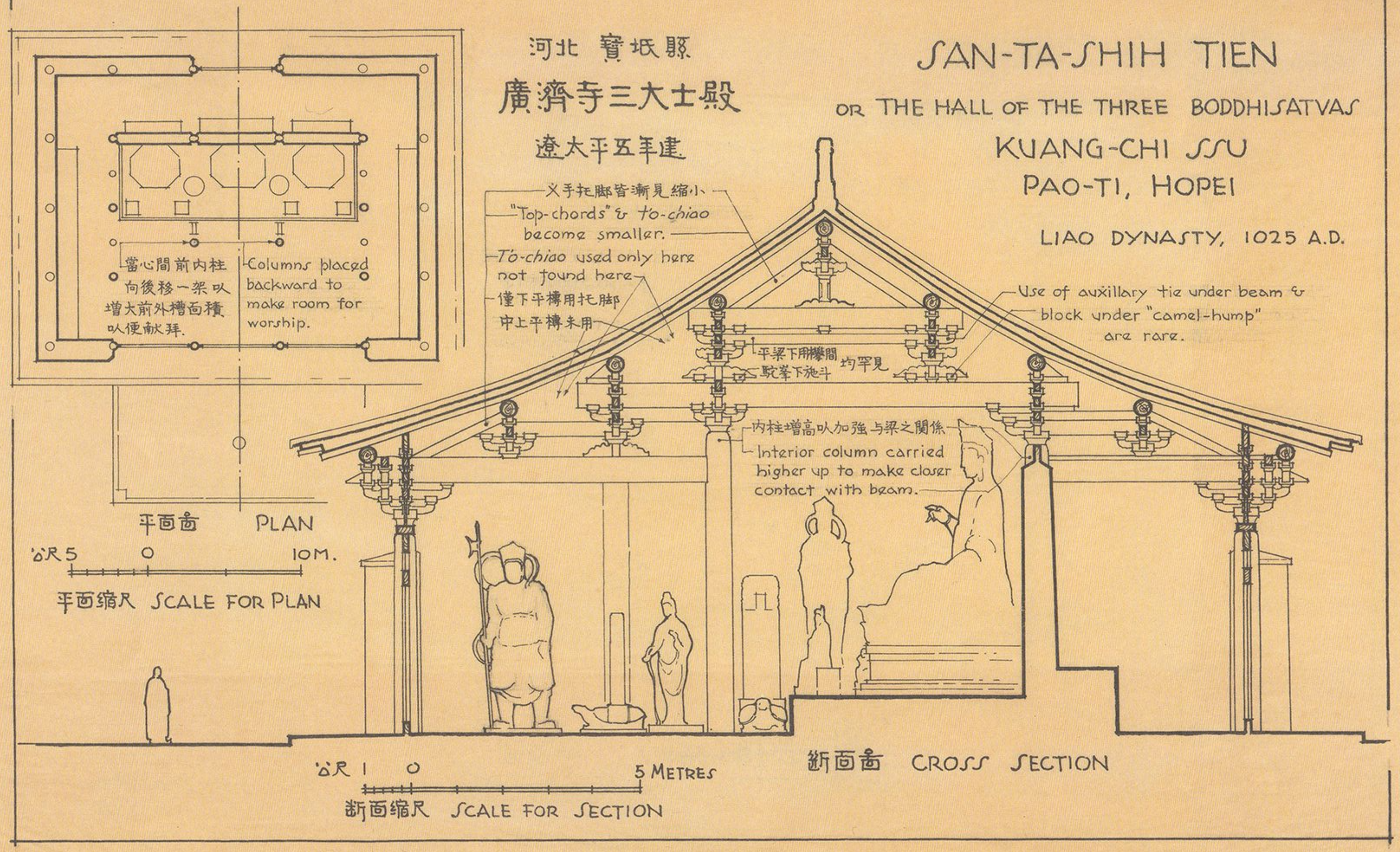
PAO-TI, HOPEI

河北 寶坻縣

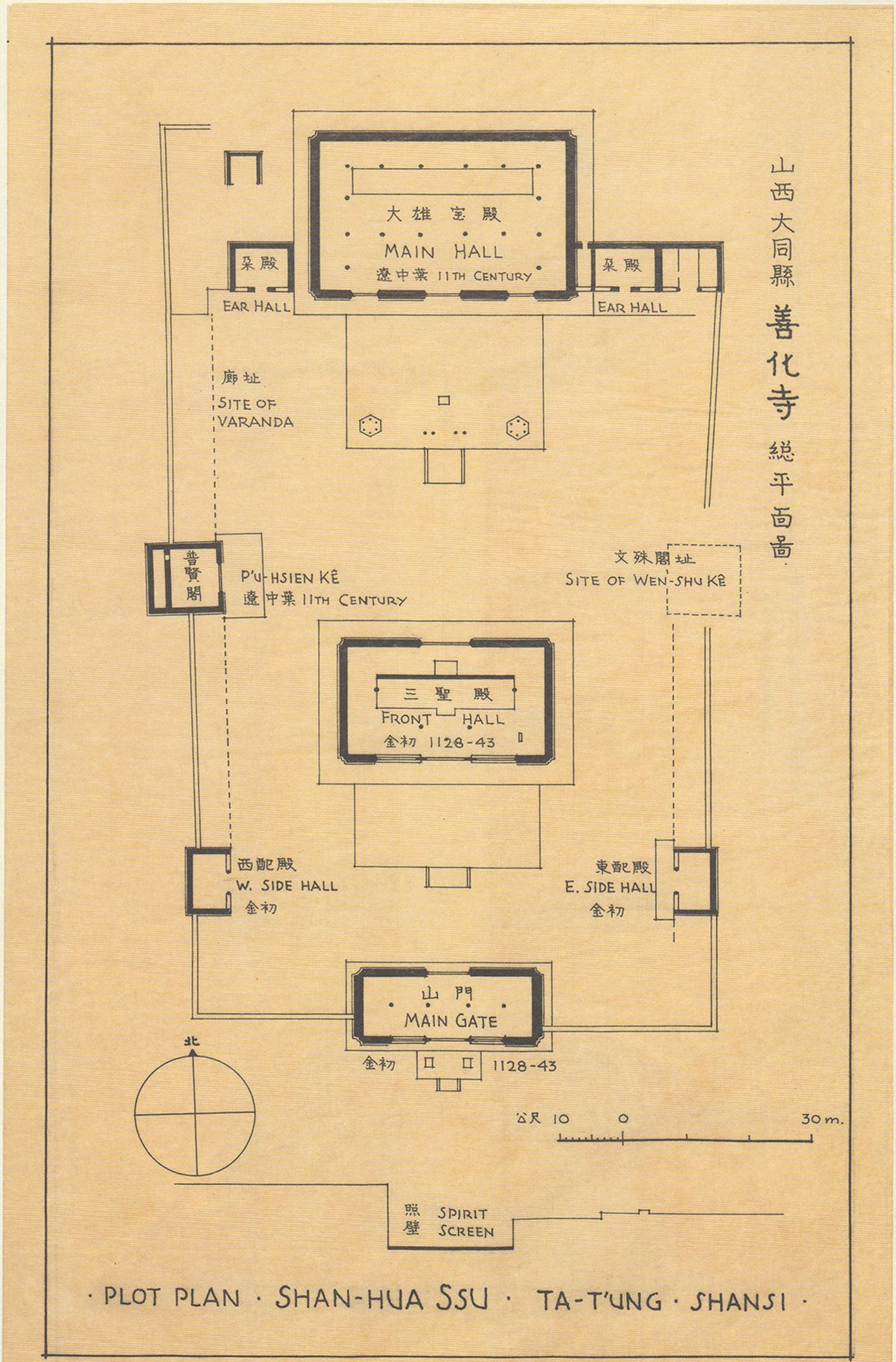
廣濟寺三大士殿

遼太平五年建

LIAO DYNASTY, 1025 A.D.



山西大同縣 善化寺 總平面圖



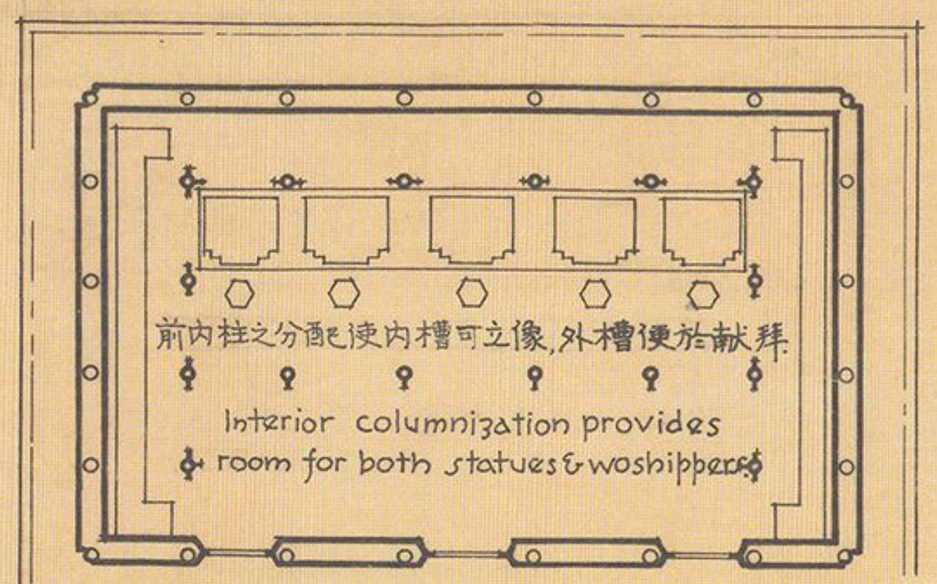
· PLOT PLAN · SHAN-HUA SSU · TA-T'UNG · SHANSI ·

山西大同縣

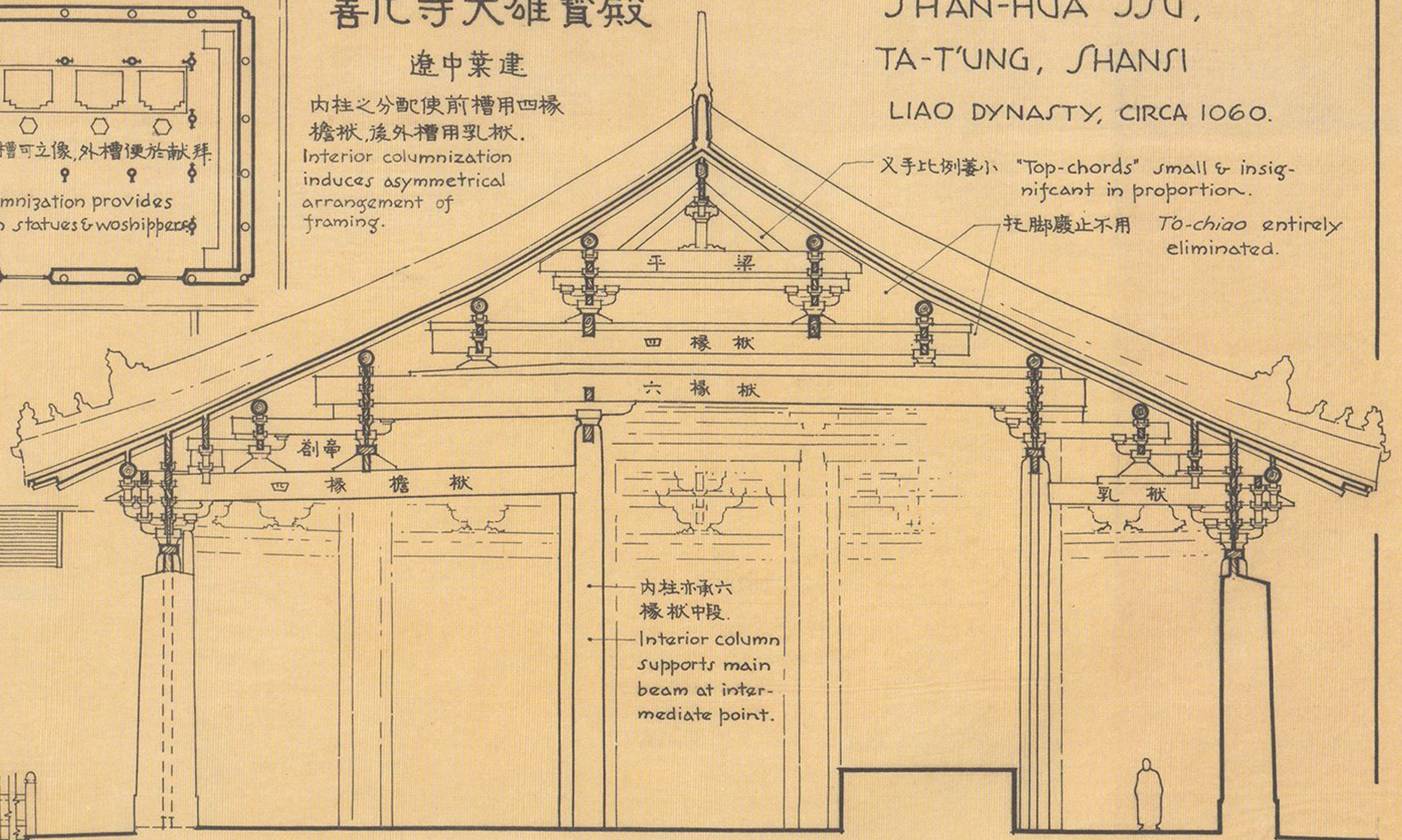
善化寺大雄寶殿

遼中葉建

MAIN HALL
SHAN-HUA SSU, TA-T'UNG, SHANSI
LIAO DYNASTY, CIRCA 1060.



內柱之分配使前槽用四椽檐栿,後外槽用乳栿.
Interior columnization induces asymmetrical arrangement of framing.



平面圖 PLAN

斷面圖 CROSS SECTION

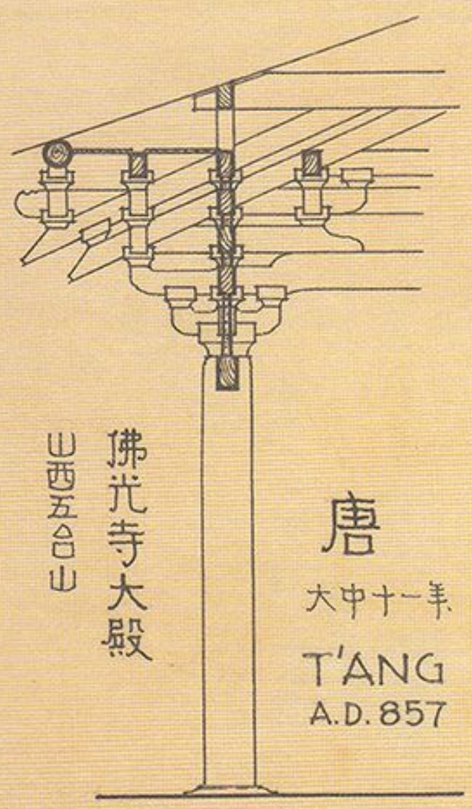
公尺10 0 20 M.

平面縮尺 SCALE FOR PLAN

公尺1 0 5 10 METRES.

斷面縮尺 SCALE FOR SECTION

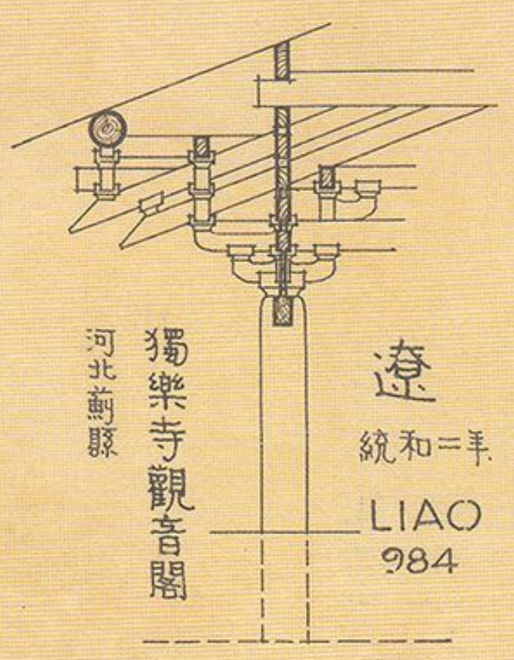
15220



佛光寺大殿
山西五台山

唐
大中十一年
T'ANG
A.D. 857

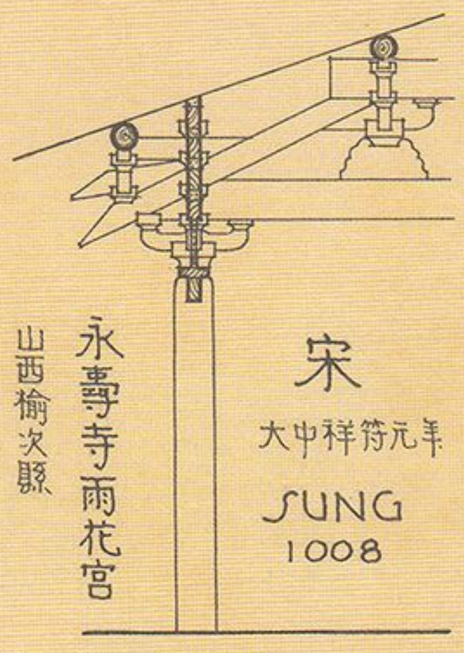
MAIN HALL, FO-KUANG SSU,
WU-T'AI SHAN, SHANSI



獨樂寺觀音閣
河北薊縣

遼
統和二年
LIAO
984

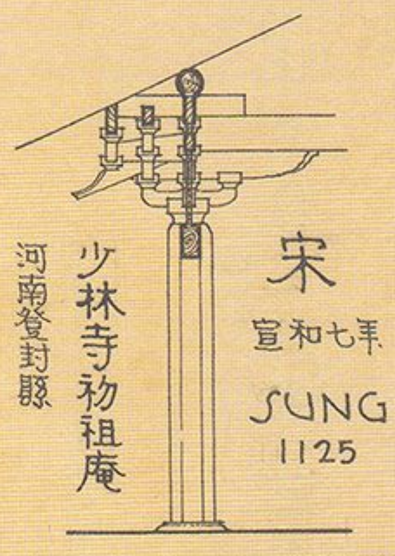
KUAN-YIN KÊ, TU-LÊ SSU
CHI HSIEN, HOPEI



永壽寺雨花宮
山西榆次縣

宋
大中祥符元年
SUNG
1008

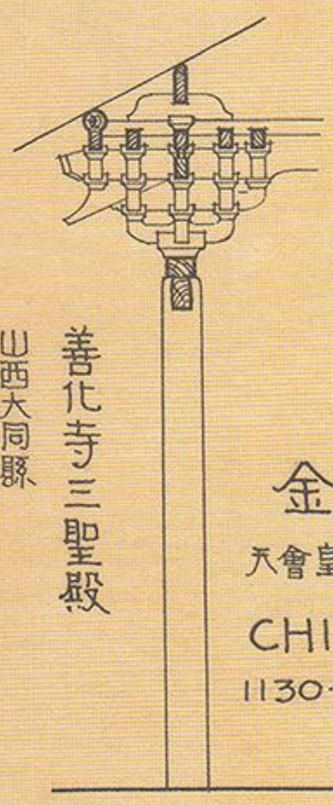
YÜ-HUA KUNG, YUNG-SHOU SSU
YÜ-TZ'Ŭ, SHANSI



少林寺初祖庵
河南登封縣

宋
宣和七年
SUNG
1125

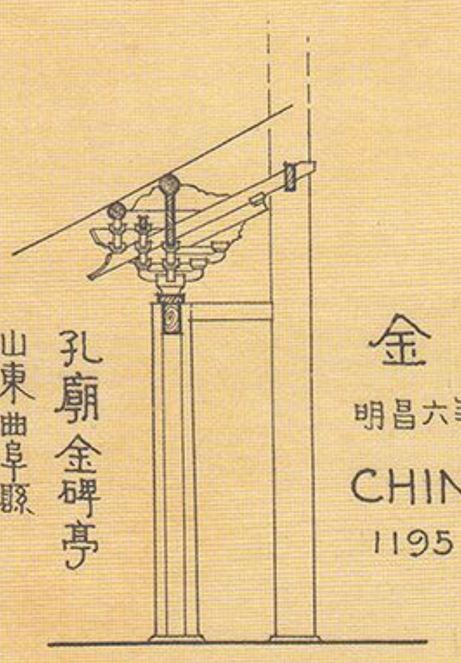
CH'U-TSU AN, SHAO-LIN SSU
TENG-FENG, HONAN.



善化寺三聖殿
山西大同縣

金
天會皇統間
CHIN
1130-43

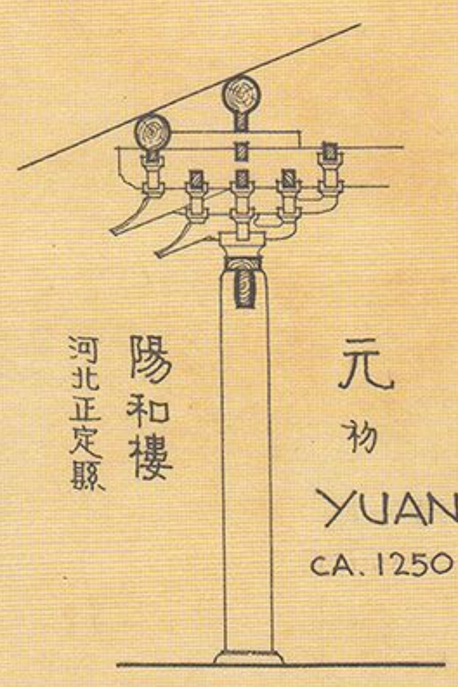
SHAN-SHENG TIEN, SHAN-HUA SSU
TA-T'UNG, SHANSI



孔廟金碑亭
山東曲阜縣

金
明昌六年
CHIN
1195

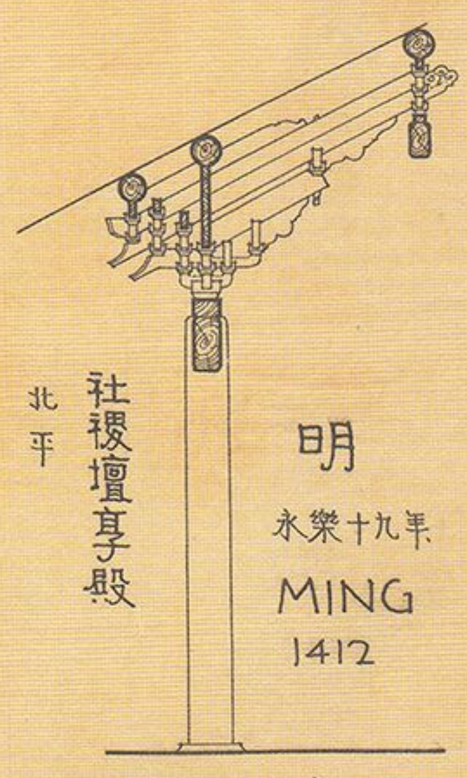
STELE PAVILION
CONFUCIUS' TEMPLE
CH'Ü-FOU, SHANTUNG



陽和樓
河北正定縣

元
初
YUAN
CA. 1250

YANG-HO LOU
CHENG-TING, HOPEI

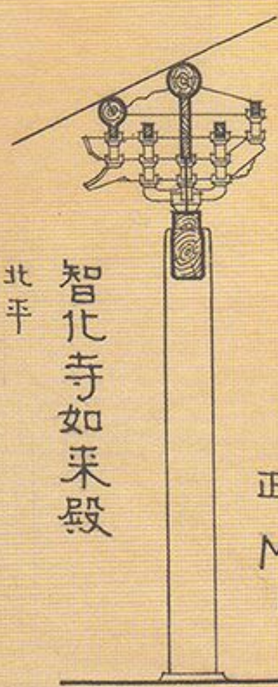


社稷壇享殿
北平

明
永樂十九年
MING
1412

MAIN HALL, SHÊ-CHI T'AN
PEIPING

公尺 1 0 5M.
縮尺 SCALE



智化寺如來殿
北平

明
正統八年
MING
1443

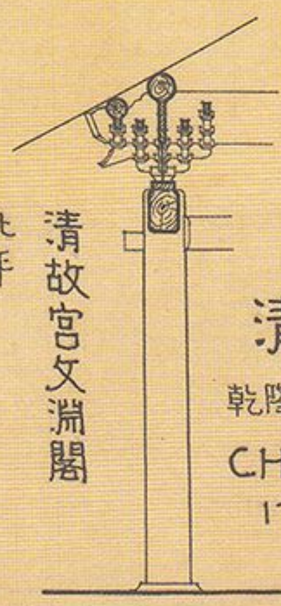
JU-LAI TIEN, CHIH-HUA SSU
PEIPING



孔廟奎文閣
山東曲阜縣

明
弘治十七年
MING
1504

LIBRARY, CONFUCIUS' TEMPLE
CHÜ-FOU, SHANTUNG



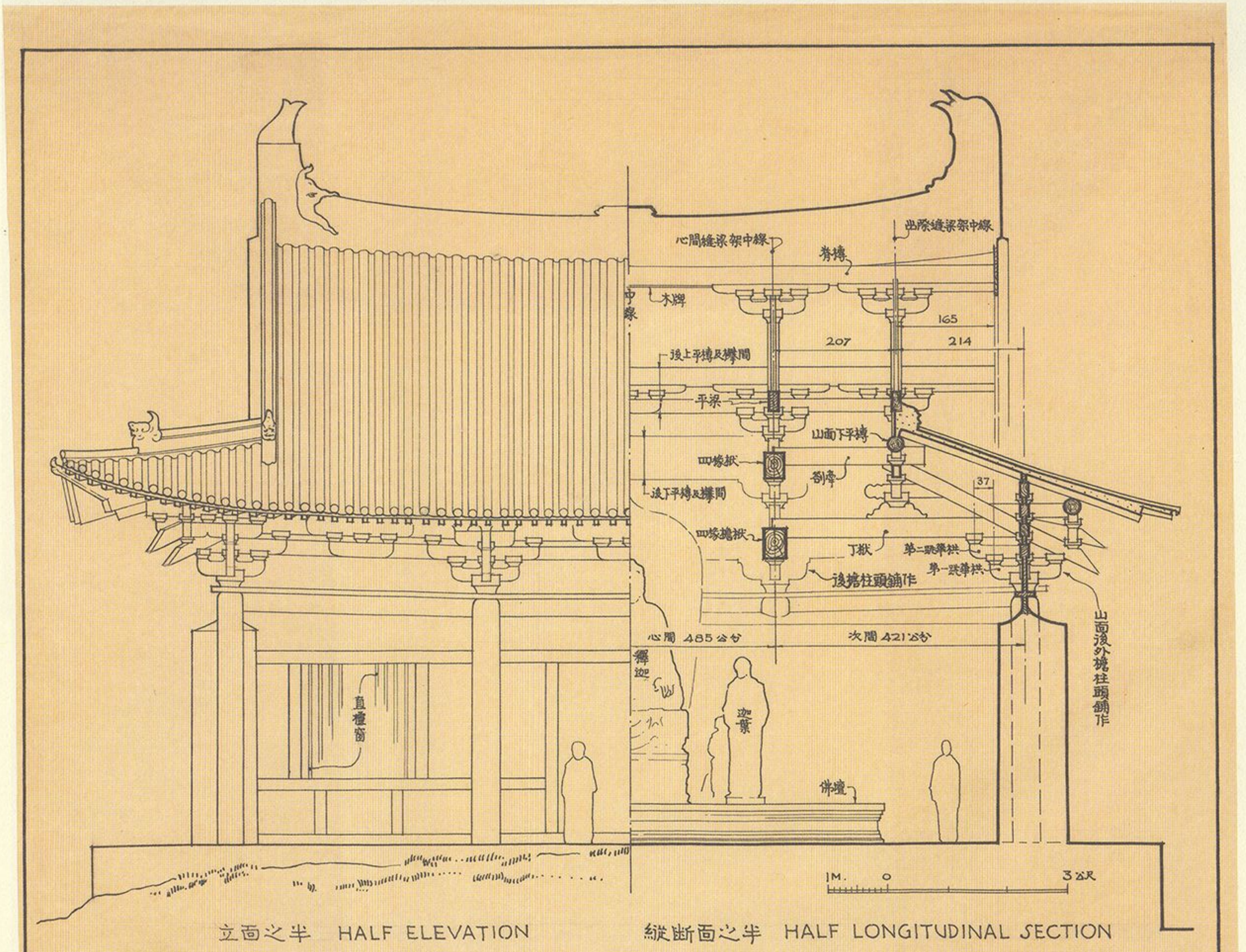
清故宮文淵閣
北平

清
乾隆四十年
CH'ING
1776

IMPERIAL LIBRARY,
IMPERIAL PALACES, PEIPING.

歷代斗栱演變
EVOLUTION OF THE CHINESE "ORDER"

228



立面之半 HALF ELEVATION　　縱斷面之半 HALF LONGITUDINAL SECTION

山西榆次縣 永壽寺雨華宮 宋大中祥符元年建

YÜ-HUA KUNG · MAIN HALL OF YUNG-SHOU SSU

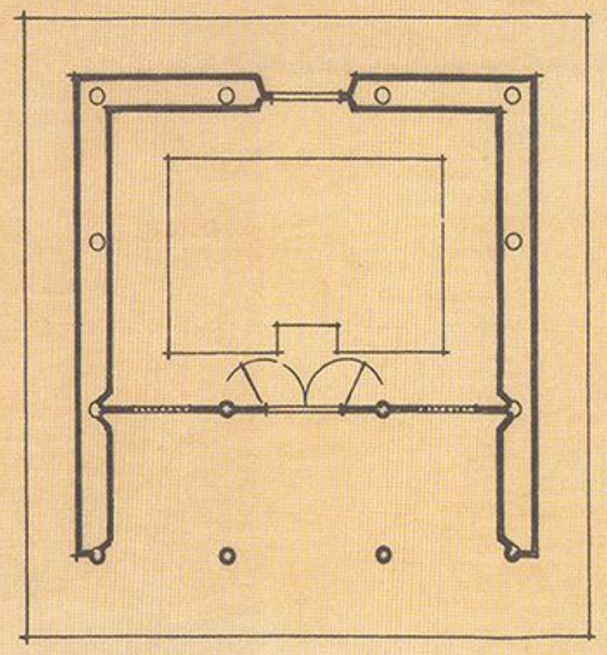
YÜ-TZ'U · SHANSI · 1008 A·D·

山西 榆次縣

永壽寺 雨花宮

宋大中祥符元年建

YÜ-HUA KUNG
YUNG-SHOU SSU
YÜ-TZ'Ŭ, SHANSI.
SUNG DYNASTY
1008 A.D.



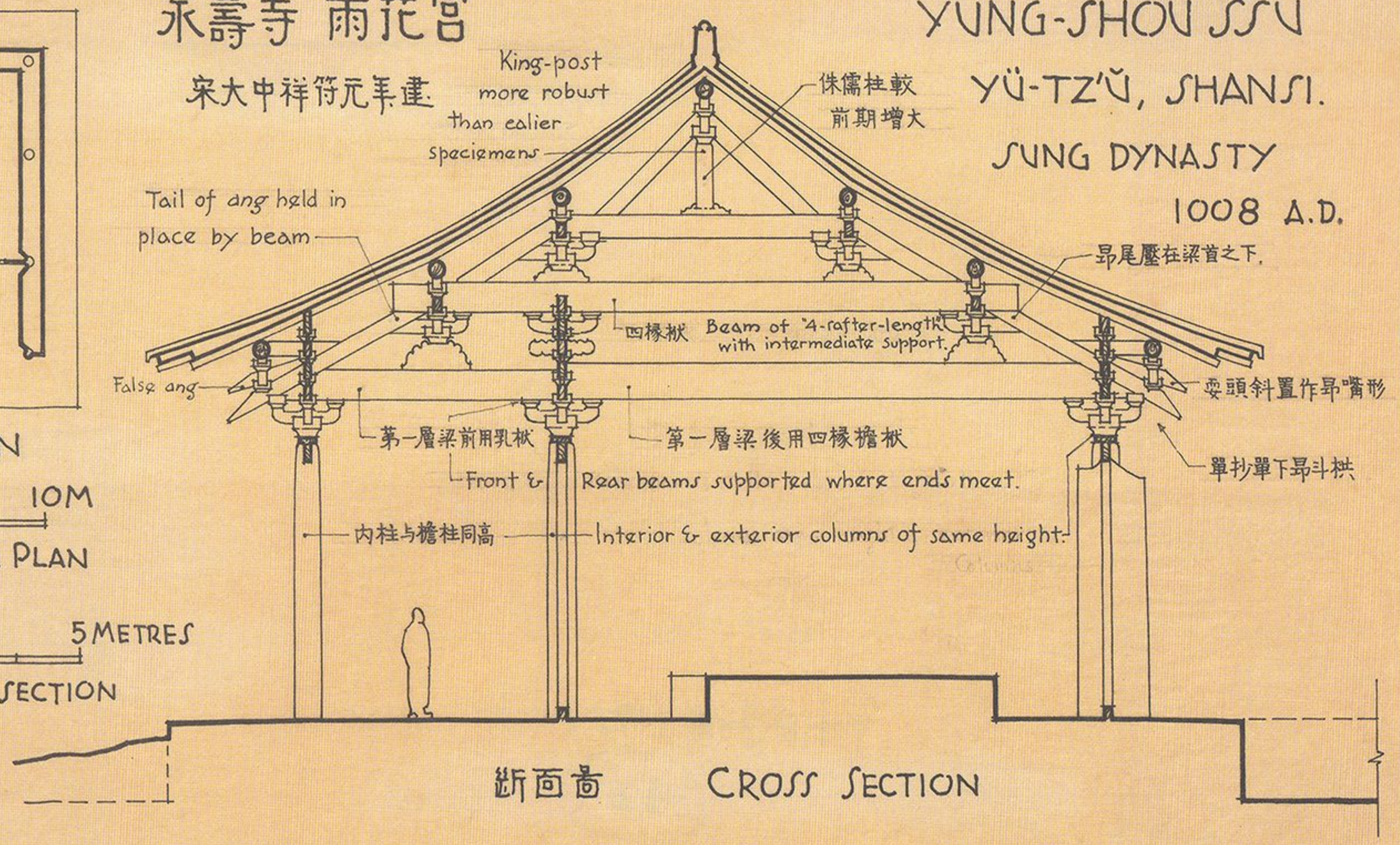
平面圖 PLAN

比尺 5 0 10M

平面縮尺 SCALE FOR PLAN

比尺 1 0 5 METRES

斷面縮尺 SCALE FOR SECTION



斷面圖 CROSS SECTION

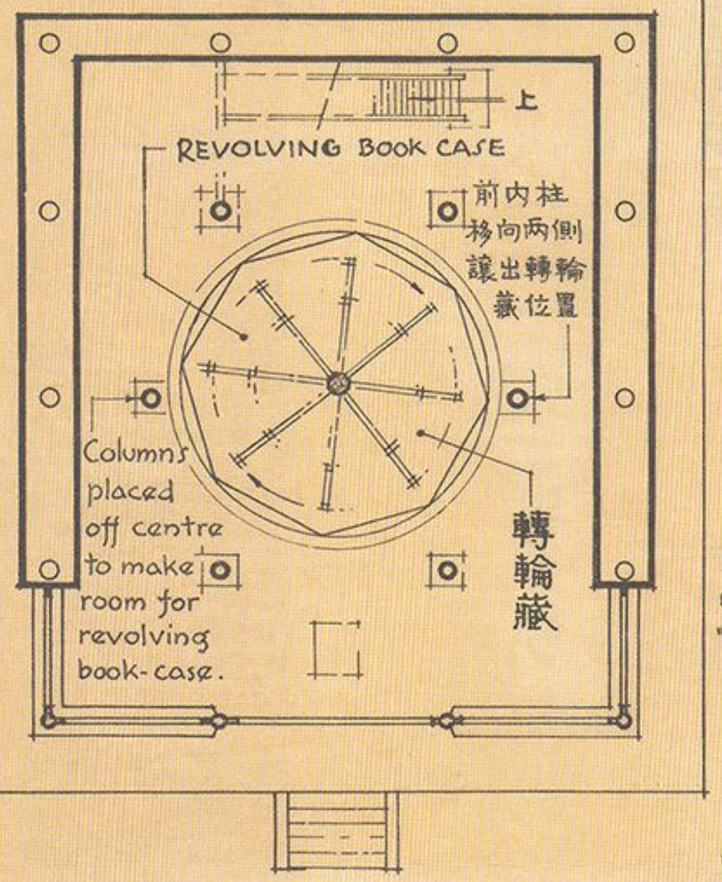
河北正定縣 龍興寺 轉輪藏殿 宋建

LIBRARY BUILDING, LUNG-HSING SSU, CHENG-TING, HOPEI

NORTH SUNG DYNASTY 960-1127.

平面及斷面畫中皆顯示特殊結構方法以適應安置轉輪藏之需要.

Both plan & section show departure from ordinary columnization & construction to accommodate housing of revolving book-case.



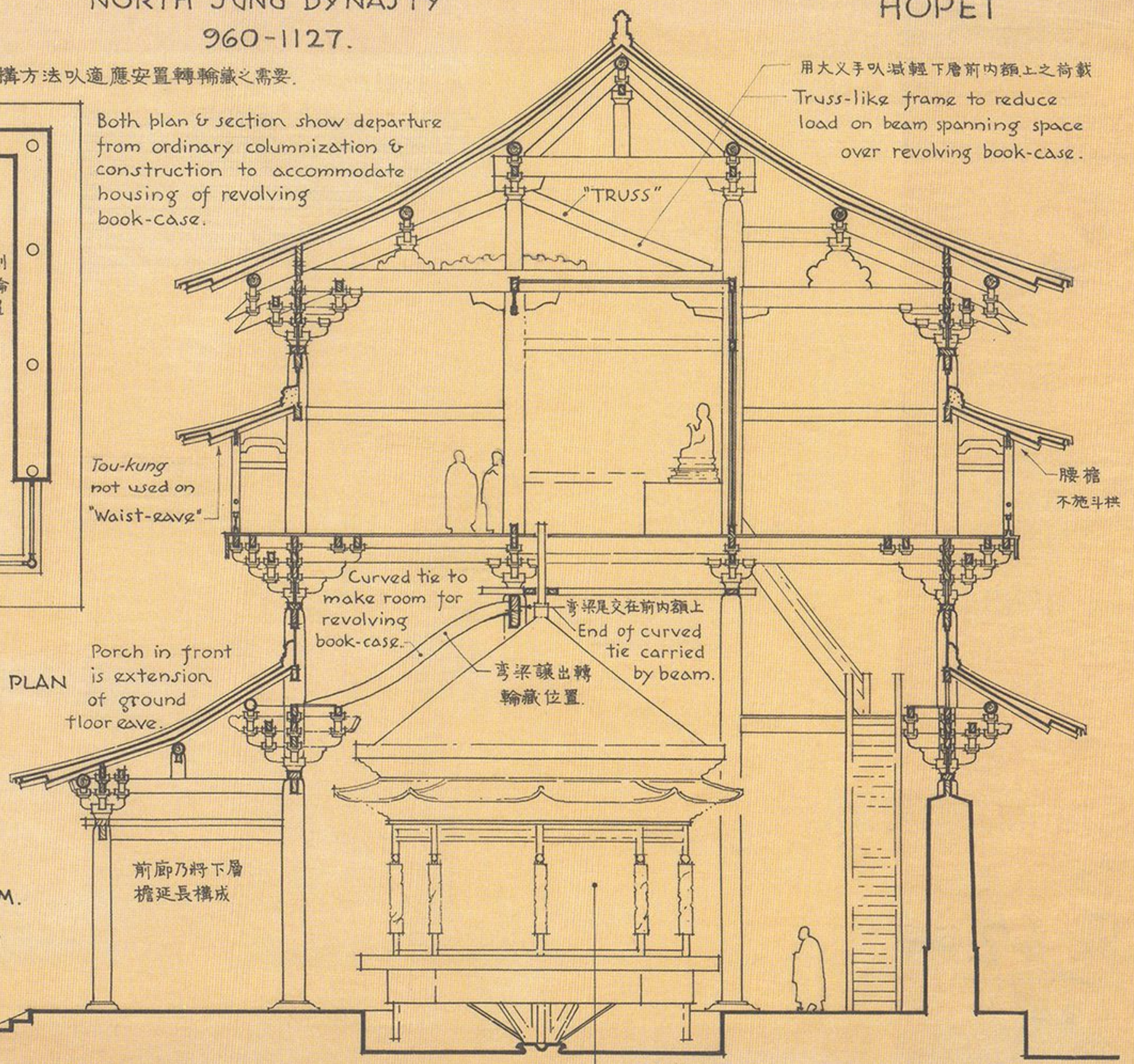
平面 GROUND FLOOR PLAN

公尺 0 5 10M.

平面縮尺 SCALE FOR PLAN

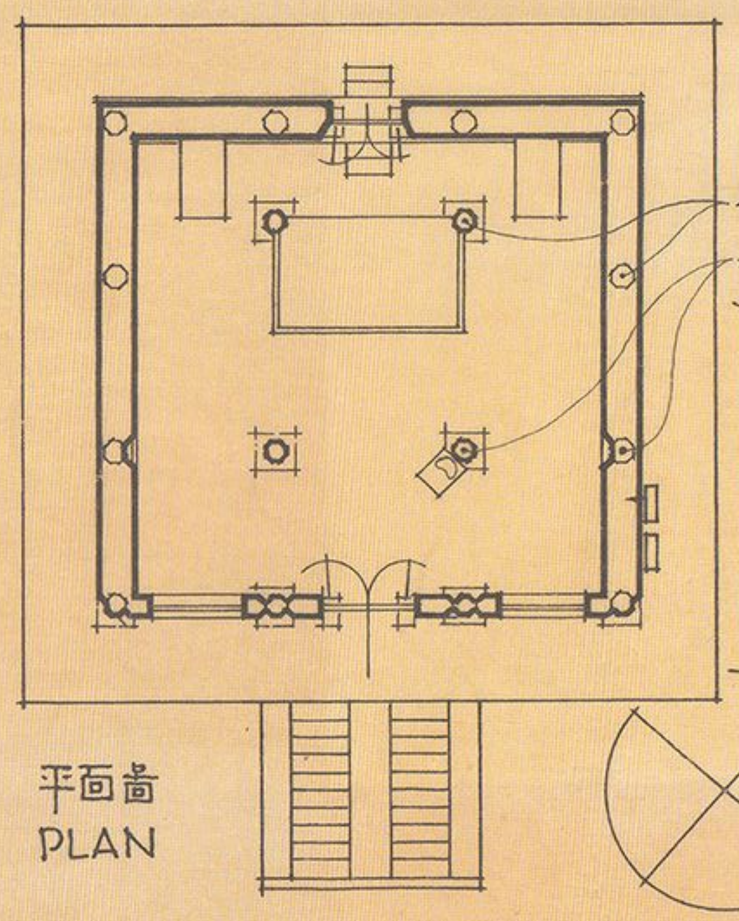
1 0 5M.

斷面縮尺 SCALE FOR SECTION



橫斷面 CROSS SECTION

河南登封縣少林寺初祖庵　CH'U-TSU AN, SHAO-LIN SSU, TENG-FENG, HONAN



平面畫
PLAN

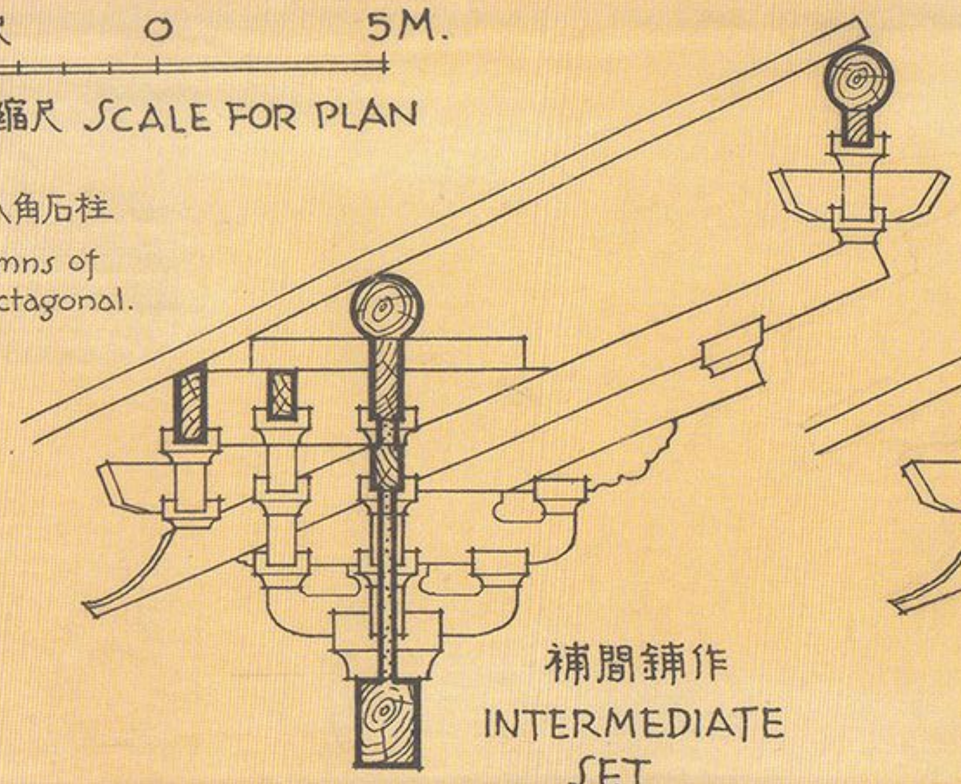
北

5公尺　0　5M.
平面縮尺 SCALE FOR PLAN

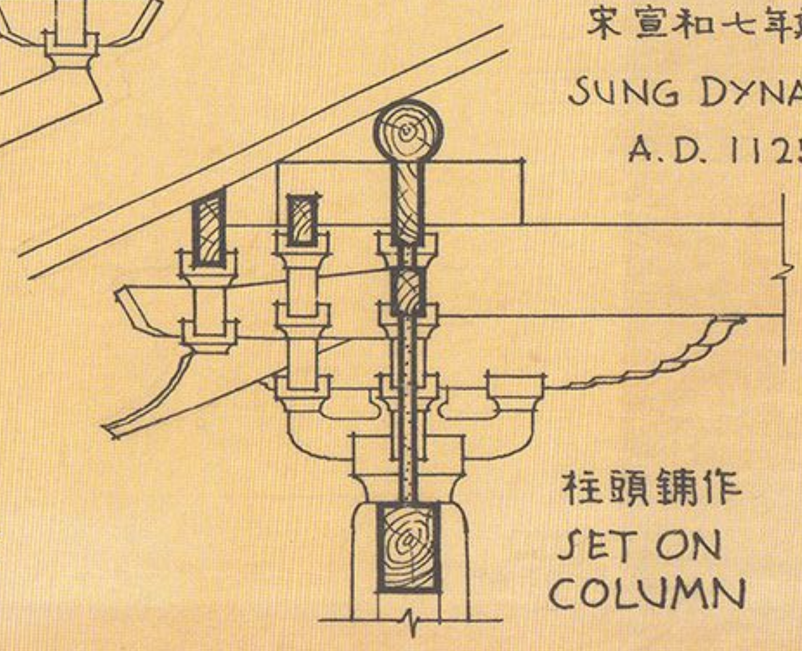
全部用八角石柱
All columns of stone, octagonal.

1公尺　0　1M
詳畫縮尺 SCALE FOR DETAIL

宋宣和七年建
SUNG DYNASTY
A.D. 1125



補間鋪作
INTERMEDIATE SET



柱頭鋪作
SET ON COLUMN

劉敦楨測繪
MEASURED BY LIU T.T.

外檐鋪作　DETAIL, TOU-KUNG SETS

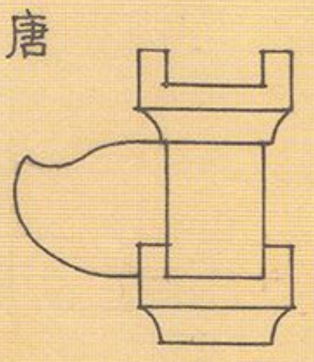
71-C

歷代耍頭(梁頭)演變圖

EVOLUTION OF THE SHUA-T'OU (HEAD OF THE BEAM)

公分 10 0 50 100 cm.

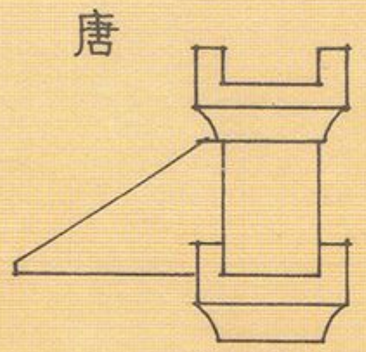
唐



857

佛光寺正殿

唐

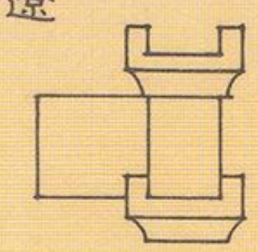


857

佛光寺正殿

MAIN HALL, FO-KUANG SSU

遼

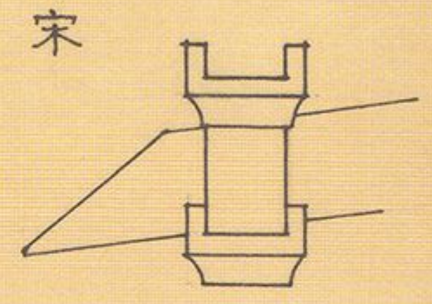


984

獨樂寺觀音閣

TU-LÊ SSU

宋

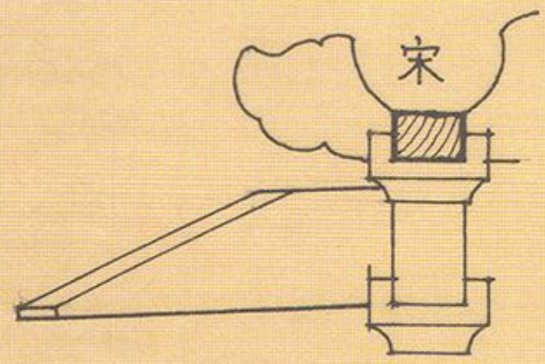


1008

永壽寺 雨華宮

YUNG-SHOU SSU

宋

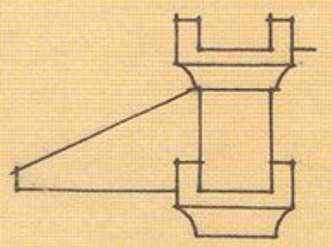


CA. 1030

佛光寺 文殊殿

WEN-SHU TIEN, FO-KUANG SSU

遼

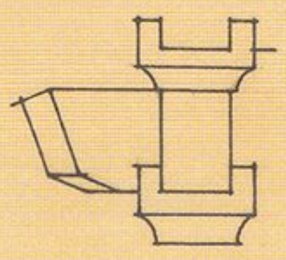


1038

薄伽教藏

LIBRARY
HUA-YEN SSU

宋

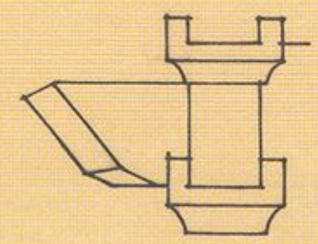


1100

營造法式

YING-TSAO
FA-SHIH

宋

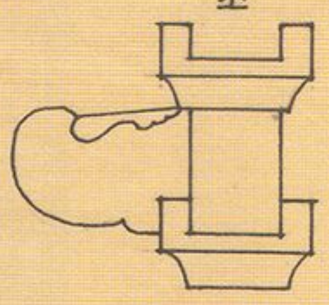


1125

初祖庵

CH'U-TSU AN

金

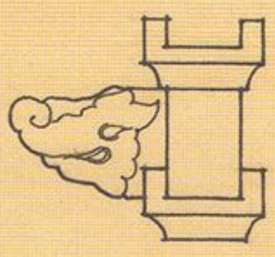


CA. 1130

華嚴寺大殿

MAIN HALL,
HUA-YEN SSU

金

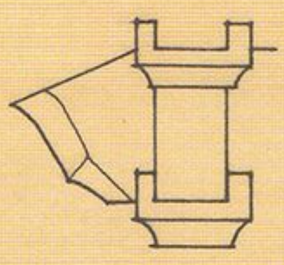


1118-43

善化寺三聖殿

FRONT HALL
SHAN-HUA SSU

金

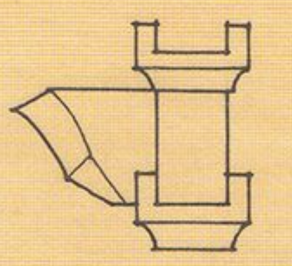


1118-43

善化寺三聖殿

FRONT HALL
SHAN-HUA SSU

金

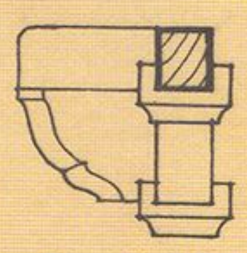


1118-43

善化寺山門

MAIN GATE
SHAN-HUA SSU

元

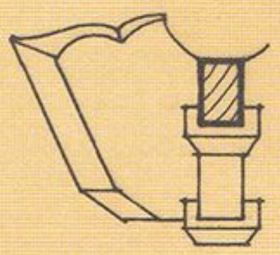


CA. 1260-80

陽和樓

YANG-HO LOU

明

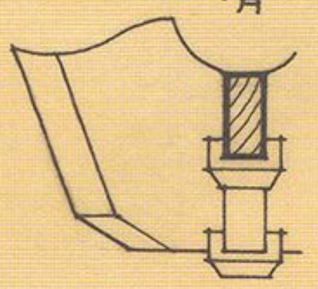


1504

奎文閣

LIBRARY
CONFUCIUS' TEMPLE

清

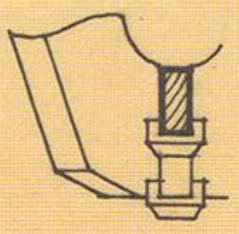


1733

工程做法

KUNG-CH'ENG
TSO-FA CHÊ-LI

清



1776

文淵閣

WEN-YUAN KÊ

唐

佛光寺

857

FO-KUANG SSU

遼

←不施普拍枋→

觀音閣

984

TU-LÊ SSU

宋

普拍枋出頭相交

闌額不出頭

雨華宮

1008

YUNG-SHOU SSU

遼

普拍枋闌額均出頭

薄伽教藏

1038

HUA-YEN SSU

宋

初祖庵

1125

CH'U-TSU AN

金

金南宋以後少不用普拍枋者

善化寺山門

1118-43

MAIN GATE
SHAN-HUA SSU

金

善化寺三聖殿

1118-43

FRONT HALL
SHAN-HUA SSU

元

陽和樓

CA. 1260-80

YANG-HO LOU

明

明清以後斗栱縮小，故普拍枋亦漸小。

社稷壇

1412

SHÊ-CHI T'AN

明

智化寺

1443

CHIH-HUA SSU

明

補間鋪作朵數加多，故闌額加大。

奎文閣

1504

K'UEI-WEN KÊ

清

工程做法則例

1733

KUN-CH'ENG
TSO-FA CHÊ-LI

公分 50 0 100 cm.

清

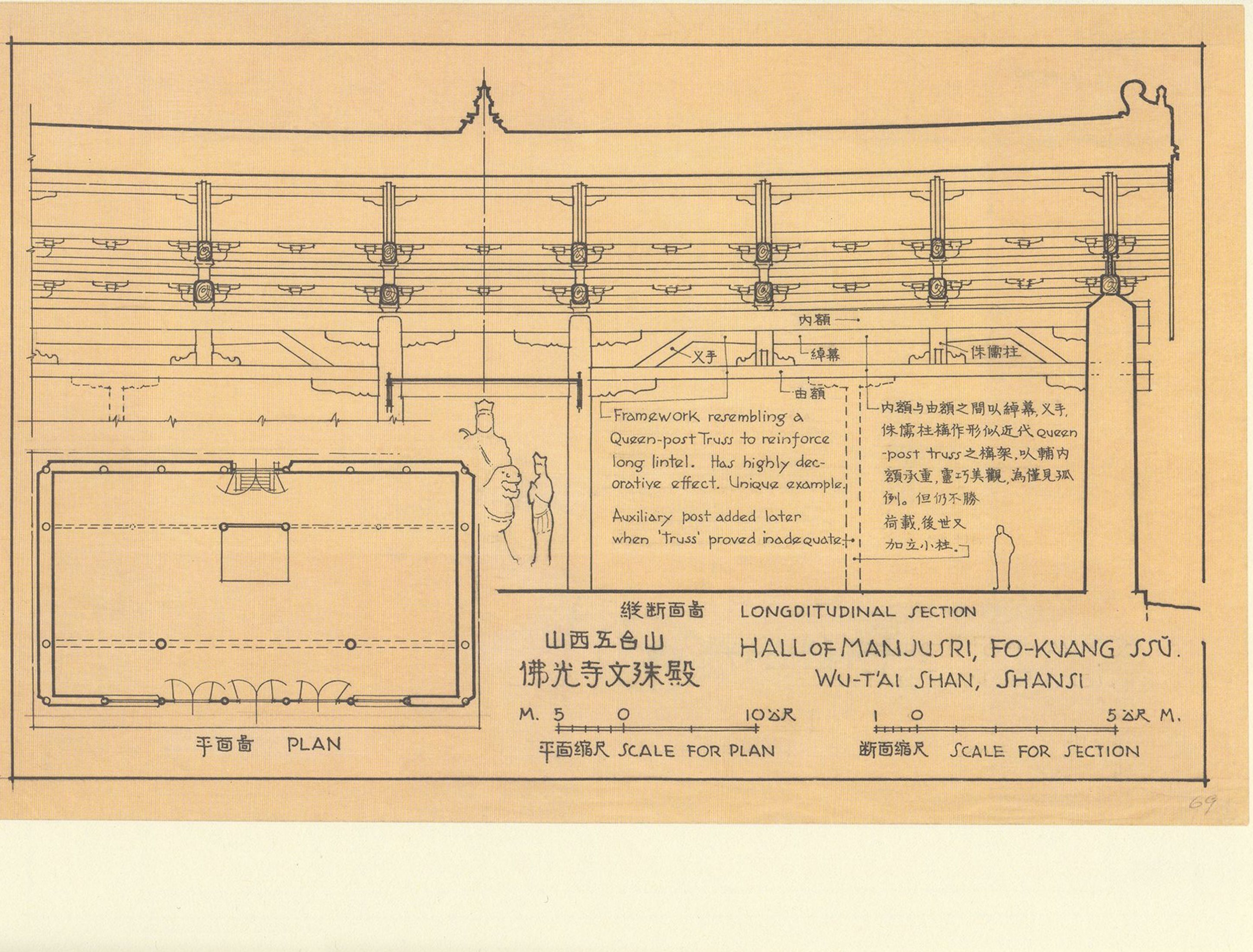
文淵閣

1776

WEN-YUAN KÊ

歷代闌額普拍枋演變圖

EVOLUTION OF THE LAN-ÊH AND P'U-P'AI-FANG (ARCHITRAVE AND PLATE)

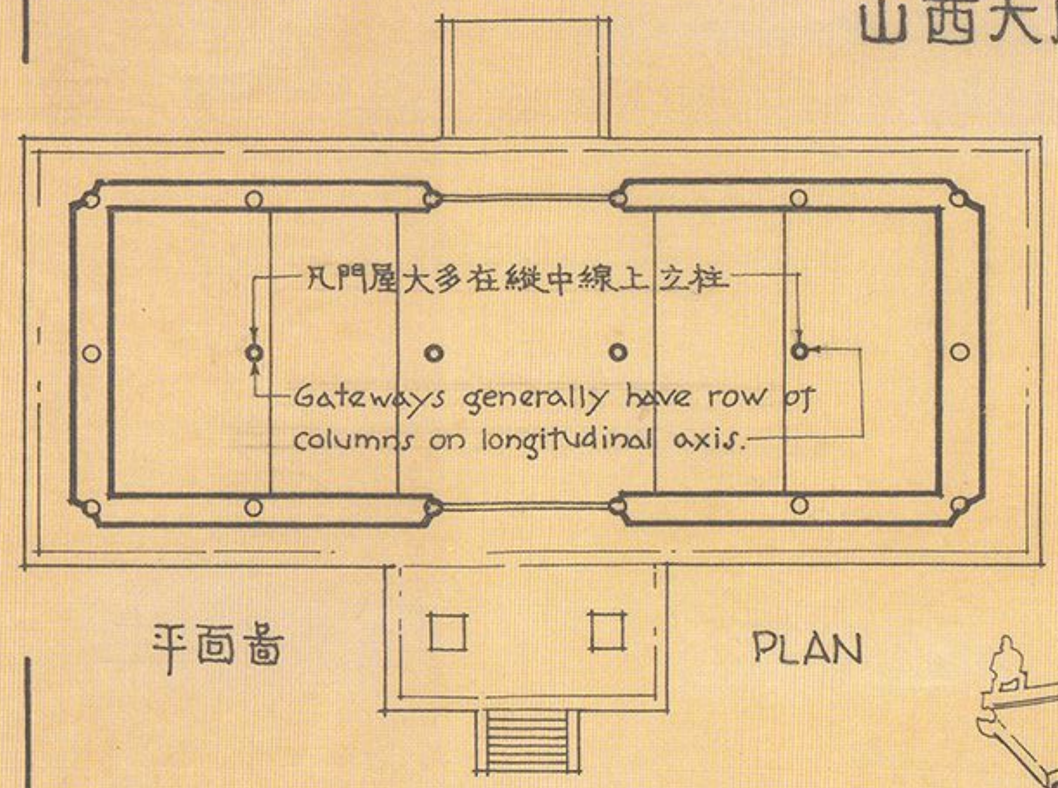

內額
義手
綽幕
侏儒柱
由額
Framework resembling a Queen-post Truss to reinforce long lintel. Has highly decorative effect. Unique example.
Auxiliary post added later when 'truss' proved inadequate.
內額与由額之間以綽幕,义手,侏儒柱構作形似近代 Queen-post truss之構架,以輔內額承重,靈巧美觀,為僅見孤例。但仍不勝荷載,後世又加立小柱。
縱斷面圖 LONGDITUDINAL SECTION
平面圖 PLAN
山西五台山
佛光寺文殊殿
HALL OF MANJUSRI, FO-KUANG SSŬ.
WU-T'AI SHAN, SHANSI
M. 5 0 10公尺
平面縮尺 SCALE FOR PLAN
1 0 5公尺 M.
斷面縮尺 SCALE FOR SECTION

山西大同縣 善化寺山門

建間年三統皇至年八會天金

ENTRANCE GATEWAY
SHAN-HUA SSU
TA-T'UNG, SHANSI

CHIN DYNASTY
BETWEEN 1130 & 1143.



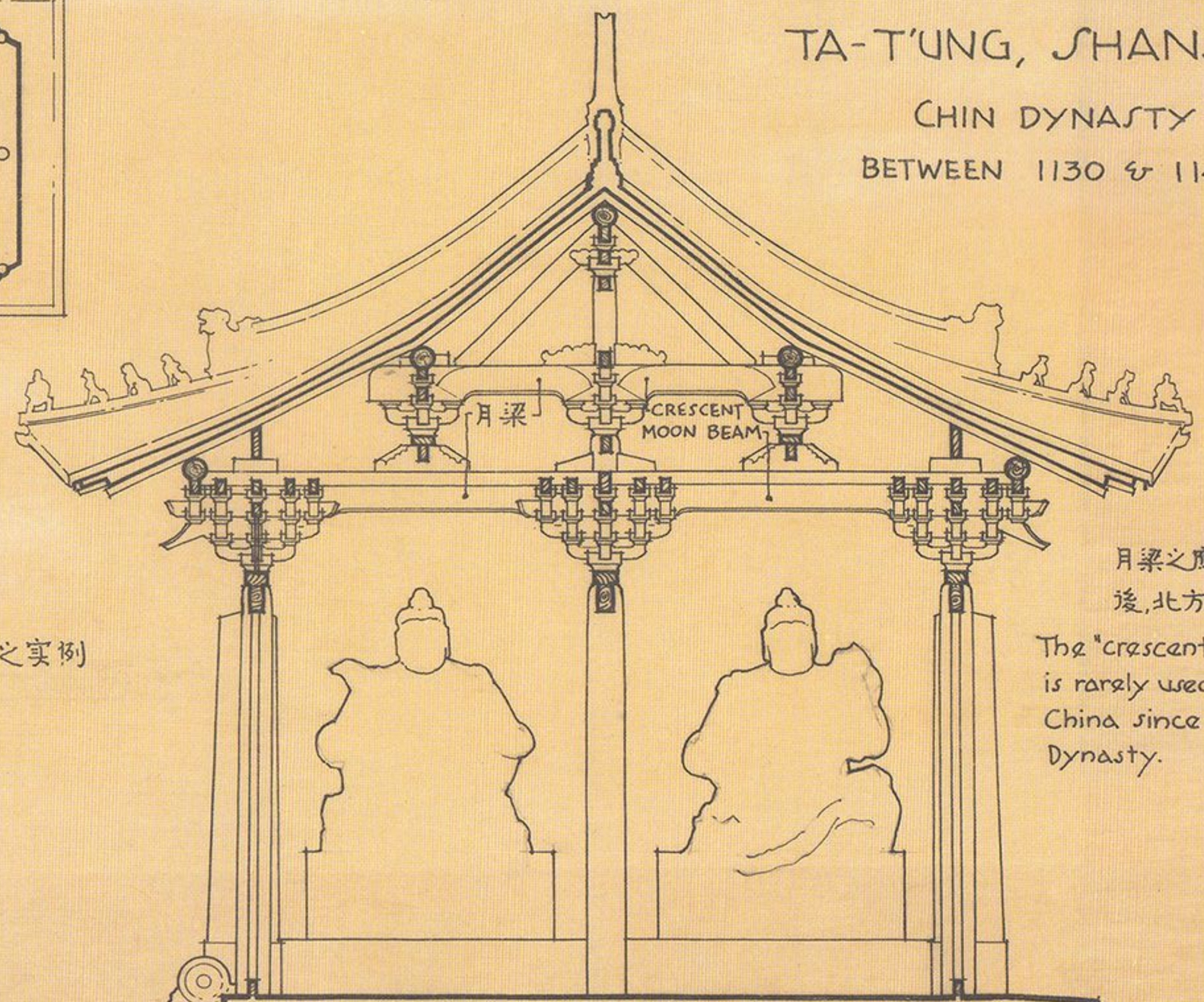
平面圖 PLAN

公尺 5 0 10M.

平面縮尺 SCALE FOR PLAN

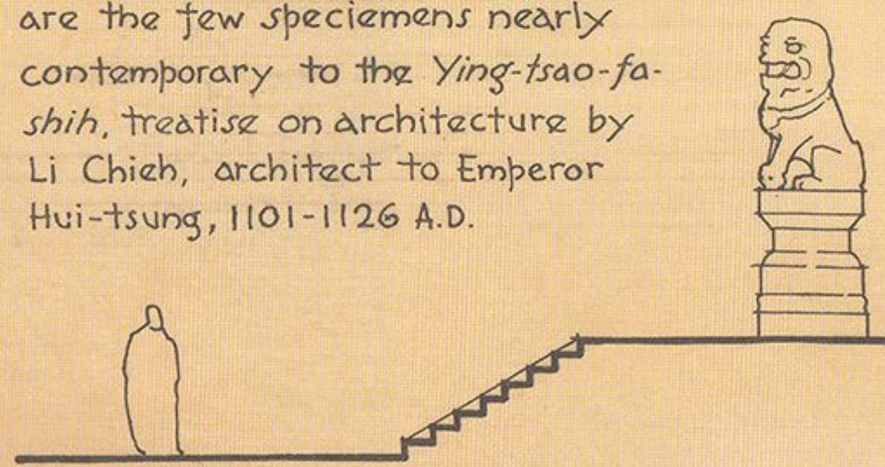
善化寺金代殿堂為与宋李誡營造法式約略同時之实例

The Chin buildings of Shan-hua Ssŭ are the few speciemens nearly contemporary to the *Ying-tsao-fa-shih*, treatise on architecture by Li Chieh, architect to Emperor Hui-tsung, 1101-1126 A.D.



月梁之應用,自宋以後,北方已極罕見。

The "crescent-moon beam" is rarely used in North China since the Sung Dynasty.



斷面圖 CROSS SECTION

斷面縮尺 1公尺 0 5M. SCALE FOR SECTION

STELE PAVILION, TEMPLE OF CONFUCIUS

CHÜ-FOU, SHAN-TUNG, CHIN DYNASTY

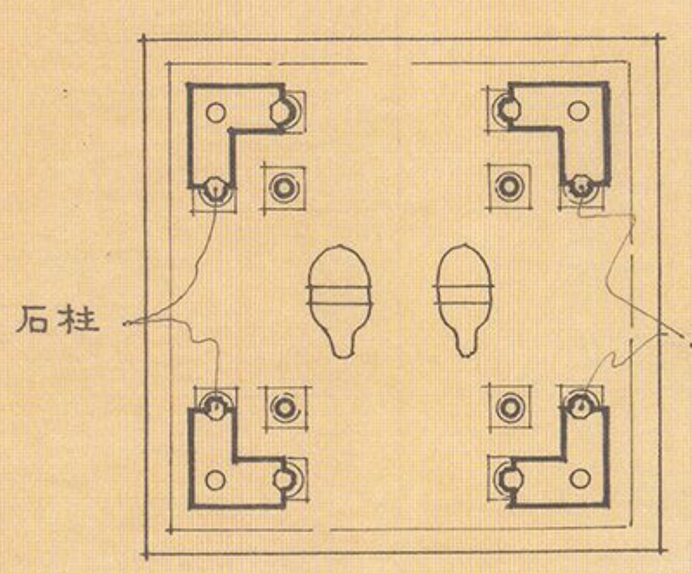
1196 A.D.

山東曲阜縣 孔廟碑亭

金明昌六年建？

為聖廟現存最古建築，後世重脩，頗有更改，尤以上層屋頂梁架為甚。柱額斗拱則大部仍保持原狀。

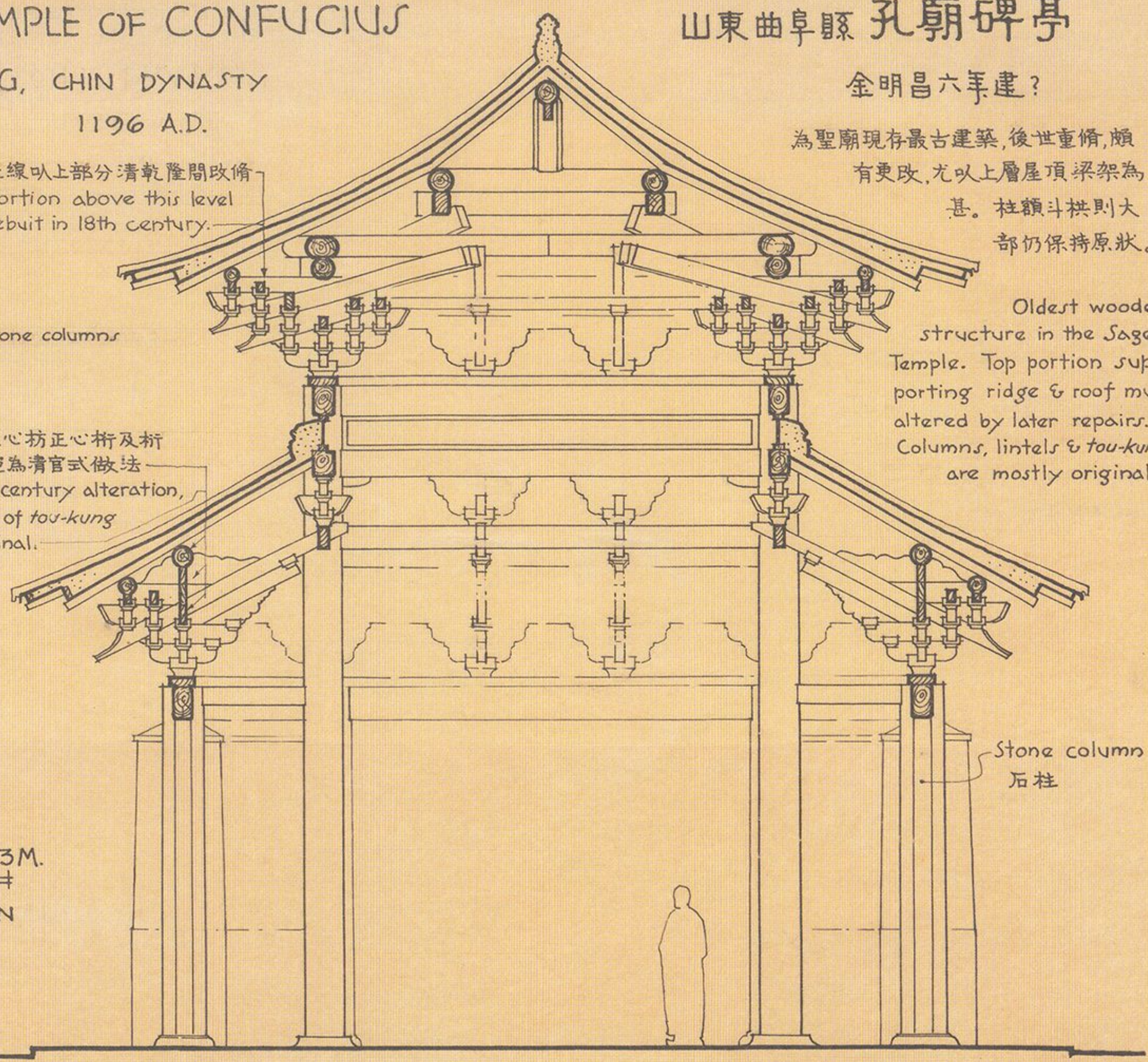
Oldest wooden structure in the Sage's Temple. Top portion supporting ridge & roof much altered by later repairs. Columns, lintels & *tou-kungs* are mostly original.



平面圖 PLAN

此線以上部分清乾隆間改脩
Portion above this level rebuit in 18th century.

正心枋正心桁及桁椀為清官式做法
18th century alteration, rest of *tou-kung* original.



斷面圖 CROSS SECTION

公尺 5 0 5M.

平面縮尺 SCALE FOR PLAN

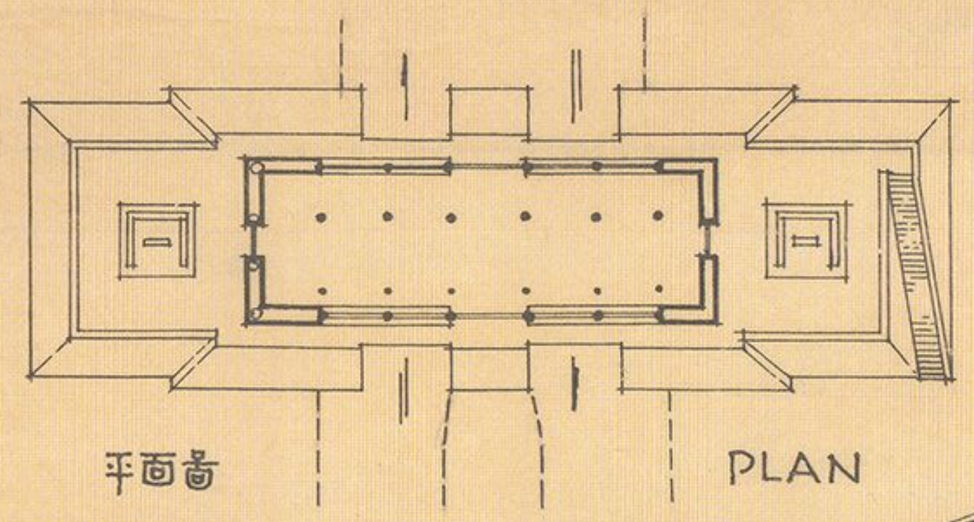
公尺 1 0 3M.

斷面縮尺 SCALE FOR SECTION

YANG-HO LOU, CHENG-TING, HOPEI.

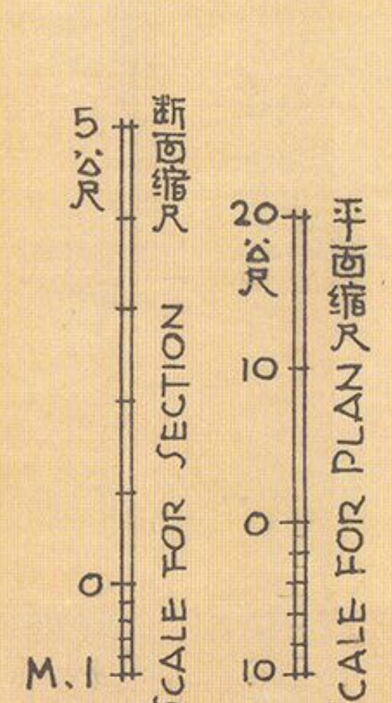
LATE CHIN OR EARLY YUAN

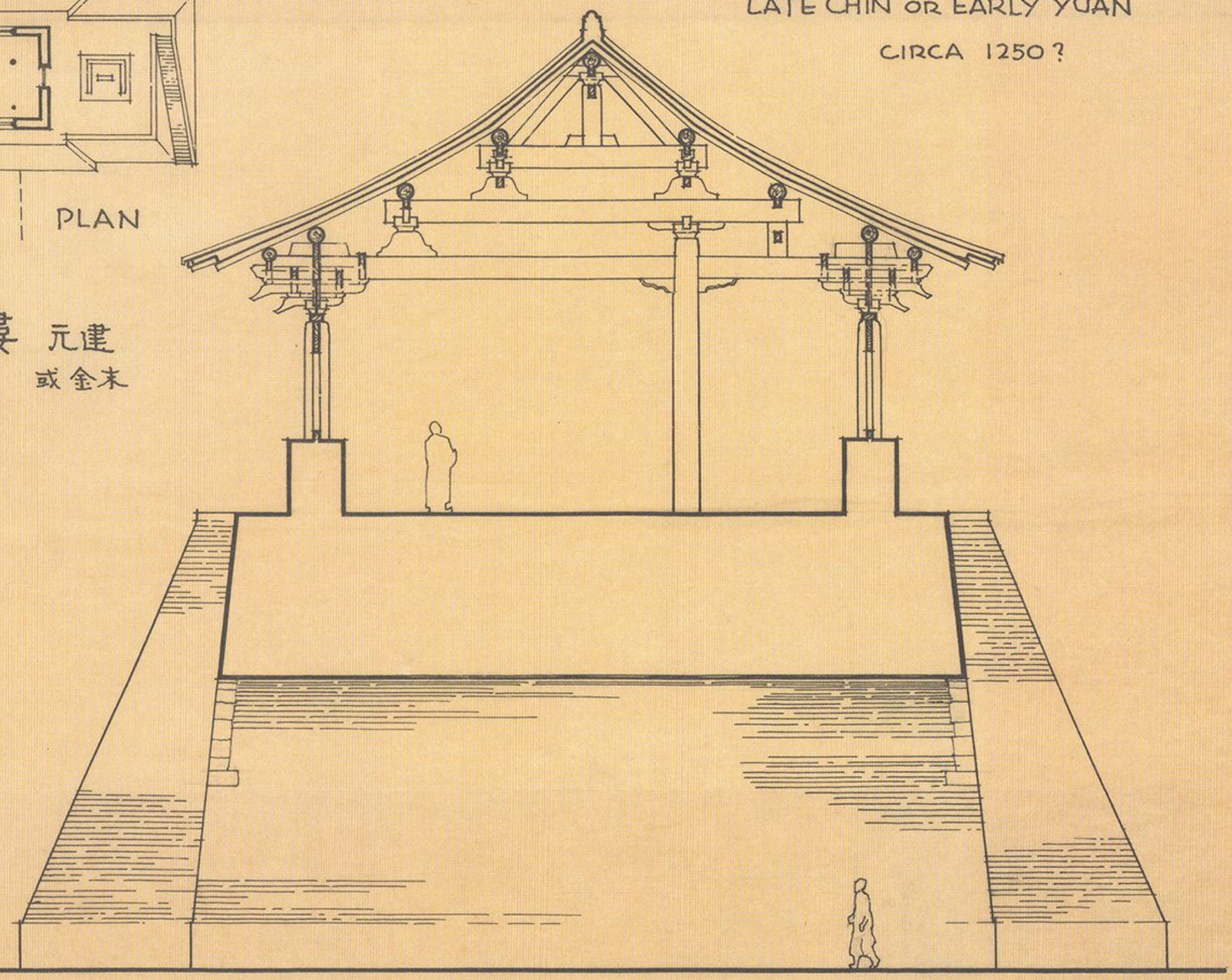
CIRCA 1250?



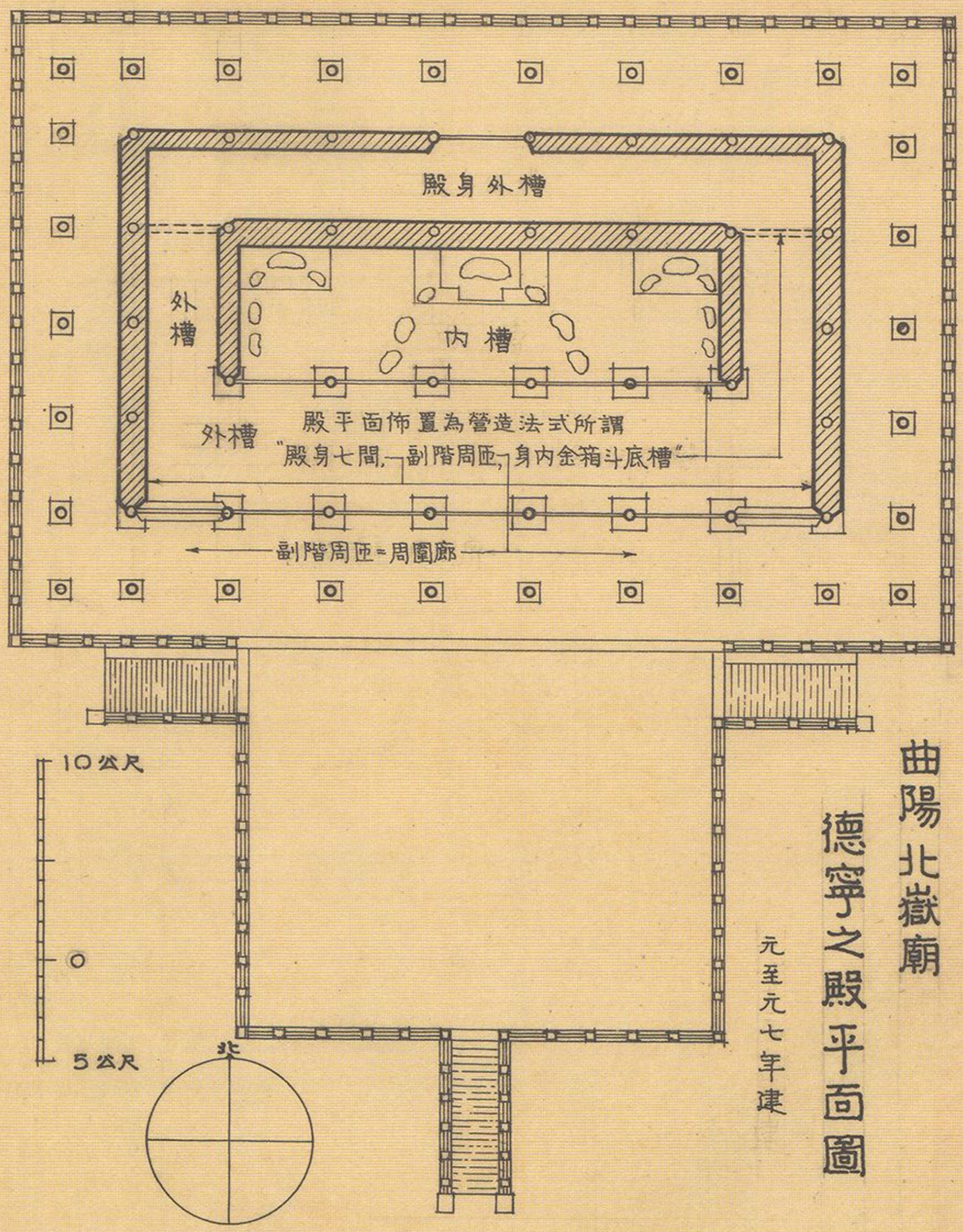
平面圖 PLAN

河北正定縣 陽和樓 元建或金末





斷面圖 SECTION



曲陽北嶽廟

德寧之殿平面圖

元至元七年建

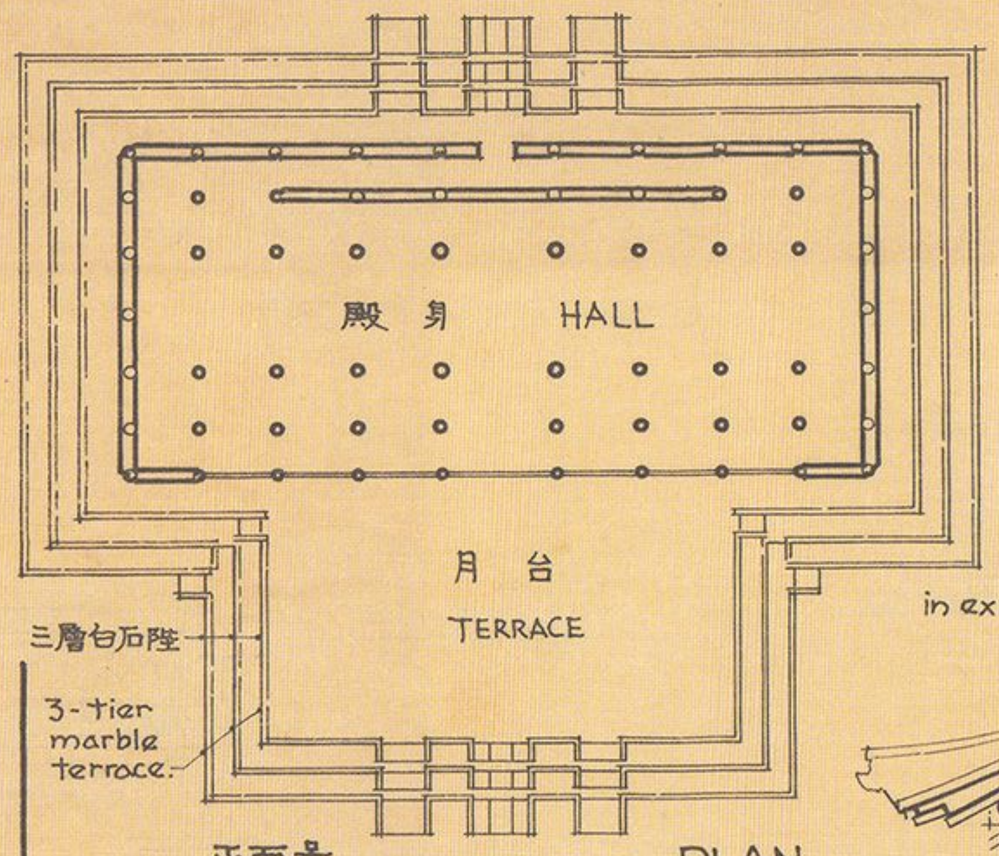
PLAN OF MAIN HALL · PEI-YUEH MIAO

CH'Ü-YANG · HOPEI · 1270

河北昌平縣 明長陵 稜恩殿 明永樂間建

SACRIFICIAL HALL
TOMB OF EMPEROR YUNG-LO
THE MING TOMBS
CH'ANG-P'ING, HOPEI.

BUILT DURING THE REIGN OF YUNG-LO, 1403-24.



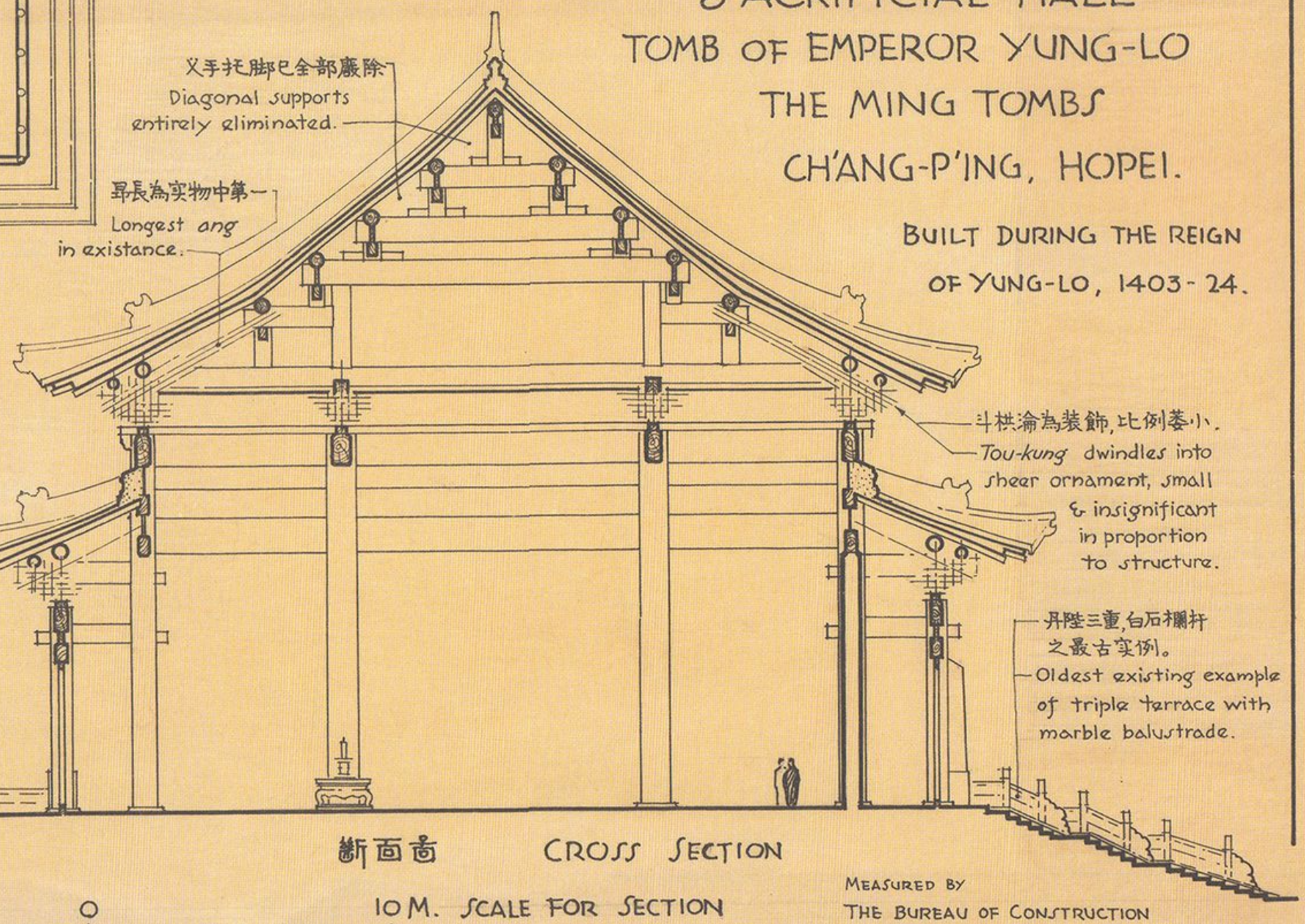
平面圖 PLAN

公尺 10 0 40 M.

平面縮尺 SCALE FOR PLAN

殿為國內最大木構之一,面積僅略遜於北平故宮太和殿。

The Hall is one of the largest wooden structures in China. Its superficies is surpassed by the Tai-ho Tien in the Imperial Palaces, Peiping. by a narrow margin.



斷面圖 CROSS SECTION

斷面縮尺 5公尺 0 10 M. SCALE FOR SECTION

MEASURED BY
THE BUREAU OF CONSTRUCTION
MUNICIPAL GOVERNMENT OF PEIPING

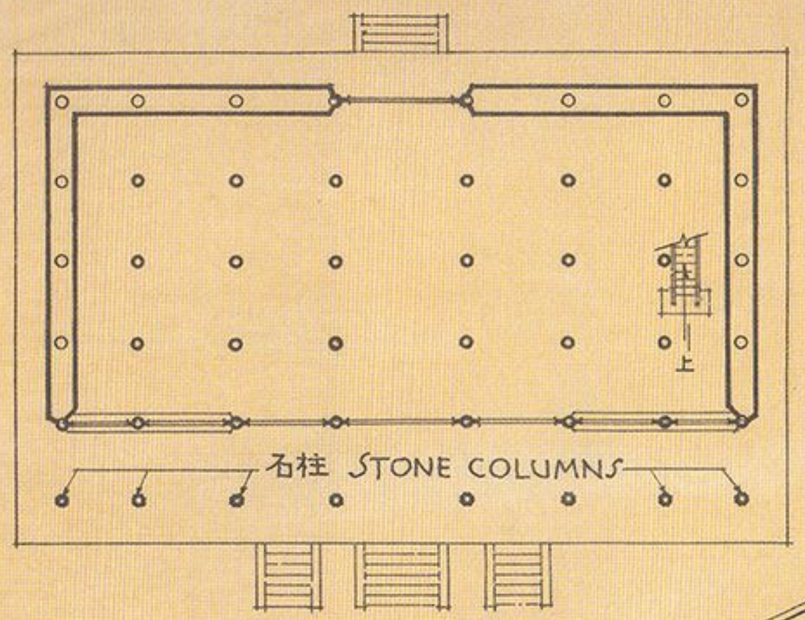
北平市政府工務局測繪

LIBRARY BUILDING
TEMPLE OF CONFUCIUS
CH'Ü-FOU, SHANTUNG.

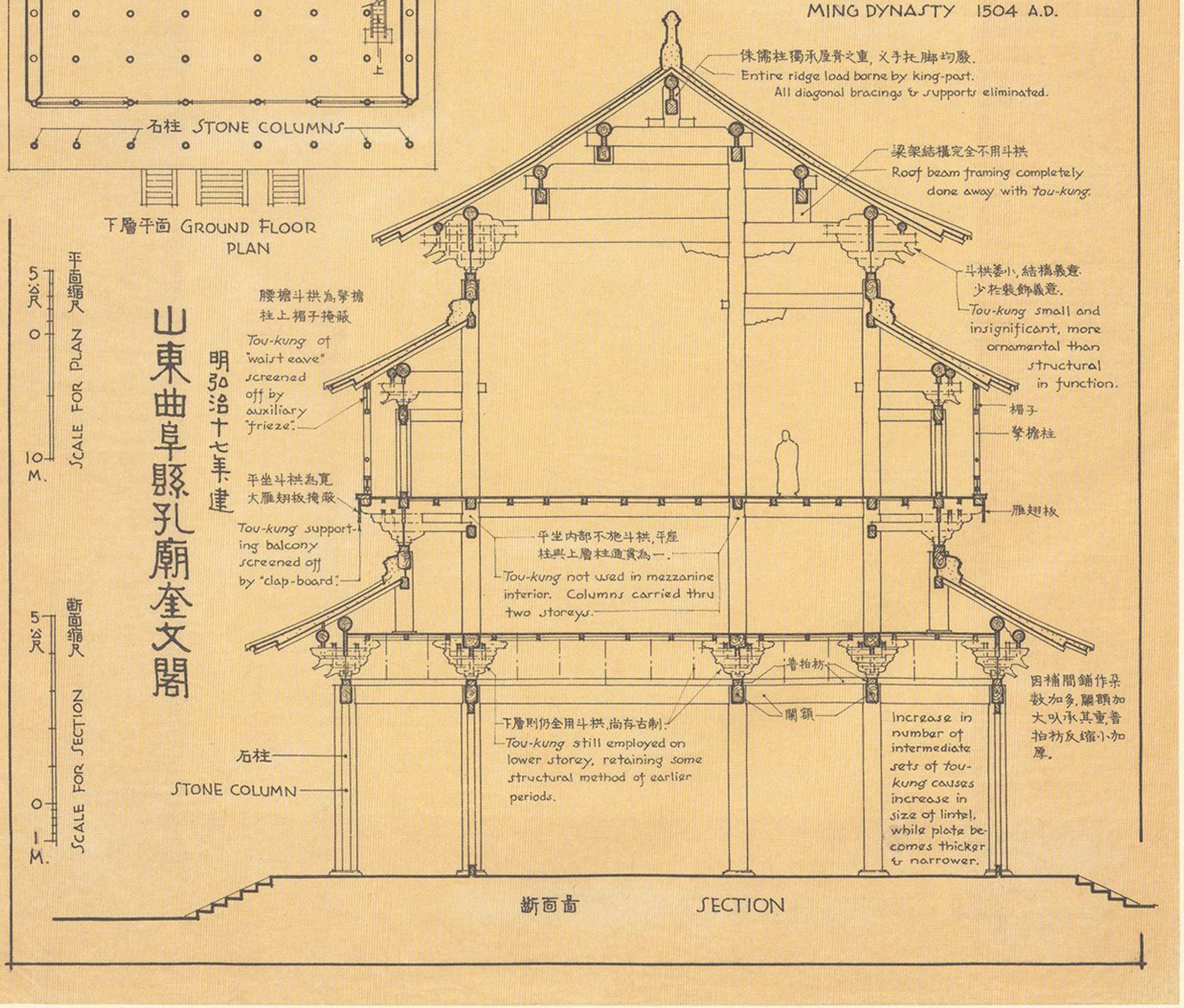
MING DYNASTY 1504 A.D.

山東曲阜縣孔廟奎文閣

明弘治十七年建



下層平面 GROUND FLOOR PLAN



SCALE FOR PLAN 平面縮尺
5公尺 0 10 M.

SCALE FOR SECTION 斷面縮尺
5公尺 0 1 M.

山東曲阜縣

孔廟大成殿

清雍正八年建

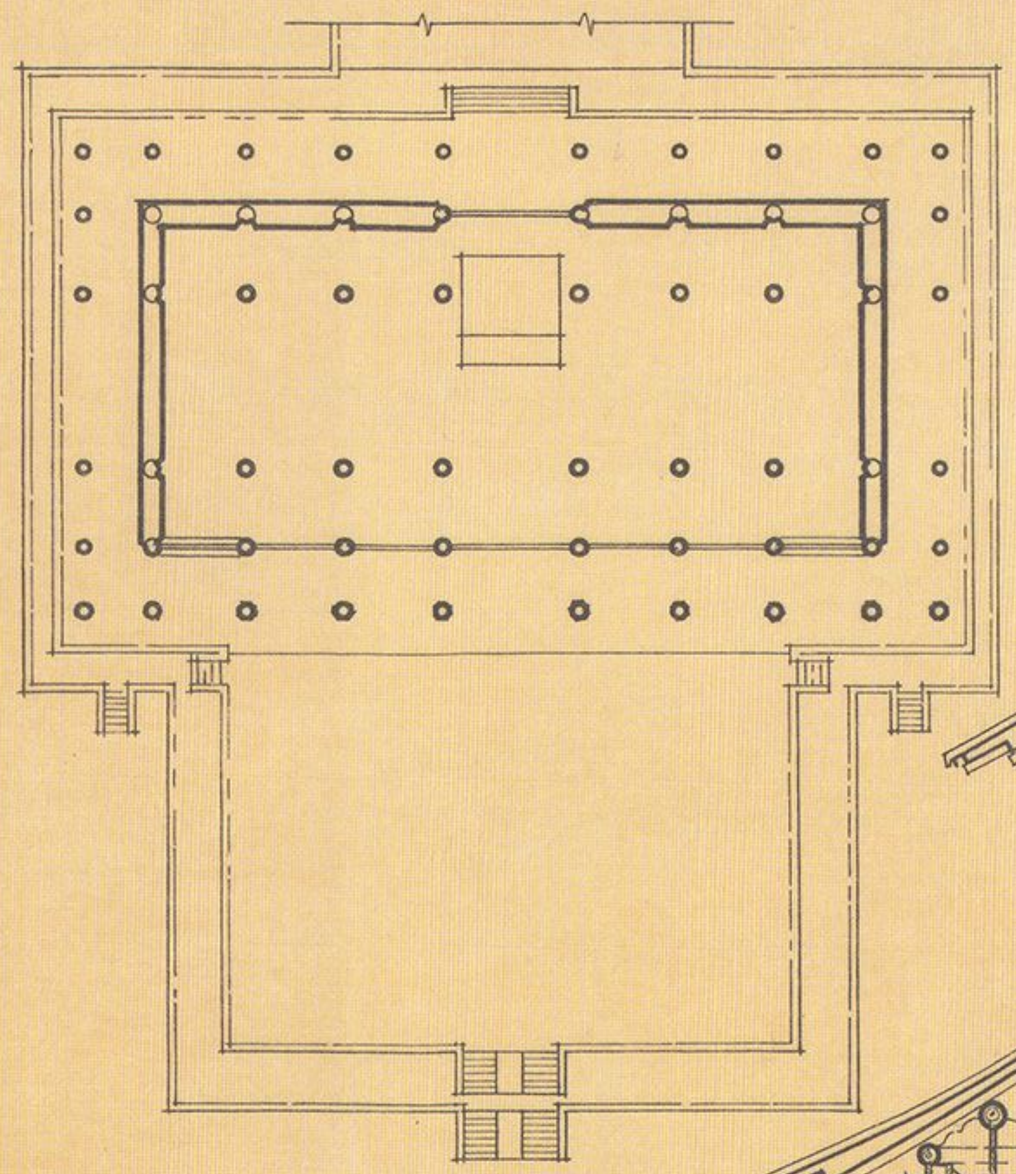
TA-CH'ENG TIEN
MAIN HALL OF THE TEMPLE OF CONFUCIUS
CH'Ü-FOU, SHANTUNG
CH'ING DYNASTY, 1730.

年代与工程做法則例最近，做法則与則例差別甚遠。舉高特甚，折甚微，屋面線生硬，缺乏圜和之感。

Nearest in date to *Kung-ch'eng-tso-fa-che-li*, but wide departure from its rules. High pitch & slight bent give roof-line appearance of clumsy rigidity.

單抄三昂斗拱
清代官式所無。

Tou-kung of 1-*kung* & 3-*angs* not in accordance with Ch'ing rules.



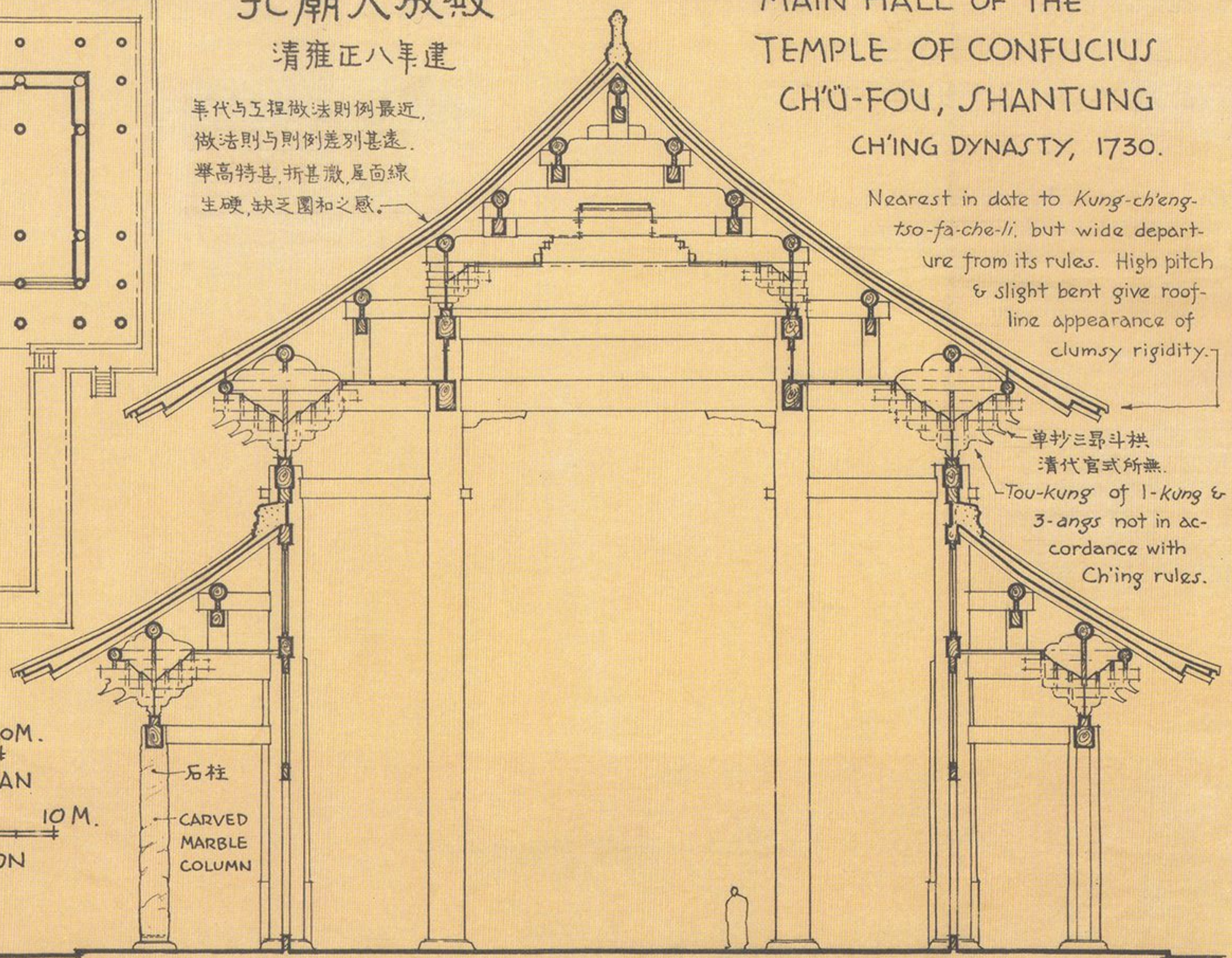
平面圖 PLAN

公尺 5 0 10 20M.

平面縮尺 SCALE FOR PLAN

公尺 1 0 10 M.

斷面縮尺 SCALE FOR SECTION

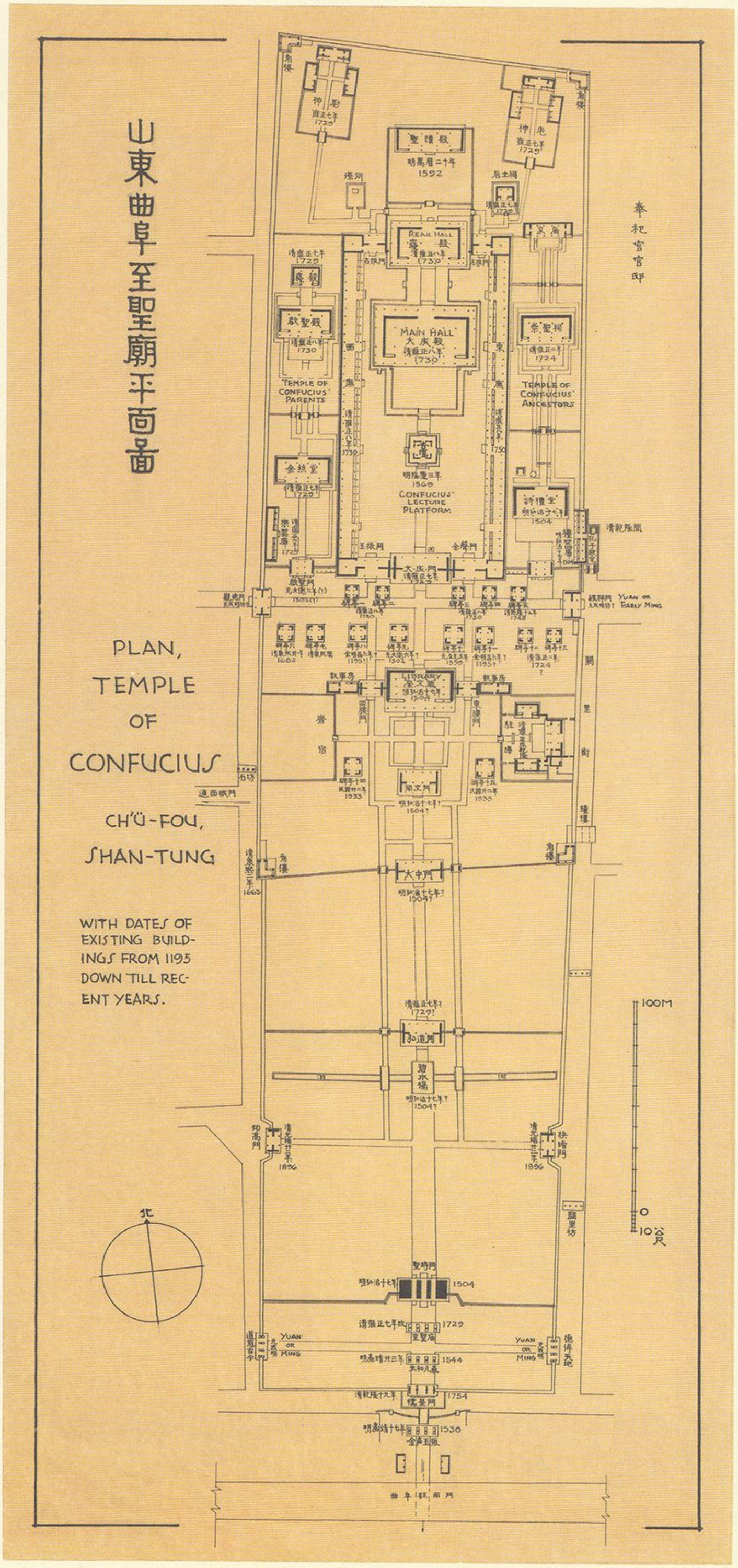


斷面圖 CROSS SECTION

山東曲阜至聖廟平面圖

PLAN, TEMPLE OF CONFUCIUS CH'Ü-FOU, SHAN-TUNG

WITH DATES OF EXISTING BUILDINGS FROM 1195 DOWN TILL RECENT YEARS.



1957

北平市清故宮三殿總平面圖

明末及清建

IMPERIAL PALACES PEIPING

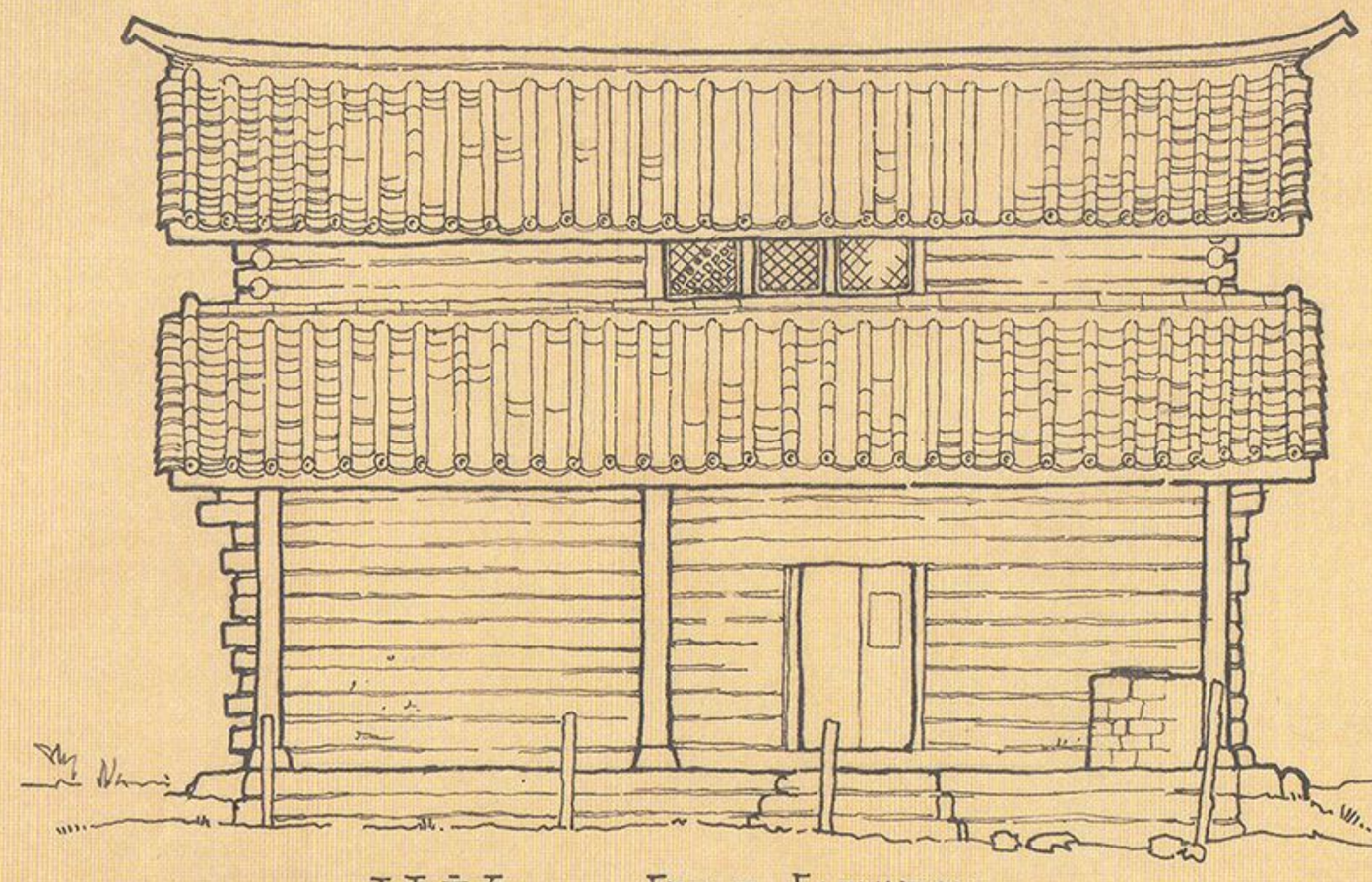
THE "THREE GREAT HALLS" AND IMMEDIATE SURROUNDING BUILDNGS

LATE MING & CHING DYNASTIES

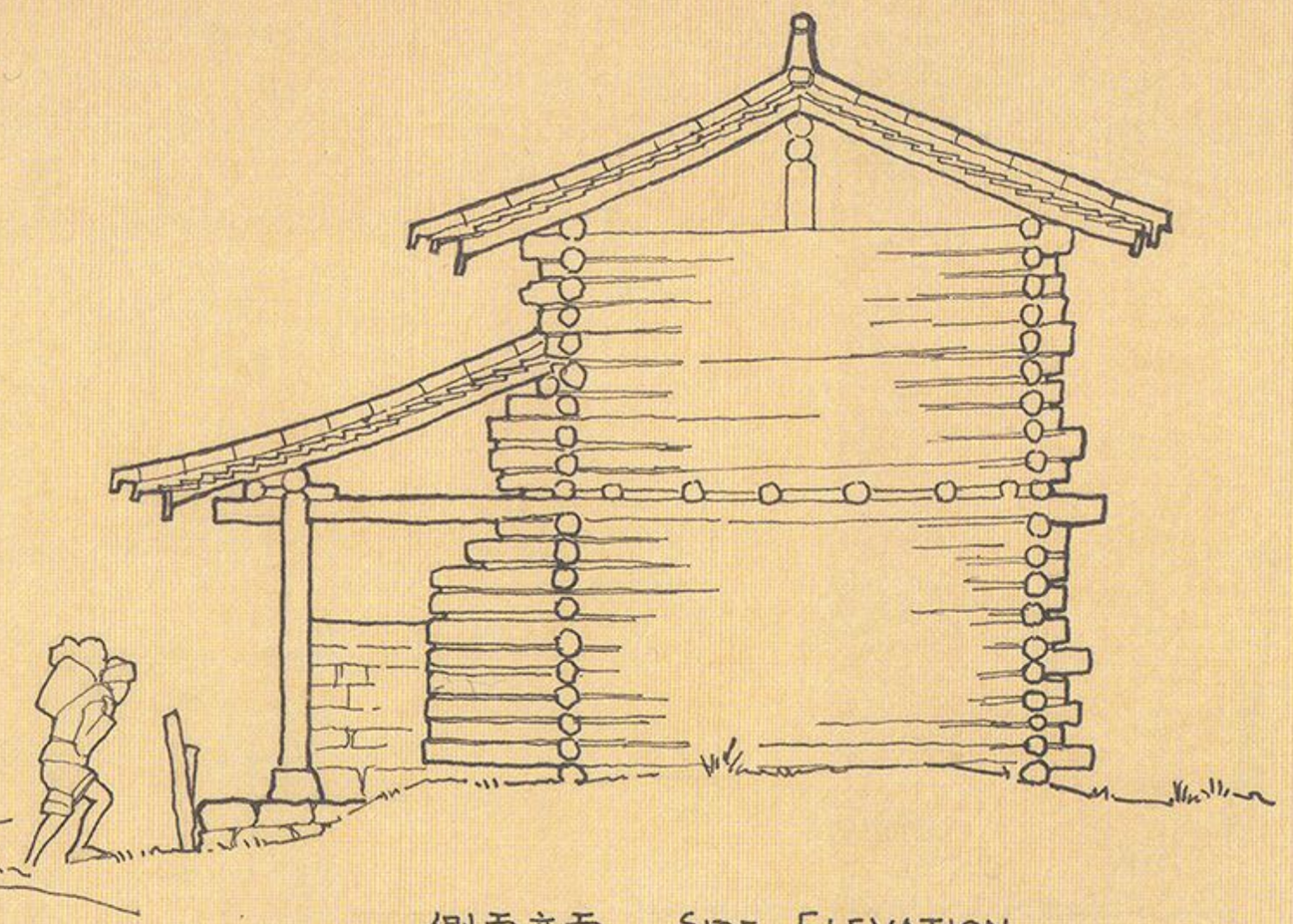
1 太和殿
2 中和殿
3 保和殿
4 太和門
5 體仁閣
6 弘義閣
7 昭德門
8 貞度門
9 左翼門
10 右翼門
11 中左門
12 中右門
13 後左門
14 後右門
15 角 樓
16 乾清門
17 景運門
18 隆宗門
19 直班房
20 軍機處

10M.
0
50
100 公尺

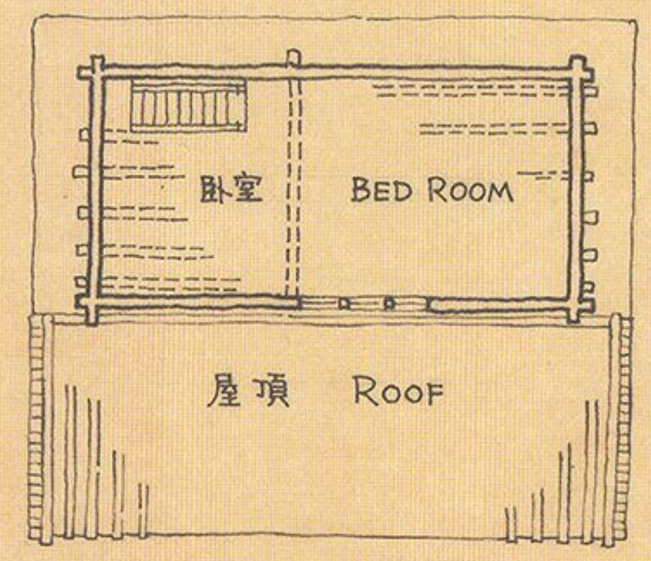
北



正面立面 FRONT ELEVATION



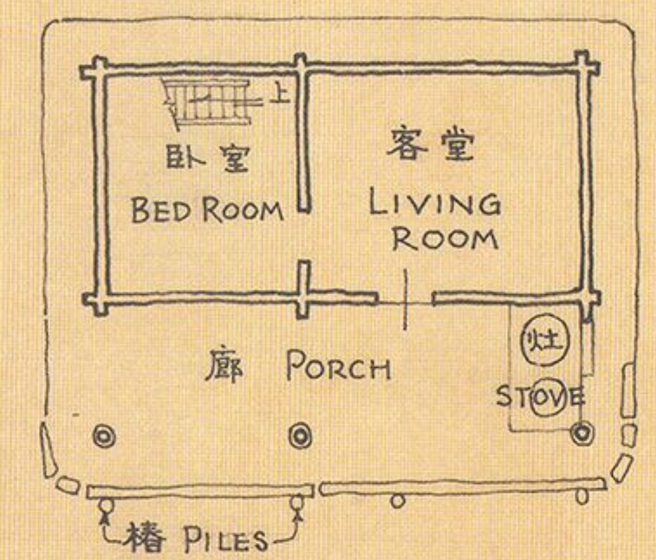
側面立面 SIDE ELEVATION



上層平面 UPPER FLOOR PLAN

公尺 1 0 5M.

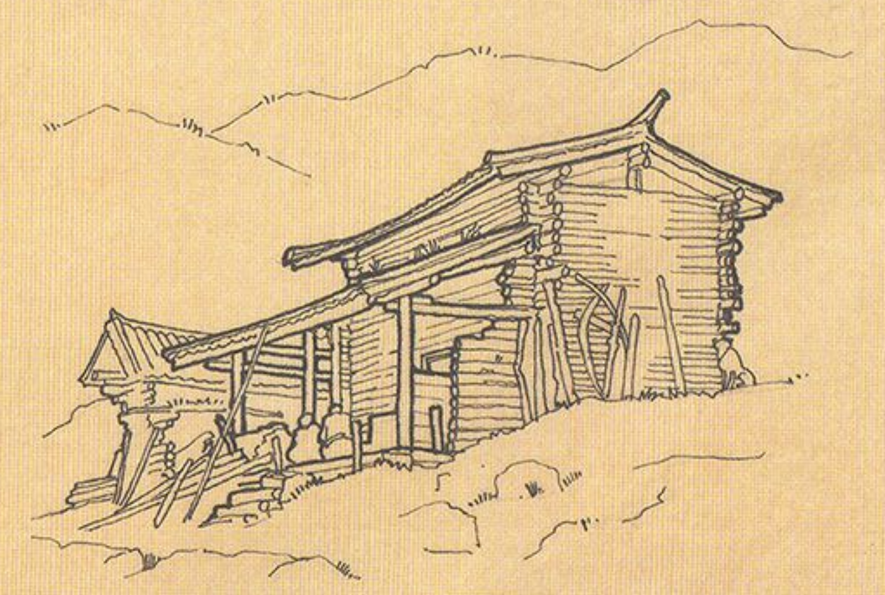
平面縮尺 SCALE FOR PLANS



下層平面 LOWER FLOOR PLAN

公尺 1 0 3 METRES

立面縮尺 SCALE FOR ELEVATIONS



LOG CABIN, MA-AN SHAN, CHEN-NAN HSIEN, YUNNAN

雲南 鎮南縣 馬鞍山

井榦構民居

MEASURED BY LIU T.T.

劉敦楨 測繪

歷代佛塔型類演變圖

各圖非用同一縮尺　NOT DRAWN TO SAME SCALE

附畫人像以示塔約略大小　HUMAN FIGURE INDICATES APPROXIMATE SCALE

各圖地平線位置表示在紀年尺中相當之年代

EVOLUTION OF TYPES OF THE BUDDHIST PAGODA

POSITION OF GROUD-LINE INDICATES DATE IN RELATION TO CHRONOLOGICAL SCALE.

CHINESE MULTI-STOREYED TOWER 中土"樓重"

先型 PROTOTYPES

印度窣堵坡 INDIAN STUPA

單層塔 UNI-STOREYED

多層塔 MULTI-STOREYED

密檐塔 MULTI-EAVED

瓶形塔 BOTTLE-SHAPED

金剛寶座塔 VAJRA-BASED

公元 A.D.

400 · 500 · 600 · 700 · 800 · 900 · 1000 · 1100 · 1200 · 1300 · 1400 · 1500 · 1600 · 1700 · 1800 · 1900

北魏 N. WEI

TS'I & CHOU 齊周

SUI 隋

唐 T'ANG

五代 5-DYN.

遼 LIAO

宋 SUNG

金 CHIN

南宋 S. SUNG

元 YUAN

明 MING

清 CH'ING

古拙時期 ARCHAIC PERIOD

繁麗時期 ELABORATION PERIOD

雜變時期 VARIETY PERIOD

雲岡后期 YÜN-KANG RELIEF 450

歷城 神通寺 SHENG-T'UNG SSU TSI-NAN 544

方 SQUARE　八角 OCTAG'L

嵩山會善寺淨藏 HUI-SHAN SSU SUNG MT. 746

嵩山少林寺同光 SHAO-LIN SSU SUNG MT. 771

嵩山少林寺行鈞 SHAO-LIN SSU SUNG-MT. 926

安陽寶泉寺靈裕 LING-CH'UAN SSU AN-YANG 1094

嵩山少林寺端師 1168 SHAO-LIN SSU

嵩山少林寺定公 1287 SHAO-LIN SSU

雲岡后期 YÜN-KANG RELIEF 450

五台佛光寺 FO-KUANG SSU WU-T'AI MT. CA. 600

西安慈恩寺 TZ'U-EN SSU SI-AN 704

無柱式 ASTYLAR

定縣開元寺 K'AI-YUAN SSU TING HSIEN 1001

太足報恩寺 PAO-EN SSU TA-TSU 1155

景縣開福寺 K'AI-FU SSU CHING HSIEN 1462?

太原永祚寺南塔 YUNG-CHAO SSU T'AI-YUAN 1595

許峰 SIN-CHIANG 清中葉 MID-CH'ING

五台普勝寺 1928 TSUN-SHENG SSU WU-T'AI MT.

倣木構式 TIMBER-FRAME

南宗 S. SCH.　北宗 N. SCHOOL

蘇州羅漢院 LO-HAN YUAN SOO-CHOU 982

杭州靈隱寺 LING-YIN SSU HANG-CHOU 960

澤縣智度寺 CHIH-TU S. TSO HSIEN CA. 1090

晉江開元寺 K'AI-YUAN SSU TSIN-CHIANG 1228-47

蓮溪慧峰寺 CHIU-FENG SSU P'ENG-HSI 南宋 S. SUNG

澄陽 CHING-YANG 明中葉 MID-MING

金華報恩寺 PAO-EN SSU CHIN-HUA 清中葉

北平玉泉山 YÜ-CH'UAN SHAN PEIPING 清末 LATE CH'ING

嵩山少林寺 SHAO-LIN SSU SUNG MT. 1736

嵩山法王寺 FA-WANG SSU SUNG MT. 盛唐 MID-T'ANG

大興崇聖寺 CH'UNG-SHENG SSU TA-LI 晚唐 LATE T'ANG

宜賓旧州城 CHIU-CHOU PA I-PIN 1102-09

洛陽白馬寺 PAI-MA SSU LO-YANG 1175

昆明妙湛寺 MIAO-CHAN SSU K'UN-MING 1458

昆明常樂寺 CH'ANG-LÊ SSU K'UN-MING 1887

方 SQUARE　八角 OCTAGONAL

嵩山嵩嶽寺 SUNG-YUEH SSU SUNG MT. 520

南京棲霞寺 CH'I-HSIA SSU NANKING 五代 5-DYN.

北平大寧寺 T'IEN-NING SSU PEIPING 遼 LIAO

正定臨濟寺 LIN-TSI SSU CHENG-TING 1185

元 YUAN 那台開元寺 K'AI-YUAN SSU HSING-T'AI

北平慈壽寺 1575 TZ'U-SHOU SSU PEIPING

五台佛光寺 FO-KUANG SSU WU-T'AI MT. 唐末 CA. 900

北平妙應寺 1271 MIAO-YING SSU PEIPING

五台塔院寺 T'A-YUAN SSU WU-T'AI MT. 1577

北海 北平 1651 PEI-HAI PEIPING

房山雲居寺 主塔遼重建 PRINCIPAL PAGODA LIAO DYN. RECONSTN.

附塔唐建 SATELLITE PAGODAS 711-27

YÜN-CHÜ SSU FANG-SHAN

正定廣惠寺 金 CHIN KUANG-HUI SSU CHENG-TING

北平真覺寺 1473 CHEN-CHUEH SSU PEIPING

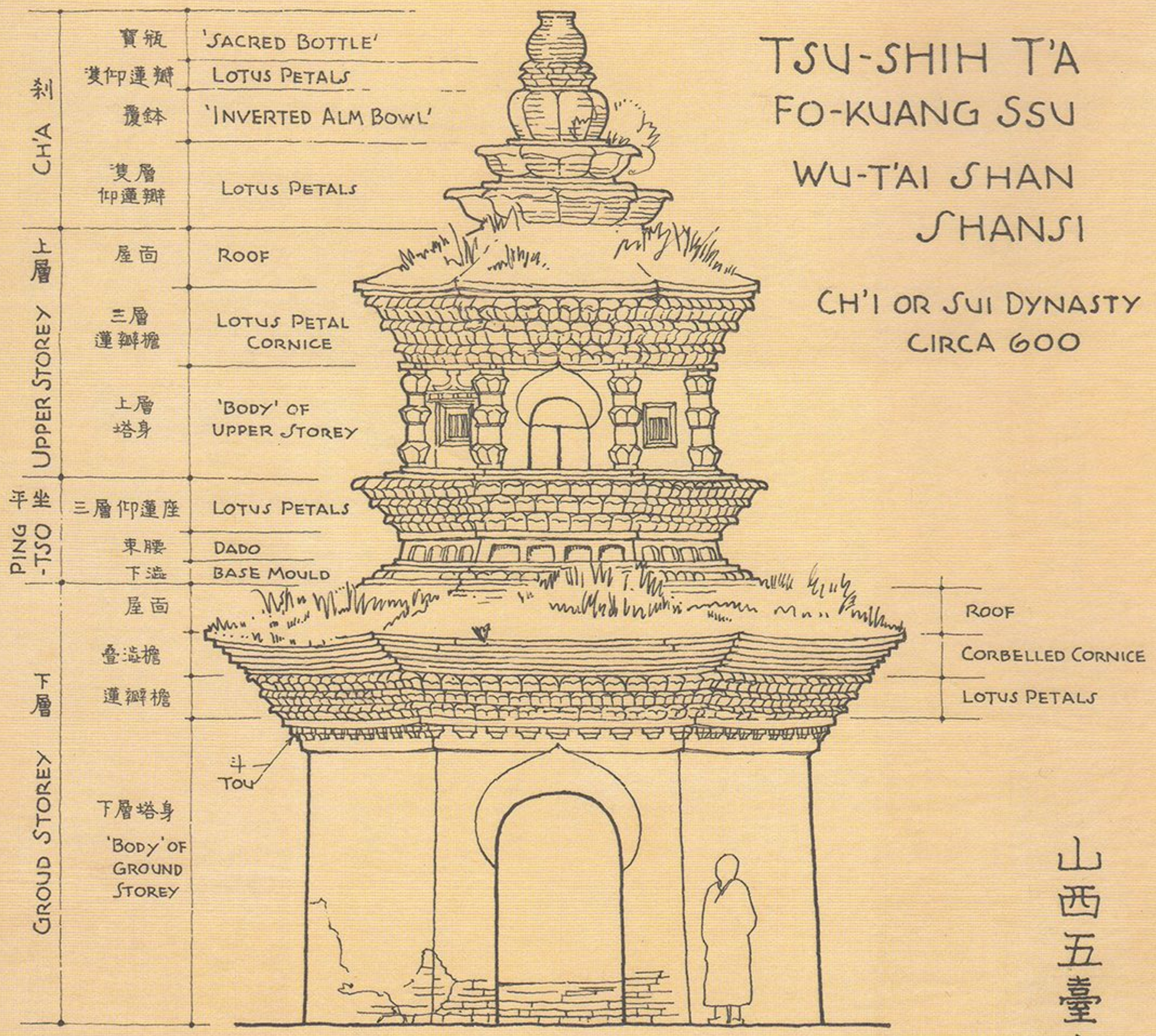
北平碧雲寺 1748 PI-YÜN SSU PEIPING

TSU-SHIH T'A
FO-KUANG SSU
WU-T'AI SHAN
SHANSI

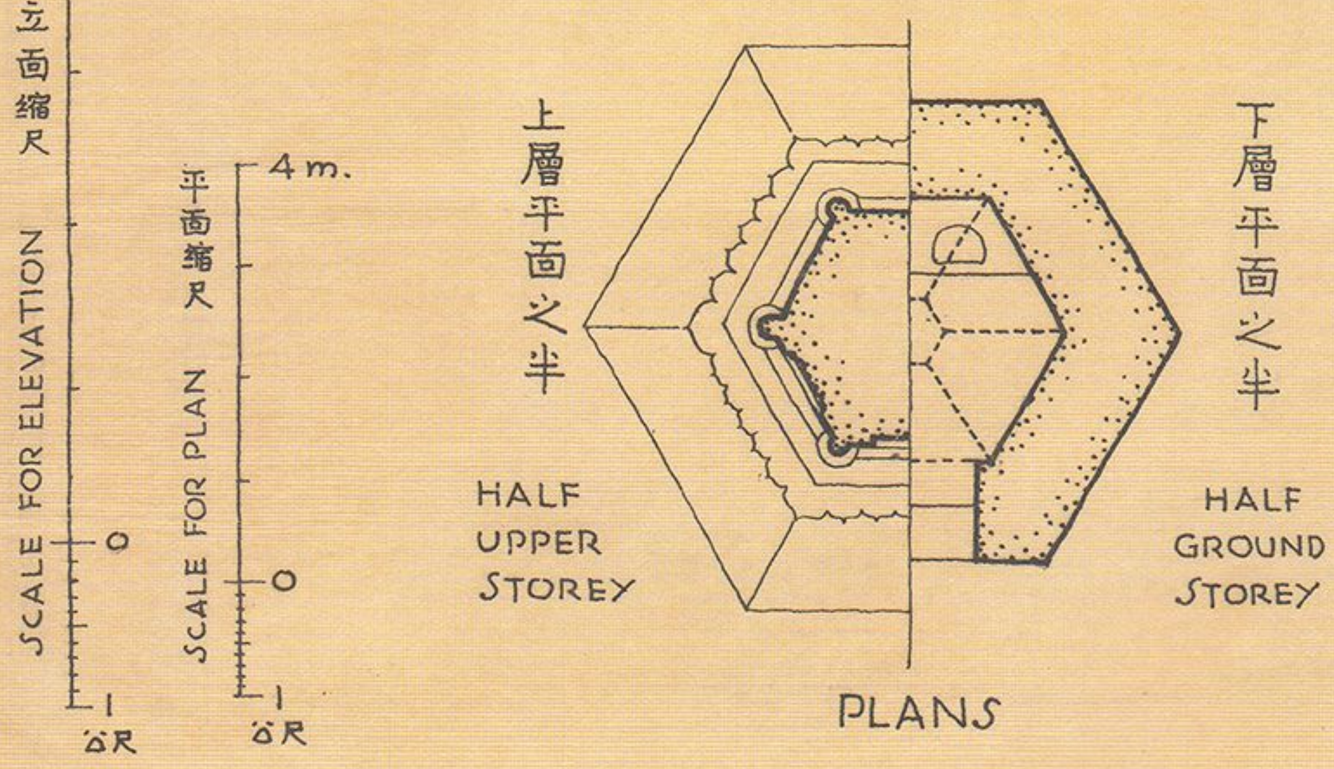
CH'I OR SUI DYNASTY
CIRCA 600

山西五臺山 佛光寺 祖師塔

北齊或隋建

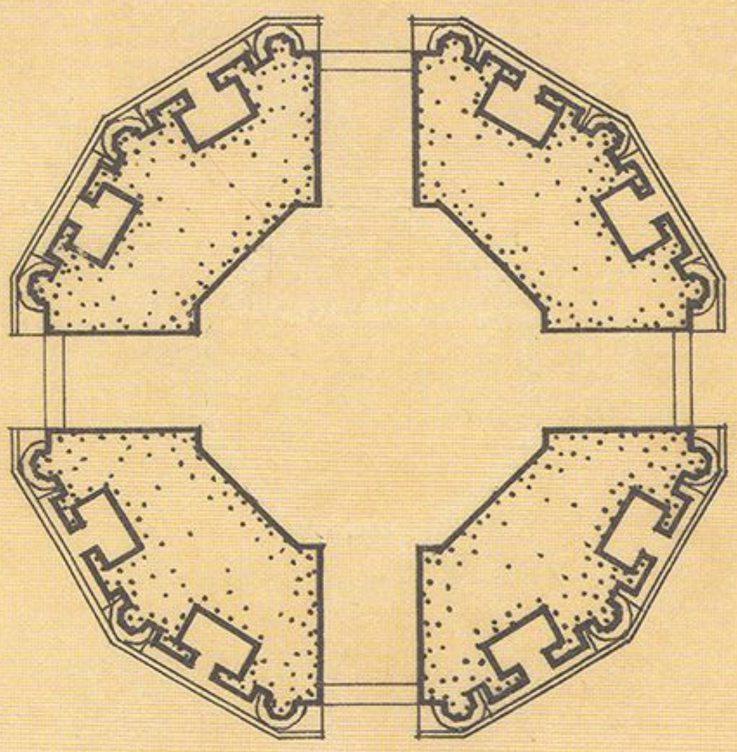


立面 ELEVATION

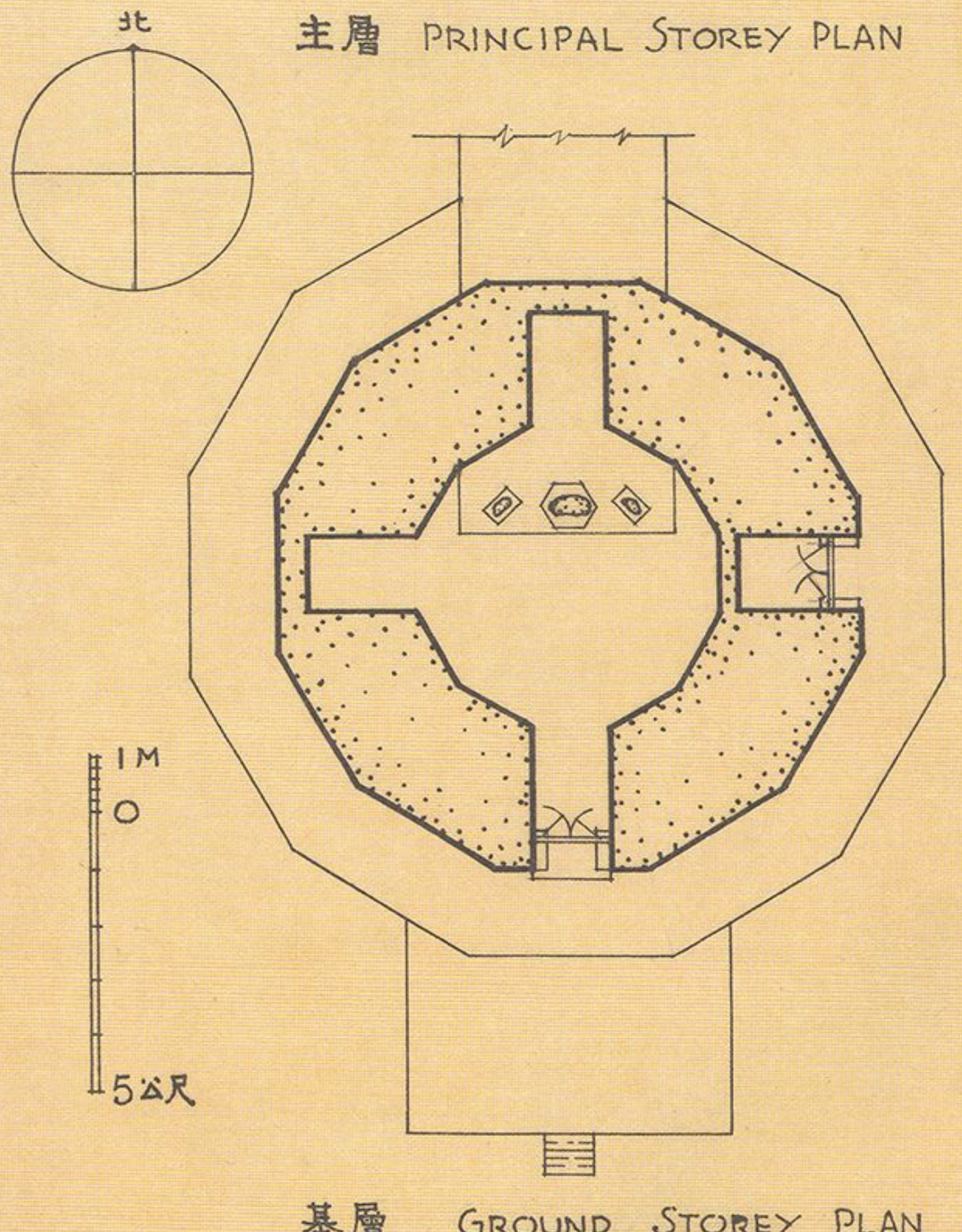


PLANS

河南嵩山嵩嶽寺塔平面



主層 PRINCIPAL STOREY PLAN



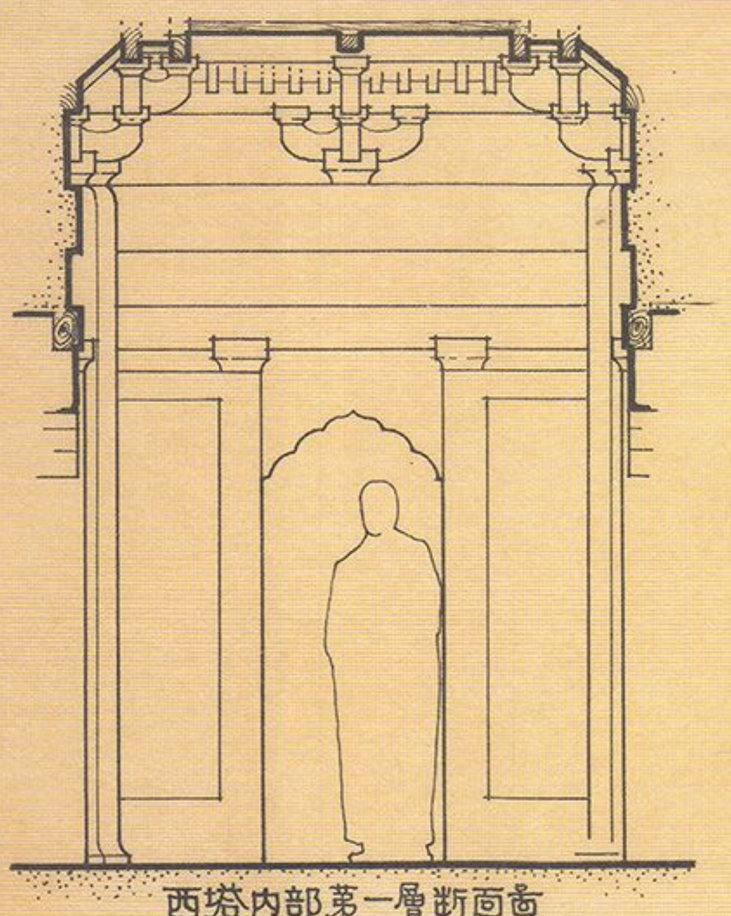
基層 GROUND STOREY PLAN

PAGODA OF SUNG-YÜEH SSU

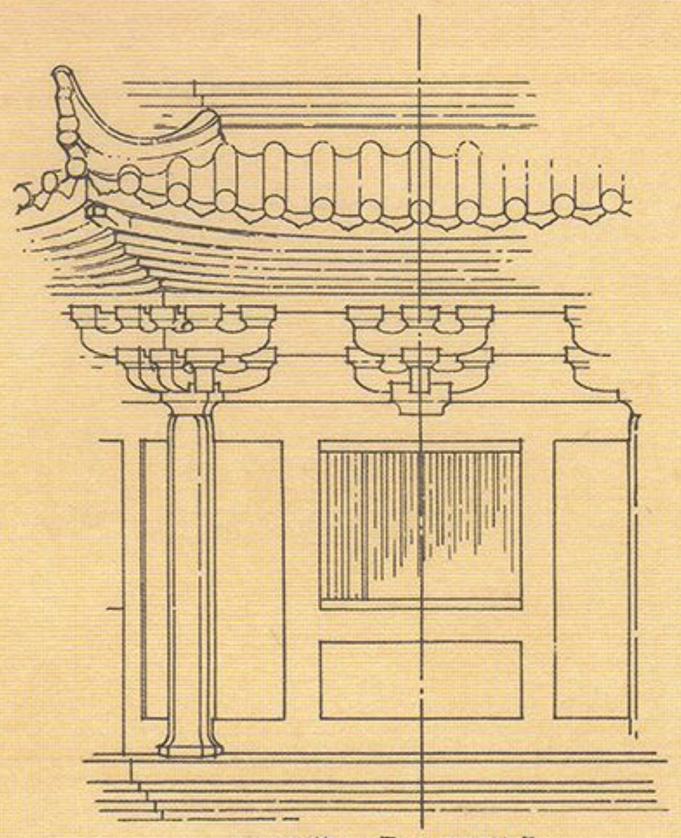
SUNG MOUNTAINS · TENG-FENG · HONAN

劉敦楨測繪 MEASURED BY LIU, T.-T.

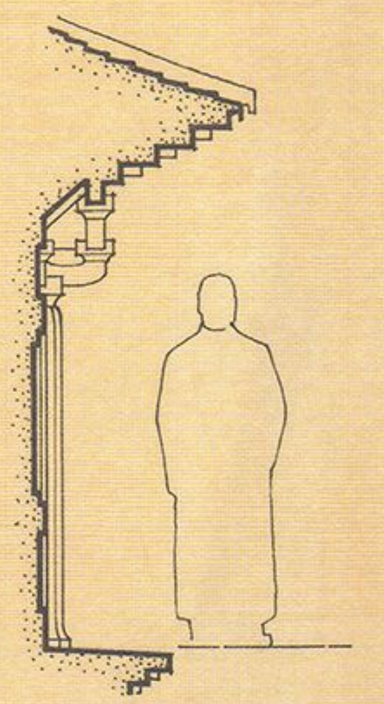
169-B



西塔內部第一層斷面圖
SECTION, 1ST FLOOR, WEST PAGODA

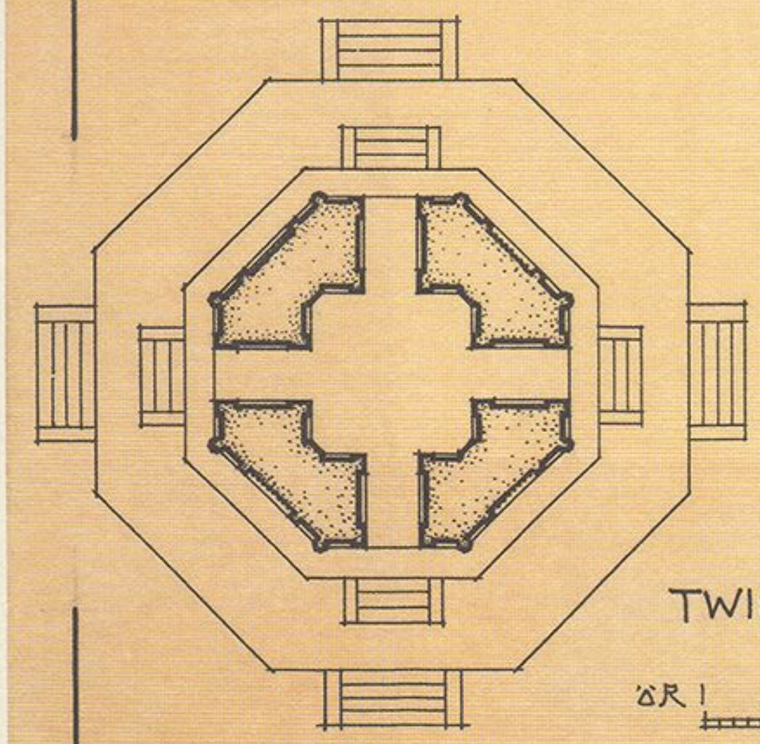


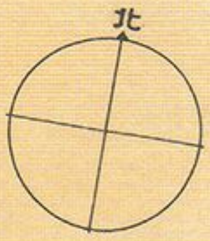
西塔第二層立面詳圖
DETAIL OF EXTERIOR, 2ND FLOOR, WEST PAGODA



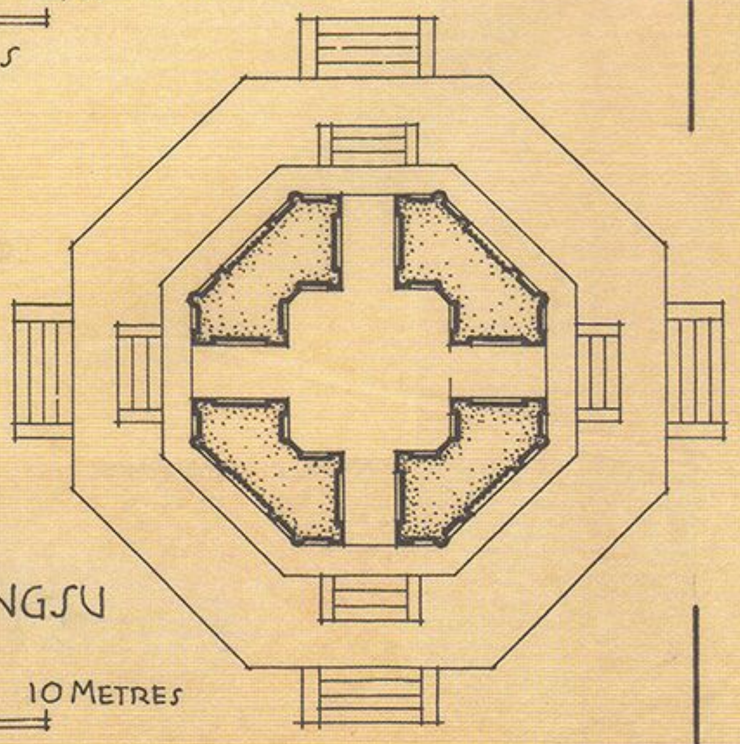
西塔第二層外層斷面圖

比尺 0.50 0 1 2M.
詳圖縮尺 SCALE FOR DETAILS





建于七國興平太宋
SUNG DYNASTY, 982 A.D.



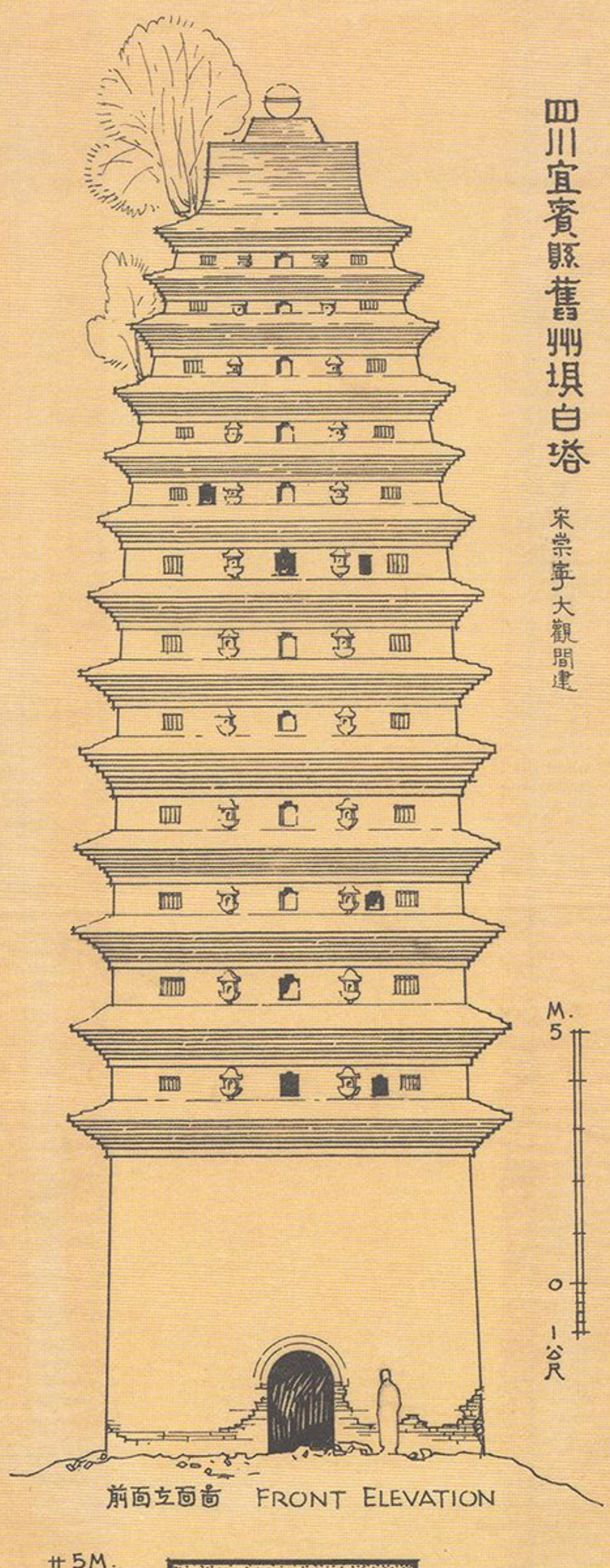
双塔平面圖
PLAN OF TWIN-PAGODAS

江蘇吳縣 羅漢院雙塔
TWIN PAGODAS, SOOCHOW, CHIANGSU

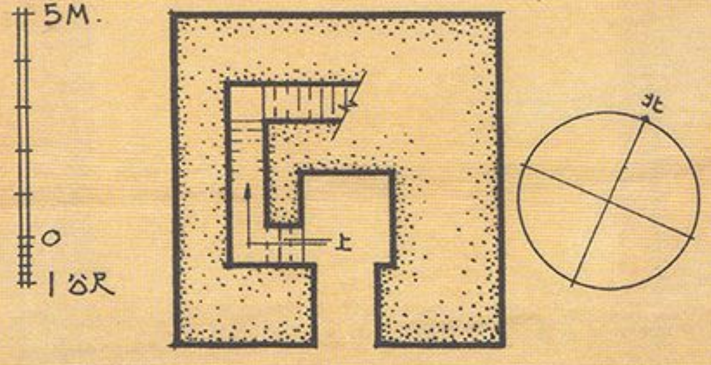
比尺 1 0 5 10 METRES
平面縮尺 SCALE FOR PLAN

四川宜賓縣舊州坝白塔

宋崇寧大觀間建



前面立面圖 FRONT ELEVATION



下層平面圖 GROUND FLOOR PLAN

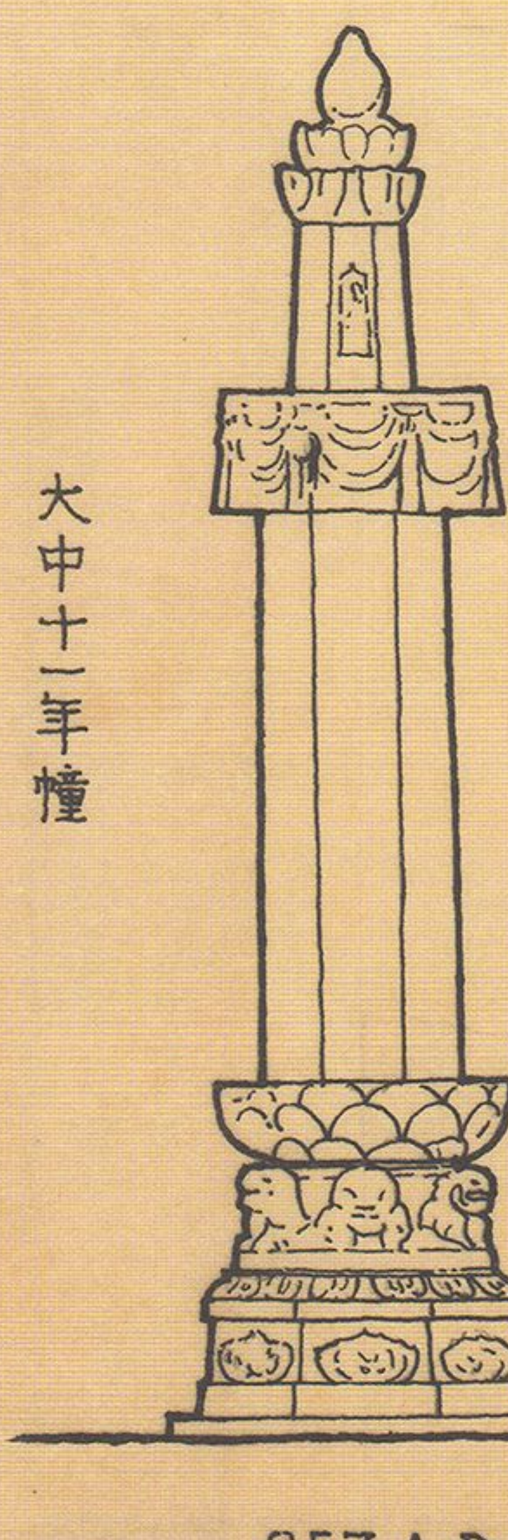
PAGODA AT CHIU-CHOU-PA,
YI-PIN, SZECHUAN

SUNG DYNASTY, 1102-09 A.D.

山西五台山 佛光寺

晚唐兩經幢

TWO DHANARI COLUMNS
OF THE LATE T'ANG PERIOD
FO-KUANG SSU
WU-T'AI SHAN · SHANSI



圓 Round

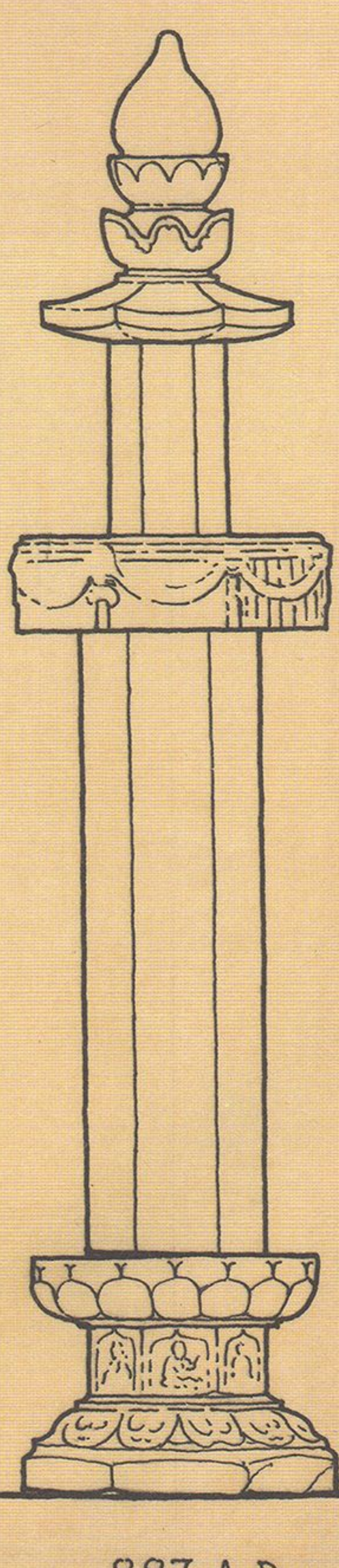
八角 Octa-gonal

大中十一年幢

圓 Round

八角 Octa-gonal

857 A.D.



圓 Round

八角 Octa-gonal

乾符四年幢

圓 Round

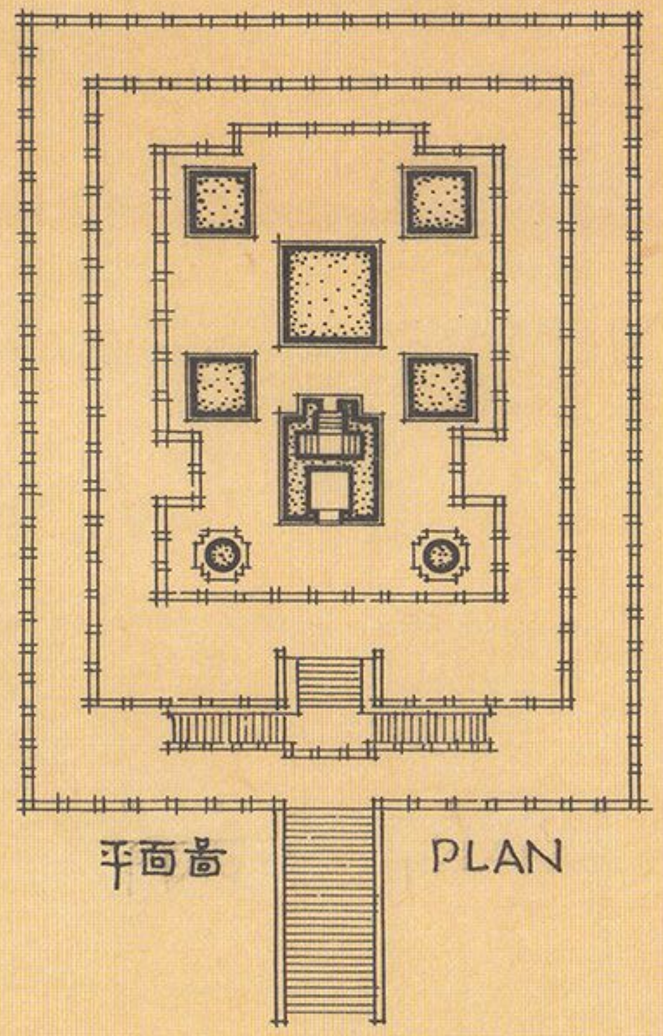
八角 Octa-gonal

887 A.D.

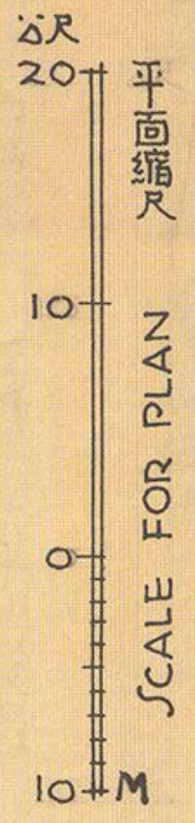
公分 50 0 100 cm.

北平西山碧雲寺金剛寶座塔

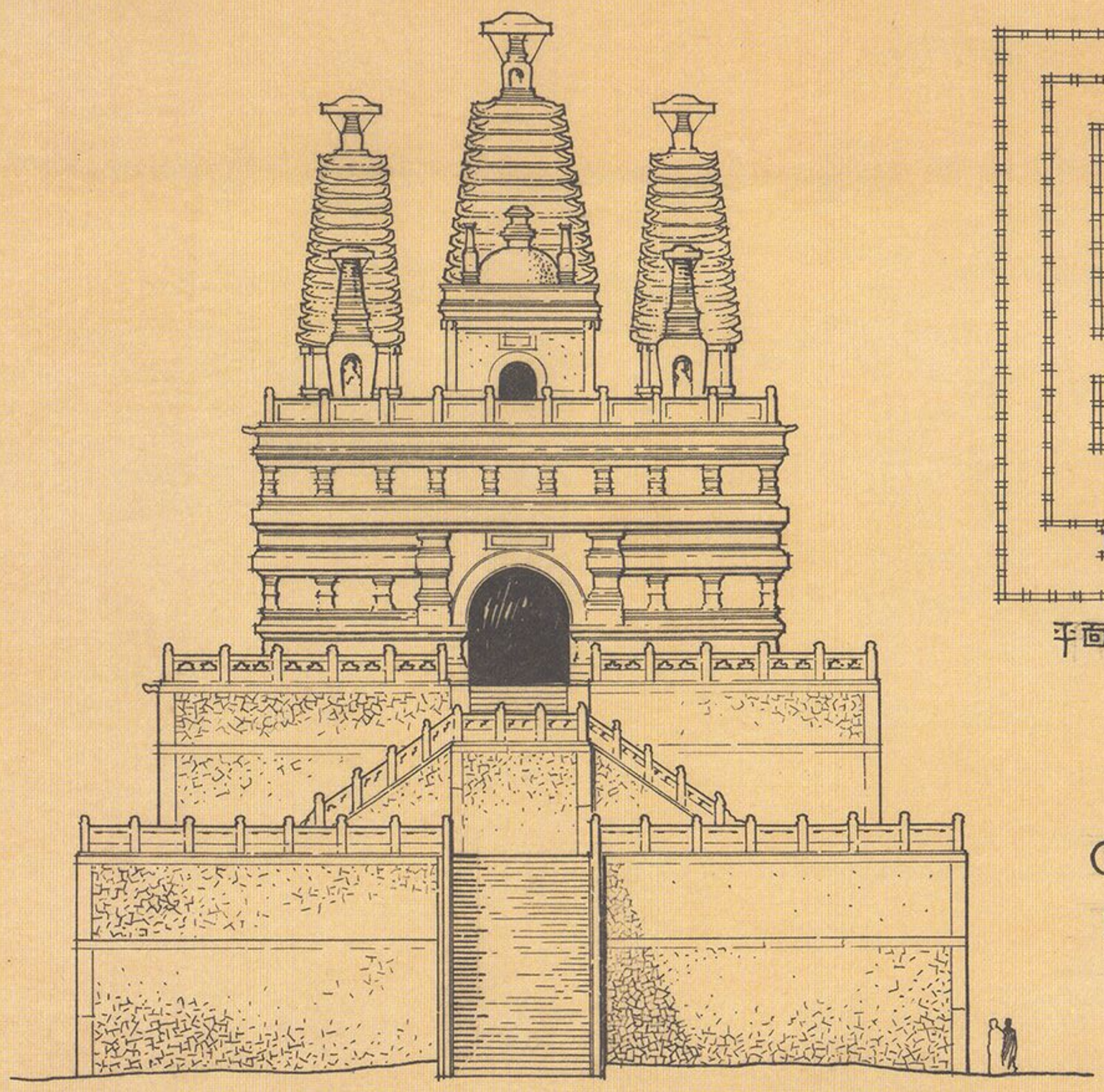
清乾隆十三年建



平面圖 PLAN



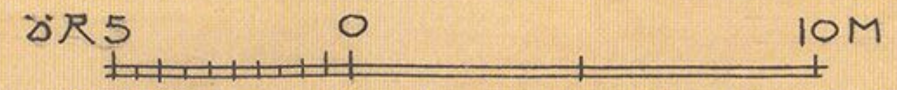
平面縮尺 SCALE FOR PLAN



前面立面 FRONT ELEVATION

CHIN-KANG-PAO-TSO T'A

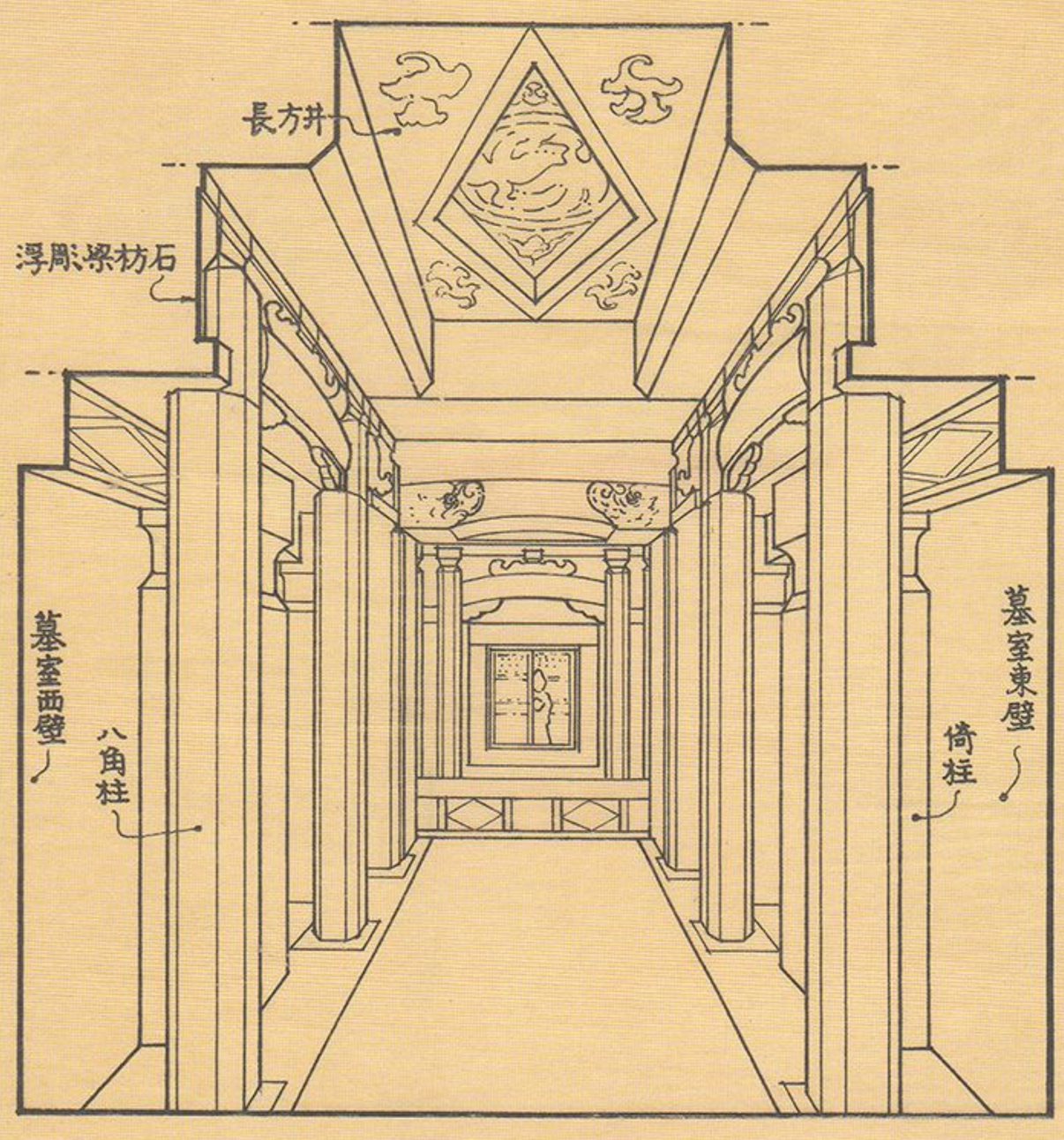
PI-YÜN SSU, WESTERN HILLS, PEIPING. CH'ING DYNASTY, 1748.



斷面縮尺 SCALE FOR SECTION

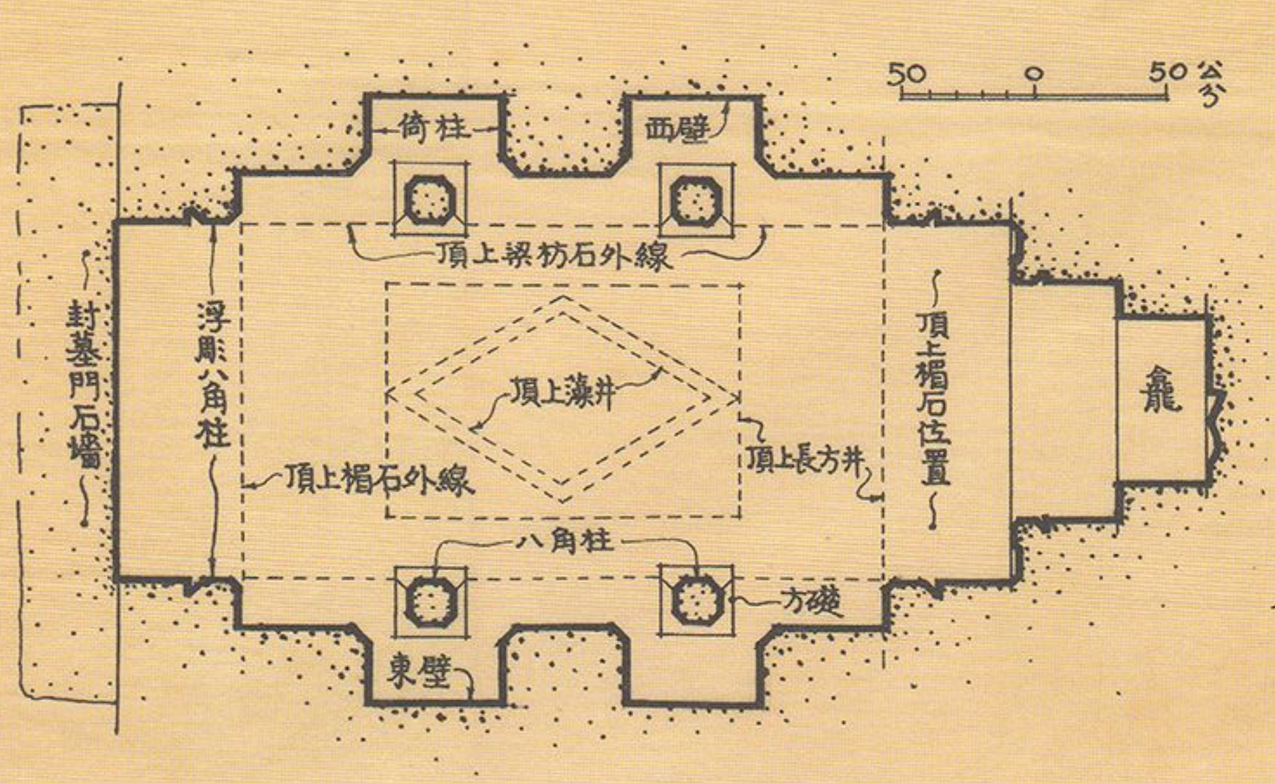
REDRAWN FROM BOERSCHMANN: CHINESISCHE ARCHITEKTUR.

四川宜賓無名墓　南宋孝宗朝(?)建



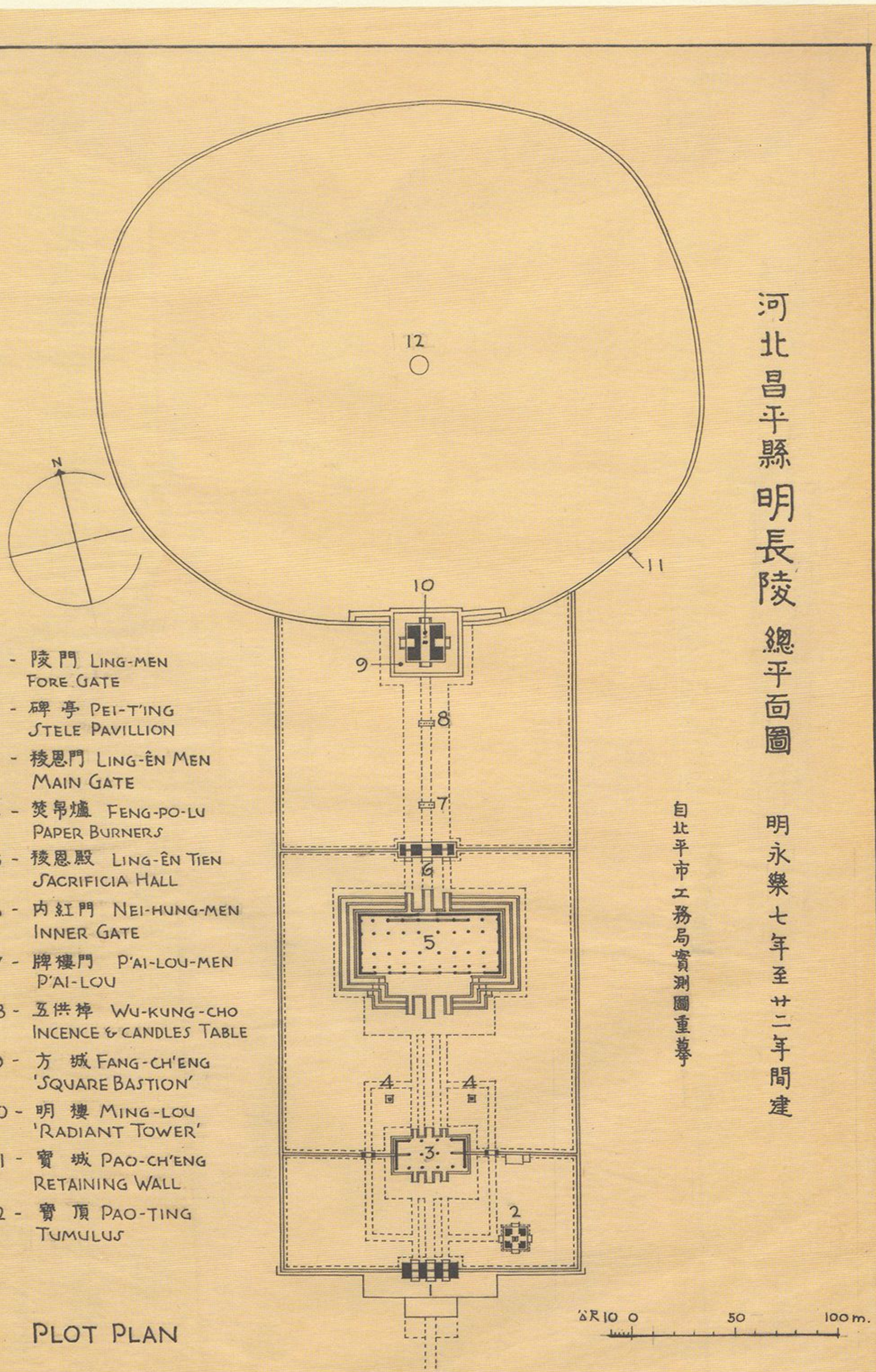
內部　INTERIOR VIEW

平面　PLAN



AN UNIDENTIFIED TOMB　C. 1170

I-PIN　SZECHUAN



1 - 陵門 LING-MEN
FORE GATE

2 - 碑亭 PEI-T'ING
STELE PAVILLION

3 - 稜恩門 LING-ÊN MEN
MAIN GATE

4 - 焚帛爐 FENG-PO-LU
PAPER BURNERS

5 - 稜恩殿 LING-ÊN TIEN
SACRIFICIA HALL

6 - 內紅門 NEI-HUNG-MEN
INNER GATE

7 - 牌樓門 P'AI-LOU-MEN
P'AI-LOU

8 - 五供桌 WU-KUNG-CHO
INCENCE & CANDLES TABLE

9 - 方城 FANG-CH'ENG
'SQUARE BASTION'

10 - 明樓 MING-LOU
'RADIANT TOWER'

11 - 寶城 PAO-CH'ENG
RETAINING WALL

12 - 寶頂 PAO-TING
TUMULUS

PLOT PLAN

0 50 100 m.

河北昌平縣 明長陵 總平面圖

明永樂七年至廿二年間建

自北平市工務局實測圖重摹

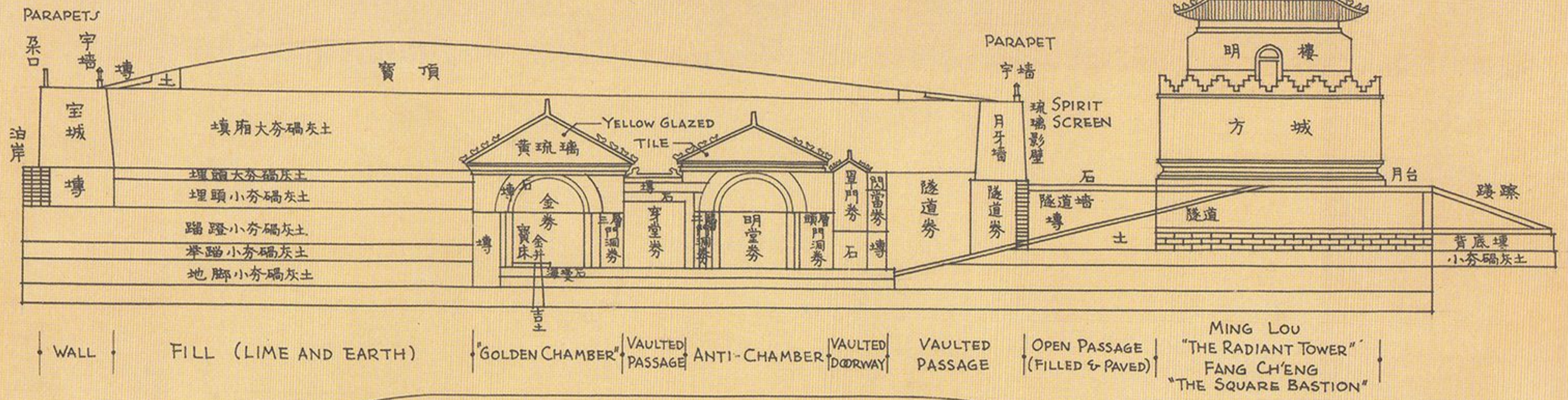
CH'ANG-LING · TOMB OF EMPEROR YUNG-LO

CH'ANG-P'ING · HOPEI · MING DYNASTY · 1409-24

REDRAWN AFTER PLAN BY THE BUREAU OF CONSTRUCTION · MUNICIPAL GOVERNMENT OF PEIPING

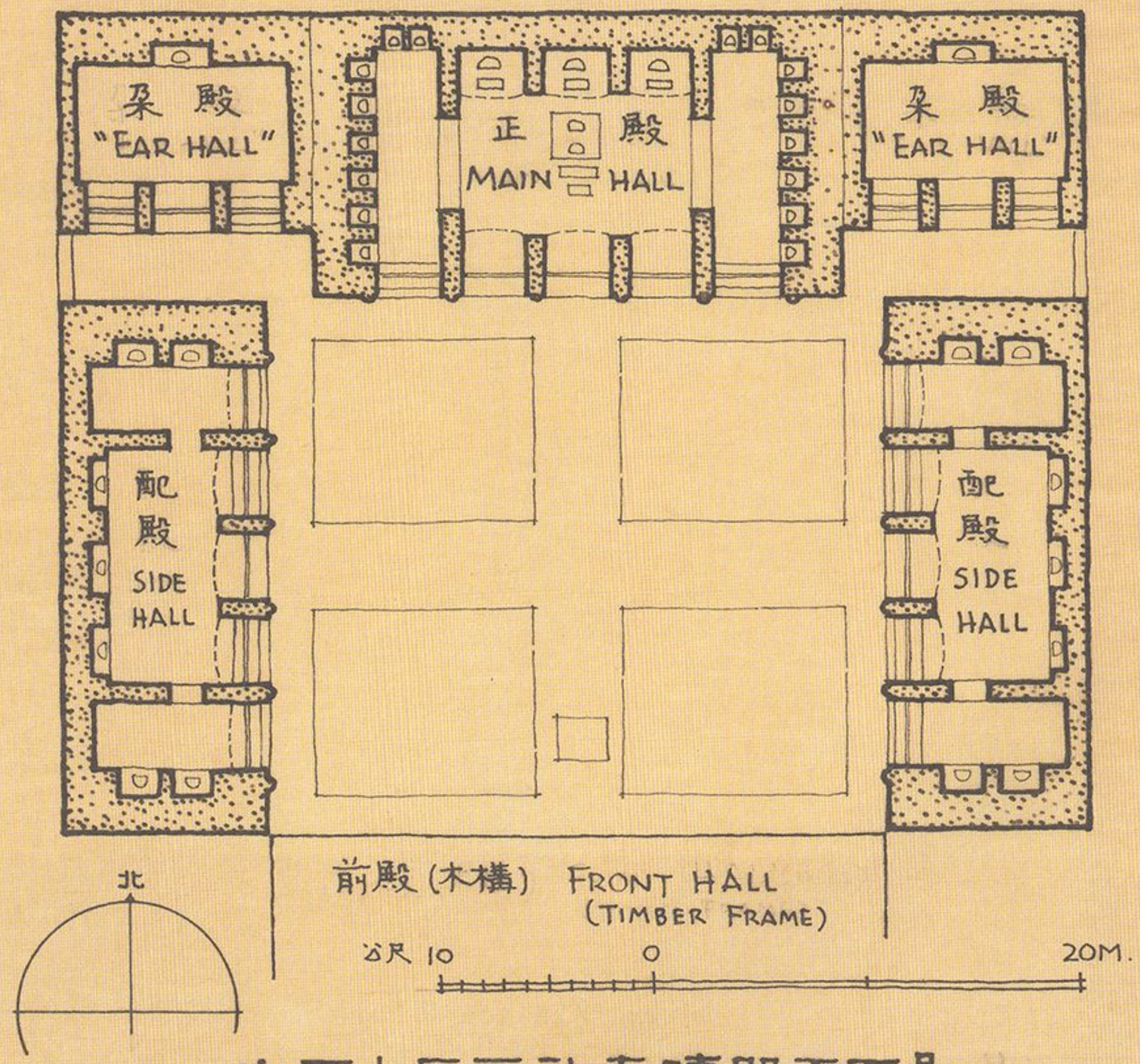
清昌陵地宮斷面及平面圖（陵在河北省易縣）

自國立北平圖書館藏 樣式房雷氏圖房式樣 摹重圖



CH'ANG LING, TOMB OF EMPEROR CHIA-CH'ING, 1796-1820, CH'ING DYNASTY

PLAN AND SECTION OF SUBTERRANEAN TOMB CHAMBERS, REDRAWN AFTER ORIGINAL DRAWINGS BY THE LEI FAMILY, HEREDITARY OFFICIAL ARCHITECTURAL DESIGNERS. (COLLECTION, NATIONAL PEIPING LIBRARY).



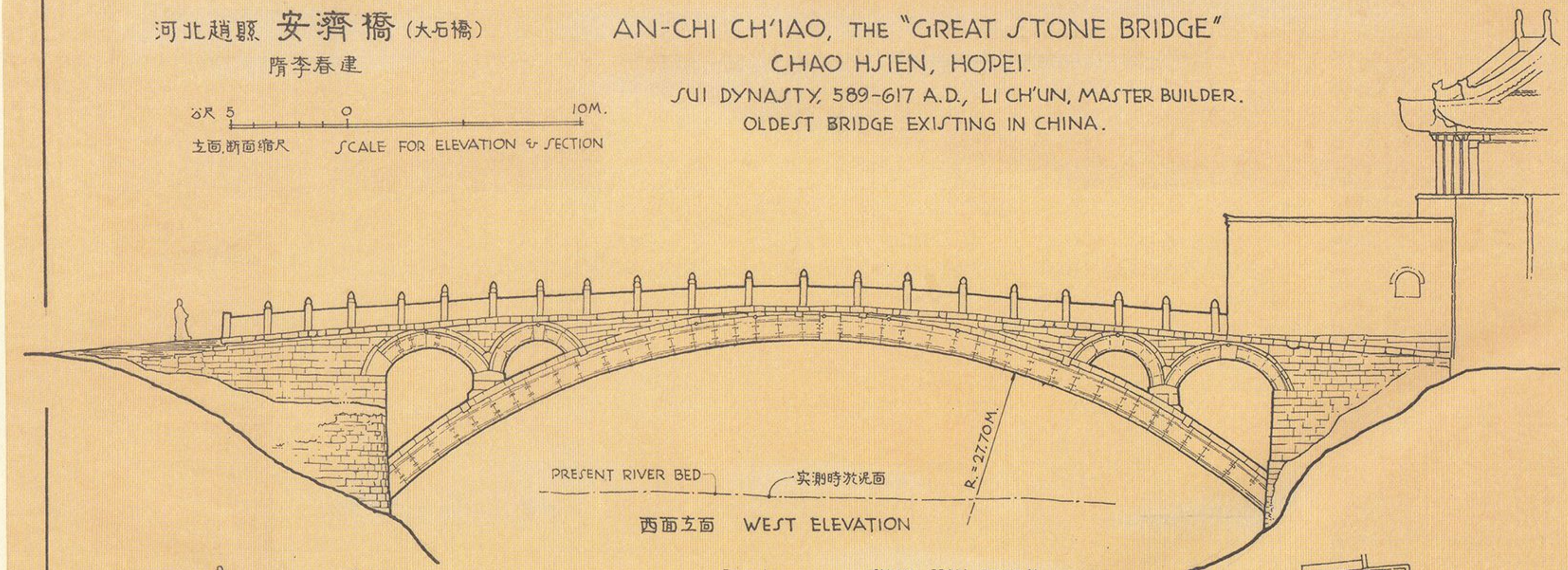
山西太原永祚寺磚殿平面圖

YUNG-CHAO SSU, T'AI-YUAN, SHANSI, PLAN OF BRICK VAULTED HALLS, 明萬曆二十五年建, A.D. 1597.

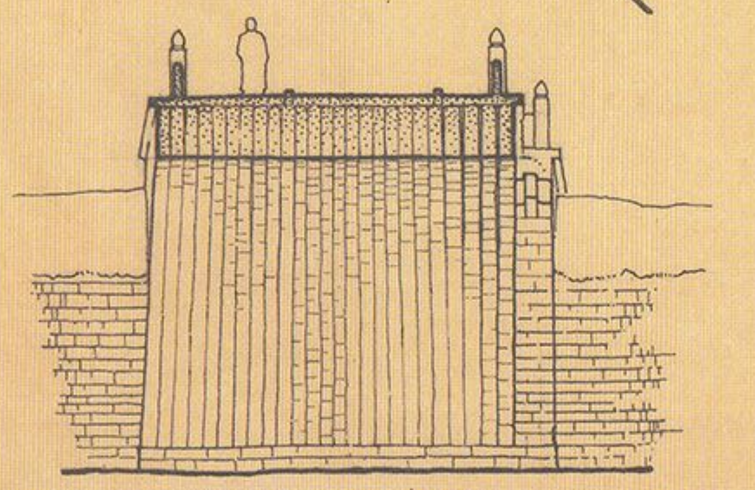
河北趙縣 安濟橋 (大石橋)
隋李春建

AN-CHI CH'IAO, THE "GREAT STONE BRIDGE"
CHAO HSIEN, HOPEI.
SUI DYNASTY, 589-617 A.D., LI CH'UN, MASTER BUILDER.
OLDEST BRIDGE EXISTING IN CHINA.

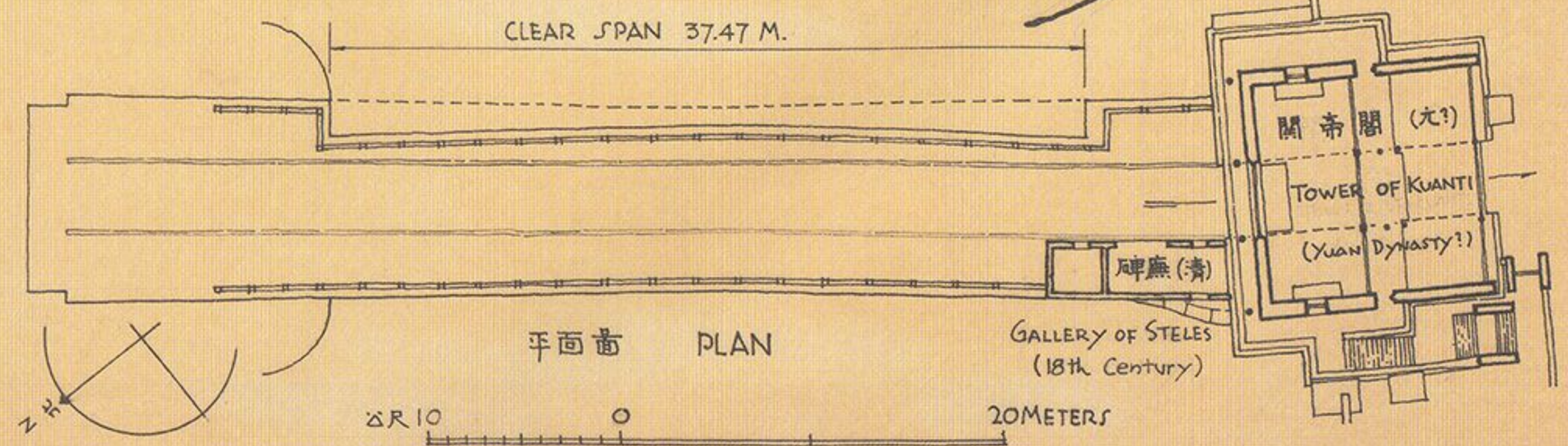
公尺 5 0 10M.
立面,断面縮尺 SCALE FOR ELEVATION & SECTION



PRESENT RIVER BED 实測時洪泥面
R.=27.70M.
西面立面 WEST ELEVATION

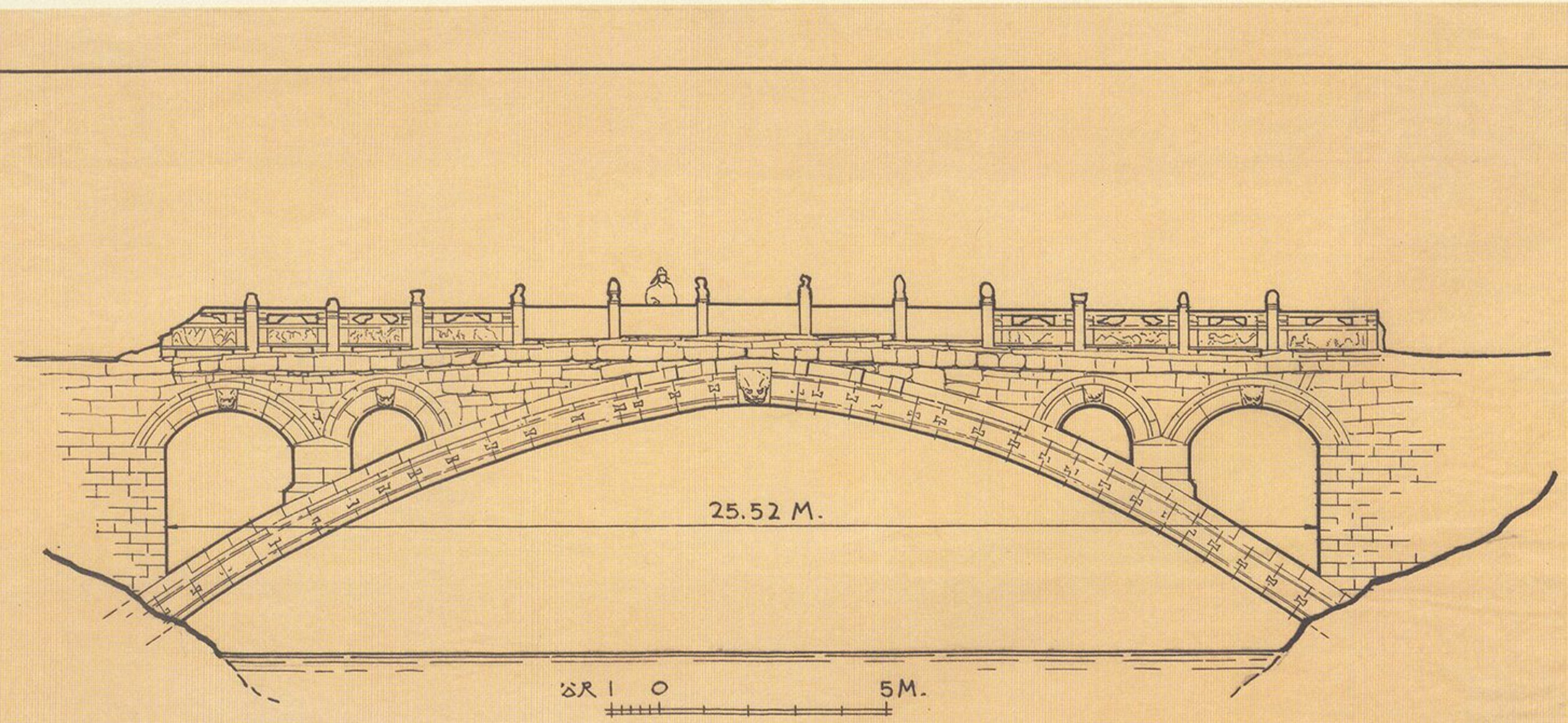


断面畫 SECTION



平面畫 PLAN

公R 10 0 20METERS
平面縮尺 SCALE FOR PLAN



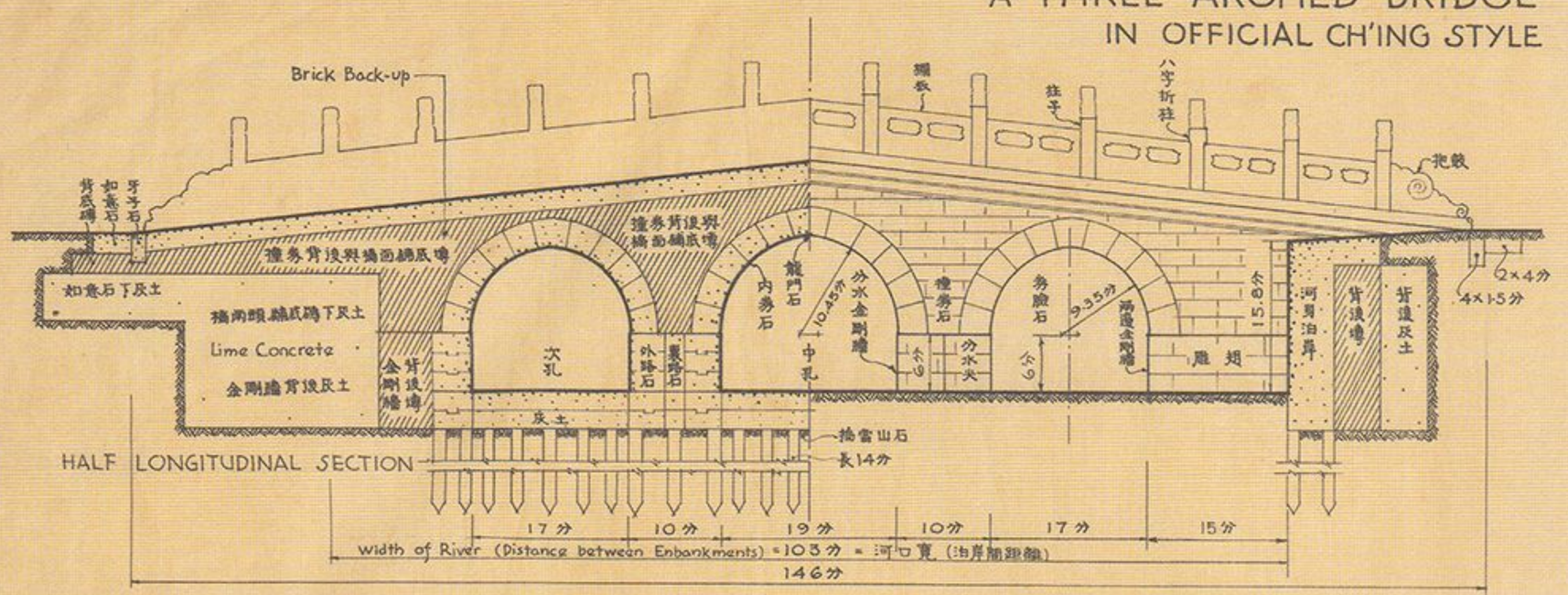
河北趙縣 永通橋

俗呼小石橋金明昌間
建而錢襄

YUNG-T'UNG CH'IAO OR LITTLE STONE BRIDGE

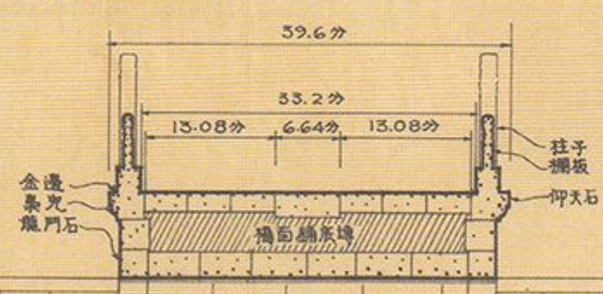
CHAO HSIEN, HOPEI, MING-CH'ANG PERIOD, 1190-95, CHIN DYNASTY.

清官式三孔石橋做法要略 RULES FOR DESIGN AND CONSTRUCTION OF A THREE-ARCHED BRIDGE IN OFFICIAL CH'ING STYLE



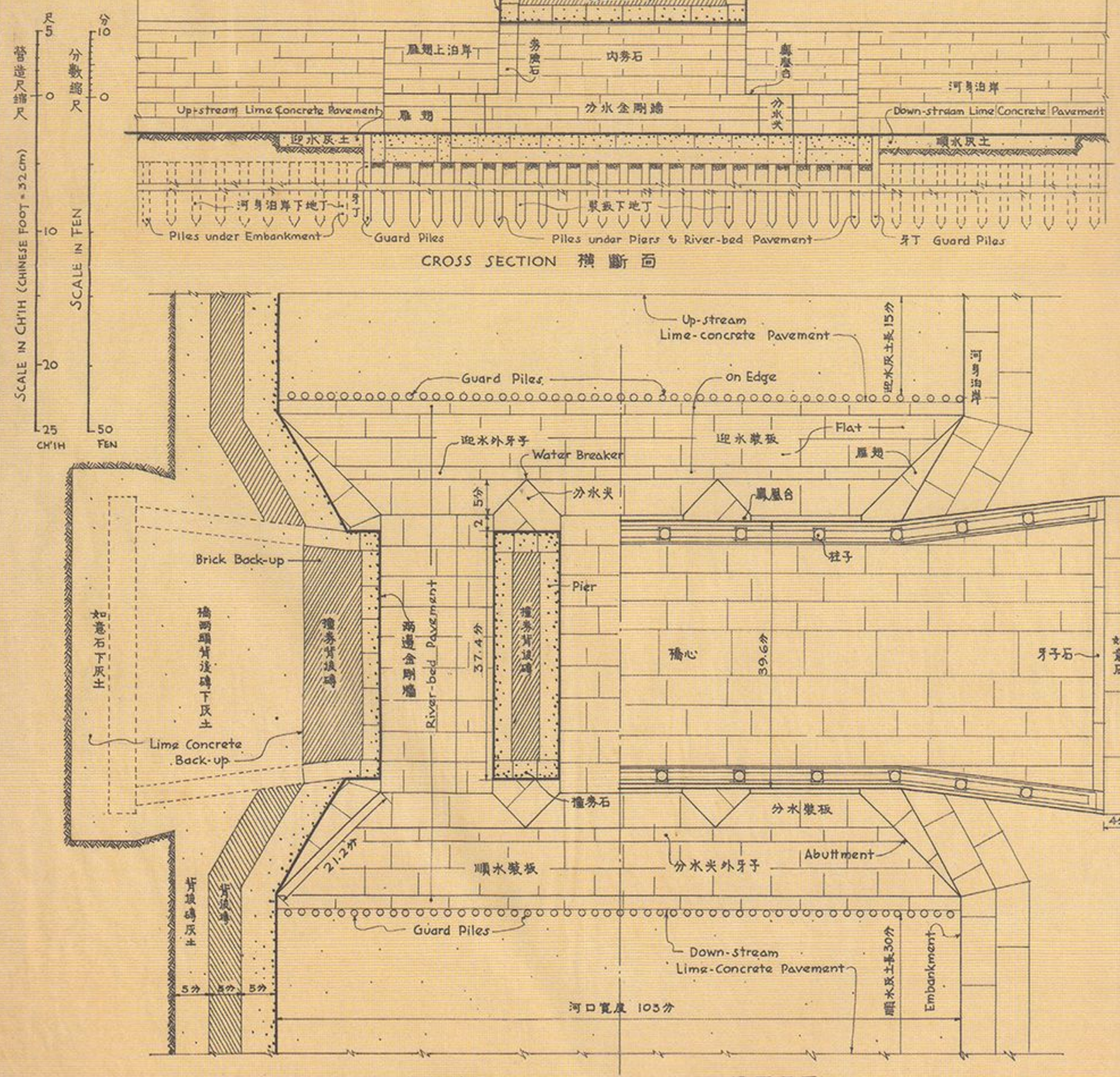
孔數 NUMBER OF ARCHES	河口寬度分作 DIST. B'TN EMBANKMENT DIVIDED INTO	(單位:分)孔寬 SPAN OF ARCHES (IN FEN) 自中孔左右 COUNTING FROM CENTRAL ARCH 中 C	I	II	III	IV	V	VI	VII	VIII
3	103 分 fen	19	17	—	—	—	—	—	—	—
5	153 " "	19	17	15	—	—	—	—	—	—
7	199 " "	19	17	15	13	—	—	—	—	—
9	251 " "	19	17½	16	14½	13	—	—	—	—
11	294 " "	19	17½	16	14½	13	11½	—	—	—
13	355 " "	19	18	17	16	15	14	13	—	—
15	399 " "	19	18	17	16	15	14	13	12	—
17	441 分 fen	19	18	17	16	15	14	13	12	11

縱斷面 立面 HALF ELEVATION

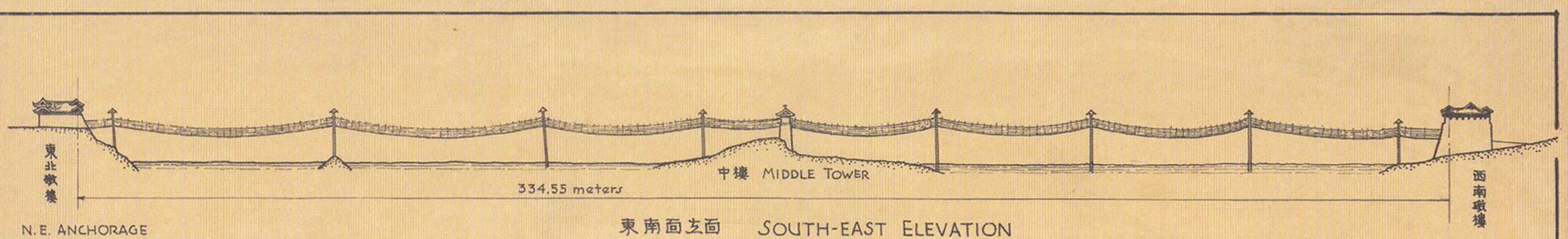


分水金剛牆一律寬 10分; 高6分.
All Piers: Width=10 fen; Height = 6 fen.
雁翅一律出 15分; 高6分.
Abutments: Width =15 fen; Height = 6 fen.
關於其他細節參閱
For Further Details, cf.
王璧文: 清官式石橋做法

CROSS SECTION 橫斷面



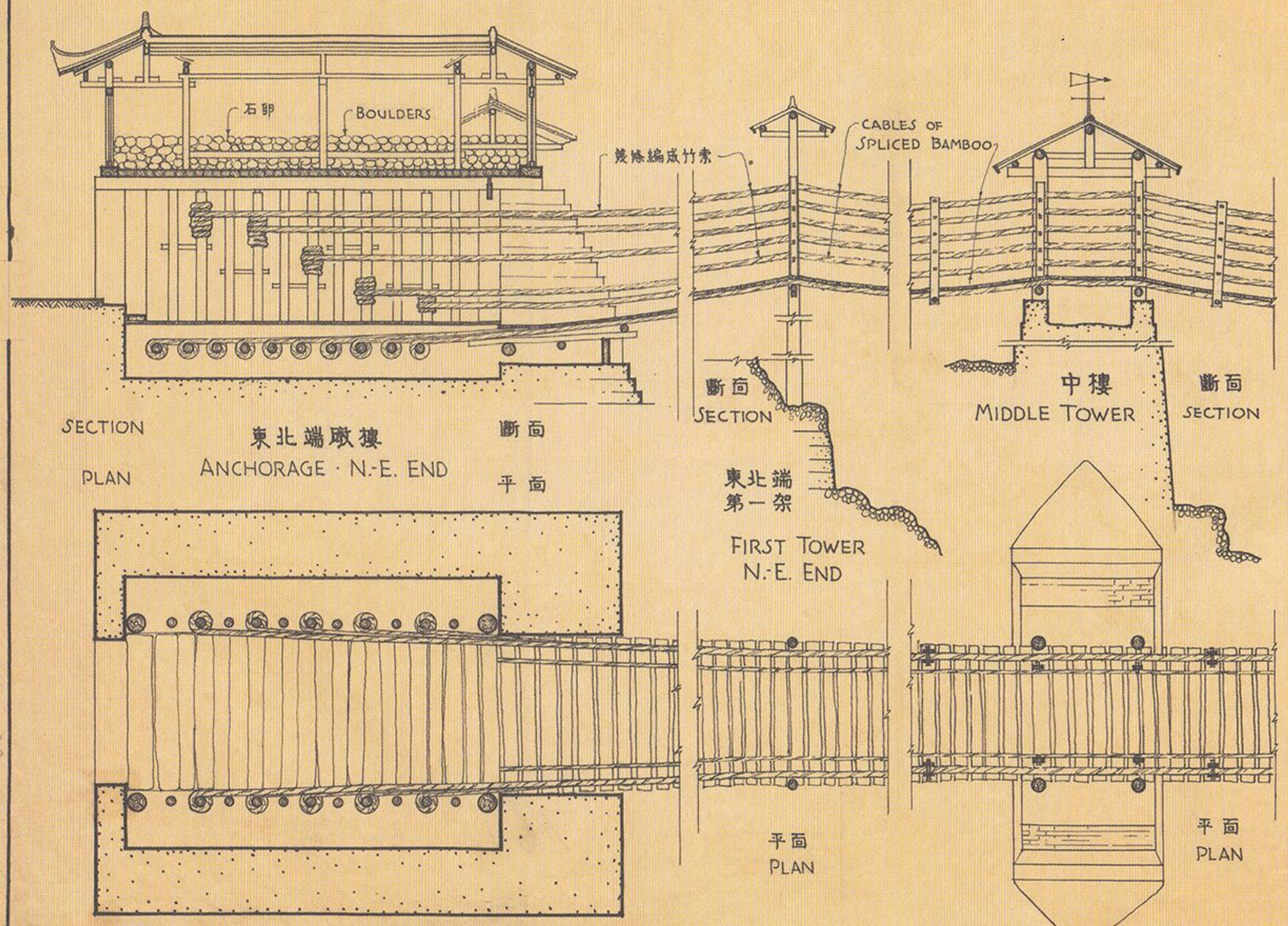
PLAN · PIER LEVEL 金剛牆平面 橋面平面 PLAN · ROADWAY LEVEL



N.E. ANCHORAGE

東南面立面 SOUTH-EAST ELEVATION

S.W. ANCHORAGE



四川灌縣

安瀾橋

(竹索橋)

清嘉慶八年倣宋舊制建
以後每年培修

AN-LAN CH'IAO

BAMBOO SUSPENSION BRIDGE

KUAN HSIEN SZECHUAN

CH'ING DYNASTY · 1803
AFTER A SUNG DYNASTY DESIGN
ANNUALLY REPAIRED

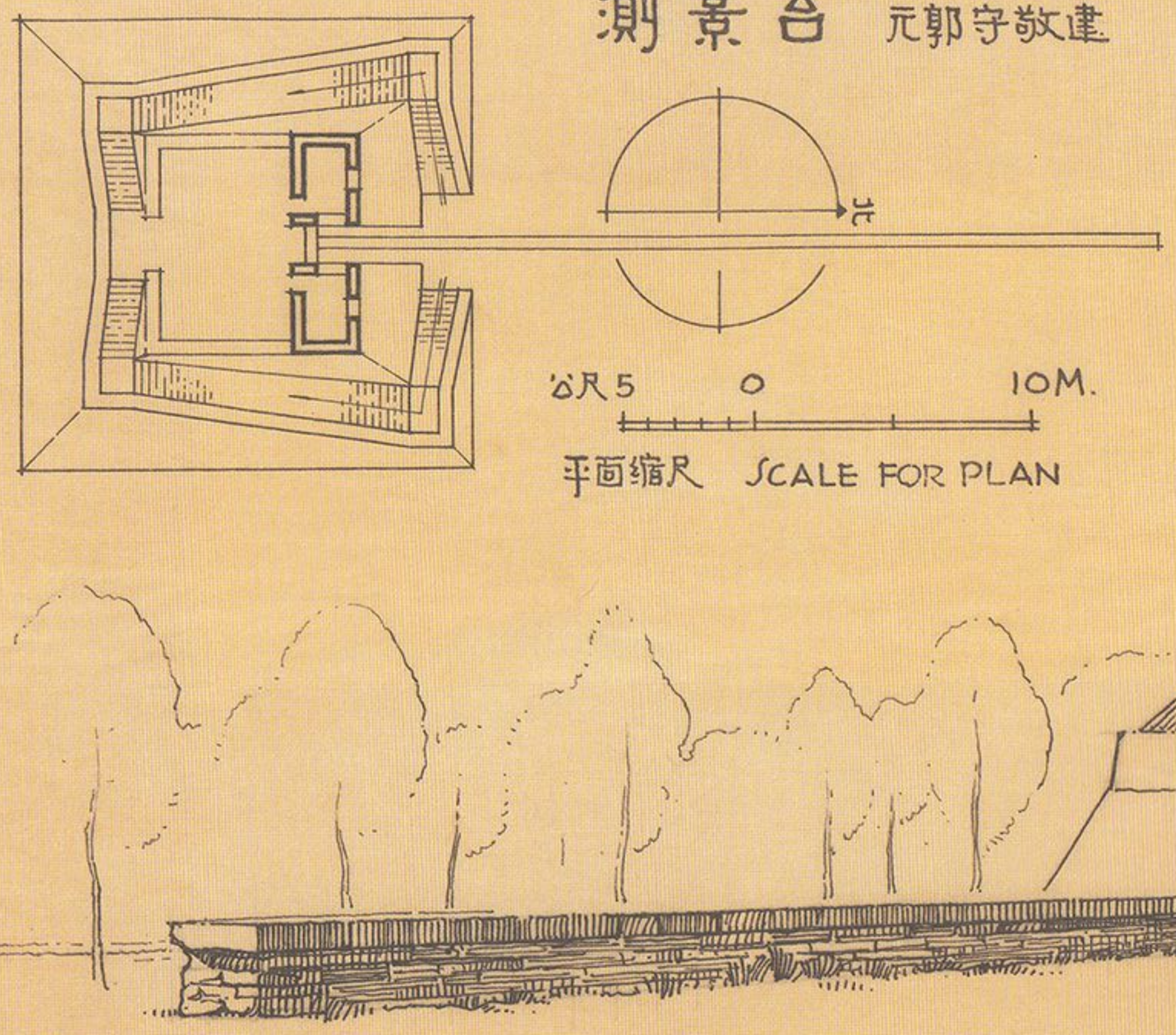
全橋立面縮尺
公R 10 0 50 meters
SCALE FOR ELEVATION

平面斷面縮尺
公R 1 0 5 meters
SCALE FOR PLAN AND SECTION

P18V 32/2

河南登封縣告成鎮 元郭守敬建 測景台

CH'ÊH-CHING T'AI, KAO-CH'ENG CHEN, TENG-FENG, HONAN
AN OBSERVATORY OF THE YUAN DYNASTY
CIRCA 1300 A.D



平面縮尺 SCALE FOR PLAN

劉敦楨測繪

MEASURED BY LIU T.T.

DUKU
讀庫

策劃統籌　張立憲
圖片編輯　黎　亮
美術編輯　耿　冰
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責任印製　田　歌

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